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TECHNICAL REPORT

Technical Report on self-interference risk evaluation and sharing mechanisms for automotive radar in the 122,25 - 130 GHz and 134 - 148,5 GHz ranges

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Foreword

This Technical Report (TR) has been produced by ETSI Technical Committee Electromagnetic compatibility and Radio spectrum Matters (ERM).

Modal verbs terminology

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Introduction

ECC Report 351 [i.1] built upon the work from ECC Report 334 [i.2] to investigate the conditions under which vehicular radars could operate in 122,25 - 130 GHz and 134 - 148,5 GHz without causing harmful interference to other radio services. Those conditions are reflected in the regulatory framework published in the ECC Decision (22)03 [i.3], CEPT Report 86 [i.4] and EC Decision 2026/883 [i.5].

While the studies in the ECC Report 351 consider sharing and compatibility with other radio services, the present document considers the risk of interference between automotive radars operating in 122,25 - 130 GHz, or 134 - 148,5 GHz according to ECC Decision (22)03 [i.3] and EC Decision 2026/883 [i.5].

1 Scope

The present document provides an evaluation of the risk of automotive radars self-interference, and a survey on interference mitigation techniques and sharing mechanisms that may be applicable to exterior and in-cabin automotive radars, for the case of transmissions in the bands studied in ECC Report 351 [i.1] (122,25 - 130 GHz, 134 - 148,5 GHz).

Within the present document, only the interference between radars on different vehicles is considered.

2 References

2.1 Normative references

Normative references are not applicable in the present document.

2.2 Informative references

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The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ECC Report 351: "UWB radiodetermination applications within the frequency range 116 GHz to 148,5 GHz for vehicular use", approved 3 February 2023.

[i.2] ECC Report 334: "UWB radiodetermination applications in the frequency range 116-260 GHz", approved January 2022, amended 3 February 2023.

[i.3] ECC/DEC/(22)03: "Technical characteristics, exemption from individual licensing and free circulation and use of specific radiodetermination applications in the frequency range 116-260 GHz", 18 November 2022, as updated 6 March 2026.

[i.4] CEPT Report 86: "In response to the EC Permanent Mandate to CEPT regarding the regular update of the technical annex of the Commission Decisions on harmonisation of radio spectrum for use by Short Range Devices (SRD), "Harmonised technical parameters for SRD radiodetermination applications in the frequency range 116-260 GHz"".

[i.5] Commission Implementing Decision (EU) 2026/883 of 21 April 2026 on the harmonisation of radio spectrum for use by radiodetermination applications in the 116-260 GHz frequency band (notified under document C(2026) 2519).

[i.6] ETSI EN 303 883-1 (V1.2.1) (2021-02): "Short Range Devices (SRD) and Ultra Wide Band (UWB); Part 1: Measurement techniques for transmitter requirements".

[i.7] FCC 2017-18463: "FCC, Permitting Radar Services in the 76-81 GHz Band".

[i.8] RSS-251: "Vehicular Radar and Airport Fixed or Mobile Radar in the 76-81 GHz Frequency Band".

[i.9] ANATEL issue Act 4776 amending Act No. 14,448 concerning the technical requirements for the conformity assessment of restricted radiation radio communication equipment, 23 September 2020.

[i.10] ERC/REC 70-03: "Relating to the use of Short Range Devices (SRD)", 6 October 1997, latest amended on 13 February 2026 (FM#112).

[i.11] ECC/DEC/(04)03: "The frequency band 77-81 GHz to be designated for the use of Automotive Short Range Radars", 19 March 2004.

[i.12] ETSI TR 103 593 (V1.1.1) (2020-05): "System Reference document (SRdoc); Transmission characteristics; Technical characteristics for radiodetermination equipment for ground based vehicular applications within the frequency range 77 GHz to 81 GHz".

[i.13] ECC Report 350: "Radiodetermination equipment for ground based vehicular applications in 77-81 GHz".

[i.14] MOre Safety for All by Radar Interference Mitigation (MOSARIM).

[i.15] MOSARIM D.6.1.1: "Final Report".

[i.16] Interference Mitigation by Cooperation in Radar for Autonomous Electric Cars (IMIKO-Radar).

[i.17] ETSI EN 302 637-2 (V1.4.1) (2019-04): "Intelligent Transport Systems (ITS); Vehicular Communications; Basic Set of Applications; Part 2: Specification of Cooperative Awareness Basic Service".

[i.18] ETSI EN 302 663 (V1.3.1) (2020-01): "Intelligent Transport Systems (ITS); ITS-G5 Access layer specification for Intelligent Transport Systems operating in the 5 GHz frequency band".

[i.19] Recommendation ITU-R M.2057-1: "Systems characteristics of automotive radars operating in the frequency band 76-81 GHz for intelligent transport systems applications".

[i.20] Simulation of Urban MObility (SUMO).

[i.21] P. A. Lopez et al.: "Microscopic Traffic Simulation using SUMO", ITSC, Maui, HI, USA, 2018.

[i.22] WiLabVIsim.

[i.23] Traffic Control Interface (TraCI).

[i.24] H. Buddendick and T. F. Eibert: "Incoherent scattering-center representations and parameterizations for automobiles [EM Programmer's Notebook]", in IEEE™ Antennas and Propagation Magazine, vol. 54, no. 1, pp. 140-148, February 2012.

[i.25] Barton, David Knox: "Radar Equations for Modern Radar", Artech House Radar Series. Boston, Mass: Artech House, 2013.

[i.26] J. Overdevest, F. Laghezza, F. Jansen and A. Filippi: "Radar Waveform Coexistence: Interference Comparison on Multiple-Frame Basis", 2020 17th European Radar Conference (EuRAD), Utrecht, Netherlands, 2021, pp. 168-171, doi: 10.1109/EuRAD48048.2021.00052.

[i.27] ETSI TS 103 613 (V1.1.1) (2018-11): "Intelligent Transport Systems (ITS); Access layer specification for Intelligent Transport Systems using LTE Vehicle to everything communication in the 5,9 GHz frequency band".

[i.28] A. Bazzi, F. Miccoli, F. Cuccoli, L. Facheris and V. Martinez: "Analysis of Proactive Uncoordinated Techniques to Mitigate Interference in FMCW Automotive Radars", in IEEE™ Transactions on Radar Systems, vol. 4, pp. 597-612, 2026, doi: 10.1109/TRS.2026.3670939.

[i.29] ETSI TS 103 789 (V1.1.1) (2023-05): "Short Range Devices (SRD) and Ultra Wide Band (UWB); Radar related parameters and physical test setup for object detection, identification and RCS measurement".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the following terms apply:

attacker: interfering transmitter, potentially disturbing the victim's signal and reducing its performance

NOTE: No hostile intent is implied.

beat frequency: product of the mixing of received signal and the transmitted signal in the receiver signal chain

car exit loss: attenuation experienced by a signal passing through a car body

chirp: linear frequency sweep transmitted by an FMCW radar

chirp slope: ratio of the chirp's signal bandwidth to sweep time

collision: occurrence when at least 50 % of an attacker's chirp falls inside the ADC bandwidth of the victim

corner radar: radar typically used for medium to short range detection, with medium FoV, mounted with ±45° azimuth angle with respect to the front and back of the vehicle, for applications such as blind spot detection

direct interferer: interfering radar in LOS conditions

duty cycle: time ratio between active and inactive transmission states

equivalent distance: distance at which a direct interferer would need to be located in order to produce the same received interference power at the victim radar as that produced by a given interferer

first reflection interferer: interfering radar reaching the sensor after a single reflection

frame: train of FMCW chirps (referred to as T_{on}) followed by an empty interval (referred to as T_{off})

frequency hopping: technique of changing the starting frequency of radar chirps, either on a chirp-by-chirp basis, or on a frame basis

front radar: radar typically used for mid to long range detection, with narrow to medium FoV, mounted with 0° azimuth angle with respect to the motion of the vehicle, for applications such as adaptive cruise control

interferer: another transmission overlapping in time and frequency with the target radar under study

penetration rate: percentage of vehicles equipped with radar sensors

potential interferer: radar sensor that could interfere with the victim radar, should they use the same frequencies at the same time

short/ultra-short range radar: radar typically used for short range detection, with medium to wide FoV, mounted all around the vehicle for instance, for applications such as 360° cocoon sensing around the vehicle

system failure: loss of more than 20 % chirps in a frame for at least 3 consecutive frames

victim: target radar under study, potentially interfered by the attacker(s)

3.2 Symbols

For the purposes of the present document, the following symbols apply:

$1_{failure}^{(k)}(i)$	Indicator function for system failure
B	Signal bandwidth of transmitted signal
B_{ADC}	Analog to digital conversion bandwidth
B_{ch}	Chirp bandwidth
B_{TOT}	Total available bandwidth
d	Total distance between vehicles
d_1	Path length from interferer to point of reflection
d_2	Path length from point of reflection to victim
d_H	Horizontal distance between vehicles
d_{ref}	Equivalent distance
d_V	Vertical distance between vehicles
f_c	Centre frequency
$f_{c,band}$	Band centre frequency
f_{max}	Maximum frequency
f_{min}	Minimum frequency
G_r	Receive antenna gain
K	Total number of radars in simulation
K_{CD}	Collision detection threshold
K_{ch}	Percentage of chirps lost per frame threshold

M_f	Number of consecutive frames lost threshold
$N^{(k)}$	Total number of frames transmitted by the k-th radar
N_{ch}	Number of chirps in a frame
N_{int}	Number of interfering radars
$N_{sys-fail}^{(k)}$	Number of system failures for the k-th radar
P_{fail}	Probability of having a system failure
P_{int}^{th}	Power threshold for collision avoidance
T_{ch}	Chirp transmitting time
T_{fail}	Average time between two system failures
T_{rch}	Chirp repetition interval (including reset time)
T_{rf}	Frame repetition time
T_{on}	Active part of the frame
T_{off}	Inactive gap between the transmissions
x_{chirp}	Percentage threshold for collision classification (set to 0,5)
α	Angle of direct line between IVR and EVR / vehicle direction
δ	Duty cycle
λ	Wavelength
σ	Radar Cross Section

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

ACC	Adaptive Cruise Control
ADC	Analog to Digital Conversion
BW	Bandwidth
CA	Collision Avoidance
CAM	Cooperative Awareness Message
CbC	Chirp-by-Chirp (frequency hopping)
CD	Collision Detection
CEPT	Conférence Européenne des Postes et Télécommunications
CFAR	Constant False Alarm Rate
CSMA-CA	Carrier Sense Multiple Access with Collision Avoidance
dBi	decibels relative to isotropic antenna
dBm	decibels relative to milliwatt
DSH	Discrete Subband Hopping
EC	European Commission
ECC	Electronic Communications Committee
EIRP	Equivalent Isotropically Radiated Power
ETSI	European Telecommunication Standards Institute
EVR	Exterior Vehicular Radar
FbF	Frame-by-Frame (frequency hopping)
FbF-CA	Frame-by-Frame with Collision Avoidance
FbF-CD	Frame-by-Frame with Collision Detection
FbF-DSH	Frame-by-Frame with Discrete Subband Hopping
FMCW	Frequency Modulated Continuous Wave
FOV	Field Of View
FSPL	Free Space Path Loss
GHz	Gigahertz
I/N	Interference-to-Noise ratio
IF	Intermediate Frequency
IMIKO	Interference Mitigation by Cooperation in Radar for Autonomous Electric Cars
ITS	Intelligent Transport Systems
ITS-G5	ETSI Intelligent Transport Systems framework series of standards
ITU-R	International Telecommunication Union - Radiocommunication sector
IVR	In-cabin Vehicular Radar
LOS	Line Of Sight
LTE	Long Term Evolution
MHz	Megahertz
MIMO	Multiple Input Multiple Output
MOSARIM	MOre Safety for All by Radar Interference Mitigation

NLOS	Non-Line Of Sight
PRI	Pulse Repetition Interval
PSD	Power Spectral Density
RCS	Radar Cross Section
RX	Receiver
SE24	Spectrum Engineering group responsible for compatibility studies related to Short Range Devices

NOTE: Within ECC.

SINR	Signal-to-Interference-plus-Noise Ratio
SNR	Signal-to-Noise Ratio
SRD	Short Range Device
SUMO	Simulation of Urban MObility
TraCI	Traffic Control Interface
TX	Transmitter
UWB	Ultra Wide Band
V2X	Vehicle-to-Everything

4 V2X Vehicle to Everything Vehicular radar applications in these bands

4.1 Exterior vehicular radar

4.1.1 Introduction

The D-band (120 - 150 GHz) exterior vehicular radar applications were introduced in ECC Report 351 [i.1], and studied for compatibility with other services operating in similar or adjacent bands. To perform different functionalities for driving assistance, cars are equipped with different types of radars that are integrated in specific positions onboard the vehicle.

Front and corner radars are used for driving assistance applications requiring long and medium range such as automatic cruise control, lane keep, lane change assist, automatic emergency braking, etc.

On the other hand, applications providing the vehicle with higher degree of autonomy require short and ultra-short-range radars for front, side and rear-view, such that 360° sensing is enabled. Those radars allow to obtain a wide field of view (elevation and azimuth) in the close proximity of the vehicle and enables features like automated parking assistance or autonomous valet parking. To perform such features, the short-range detection radars should be capable to detect a wide range of objects different in nature to those traditionally detected for driving assistance radars (pedestrian, bicycles, vehicles, etc.).

Technical characteristics of these exterior radars are listed in ECC Report 351 [i.1], section 2.1.

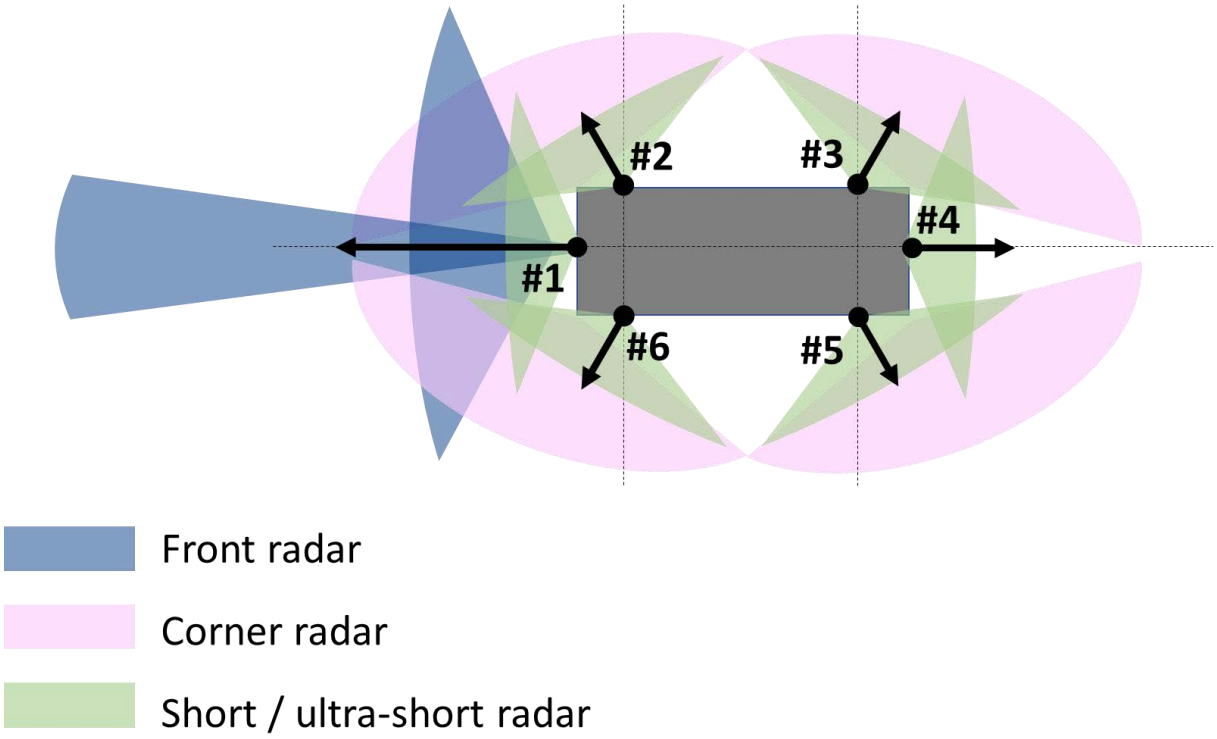


Figure 1: Exterior vehicular radar types

4.1.2 Radar cross sections

The present document uses a radar cross section of -10 dBm^2, based on Table 18 from ECC Report 334 [i.2], where a range of cross sections between -30 dBm^2 and -10 dBm^2 is quoted.

4.1.3 Antenna patterns

As in ECC Report 351 [i.1], the present document bases the EVR antenna patters on Recommendation ITU-R M.2057-1 [i.19].

4.1.4 EVR configurations

The regulations provide for three types of EVR: front, corner and short/ultra-short range radars.

Reference parameterizations are given in ECC Report 351 [i.1], Tables 3, 4 and 5. Based on those values, this report considers the following configurations:

Table 1: EVR configurations

	Mean EIRP (T_{on})	Bandwidth	Duty cycle	ECC Report 351
Front radar	+35 dBm	1 GHz	50 %	Table 3
Corner radar	+15 dBm	4 GHz	25 %	Table 4
Short/ultra-short	+15 dBm	7 GHz	25 %	Table 5

For the single-entry studies with EVR considered as victim (see clause 7.4) and for the EVR-EVR (see clause 8), three extra configurations with minimal bandwidth have been studied. Those reduce the bandwidth to 150 MHz for front radar, 1,5 GHz for corner radars and 3 GHz for short/ultra-short radars respectively.

4.1.5 Interference criterion

As in ECC Report 334 [i.2], the present document assumes a minimum SNR of 15 dB is required.

4.2 In-cabin vehicular radar

4.2.1 Introduction

The in-cabin vehicular radar applications were introduced in ECC Report 351 [i.1].

In-cabin applications include contactless gesture control, presence detection (including baby/child detection) and vital sign monitoring such as respiration rate, heart rate and heart rate variation. The use of higher frequency ranges further reduces the risk of interference with other automotive radars (e.g. 77 GHz or 79 GHz radars) or wireless communication devices using the 60 GHz band. With the increasing miniaturization, angular resolution offers the possibility to discriminate between multiple seats inside a car with a single radar sensor with beamforming or MIMO capability.

Technical characteristics of these exterior radars are listed in ECC Report 351 [i.1], section 2.2.

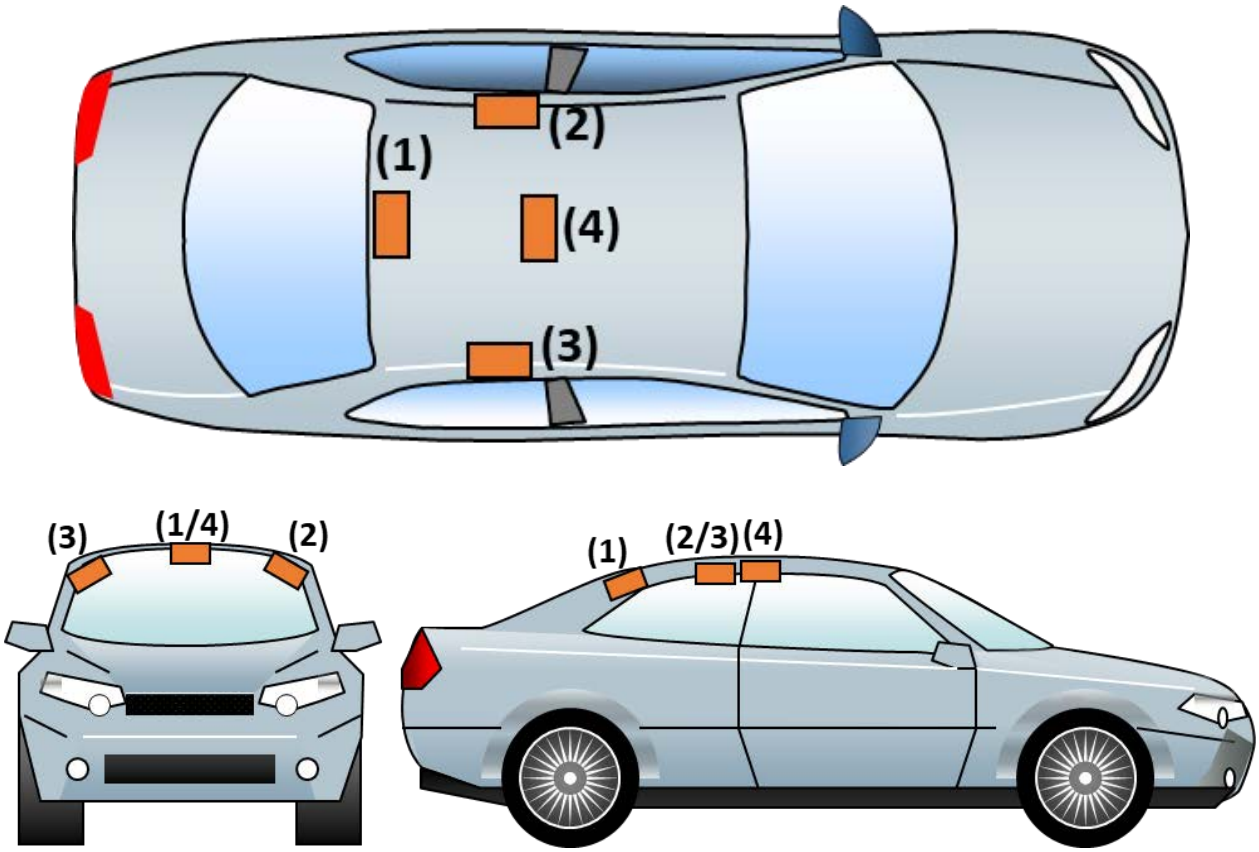


Figure 2: Examples of possible mounting positions for in-cabin radar sensors

4.2.2 Radar cross sections

To estimate the desired signal strength at the receiver, the calculations in the present document start from the radar cross sections of living objects in ETSI TS 103 789 [i.29]. For the 60 - 77 GHz range, radar cross sections there are around -15 dBm^2 for presence detection applications and around -20 dBm^2 for vital signs monitoring.

To scale the radar cross sections to the 140 GHz frequency range considered here, the list of radar targets and related RCS from Annex A in ETSI TS 103 789 [i.29] have been considered. Annex A lists the analytical formulas for the RCS of several shapes. The RCS for trihedral corner targets (whether triangular, square or round), flat plat targets and dihedral corner targets all scale by a factor of λ^2 in the denominator. Cylindrical corner targets have a factor of λ in the denominator of the RCS formula. Spherical and ellipsoid targets have an RCS independent of λ, while conic corner targets contain λ^2 in the numerator of the RCS formula.

For a frequency ratio of 80/132,25, this implies that the higher frequency would have an RCS ratio as expressed in the following table:

Table 2: RCS scaling with frequency

Wavelength dependence	RCS ratio
"λ²" in the denominator of the RCS formula	+4,366 dB
"λ" in the denominator of the RCS formula	+2,183 dB
"λ" not in the RCS formula	0 dB
"λ²" in the numerator of the RCS formula	-4,366 dB

Given most shapes have λ or λ^2 in the denominator, it was agreed to use an offset of +2 dB for the 140 GHz range compared to the values from ETSI TS 103 789 [i.29]. Based on this, the present document uses an RCS of -18 dBm^2 as worst-case for vital signs monitoring.

4.2.3 Antenna pattern

Like in ECC Report 351 [i.1], no particular antenna pattern was used. Instead, based on the measurements summarized in Table 9 of ECC Report 351 [i.1], the car exit loss of 18 dB for 0° is used here.

4.2.4 IVR configurations

Table 8 in ECC Report 351 [i.1] list representative values for parameters of IVR. In this report, the extremes of the quoted sweep time and signal bandwidth were considered, leading to the following four configurations:

Table 3: IVR configurations

IVR configuration	config 1	config 2	config 3	config 4
sweep-time	10 µs	15 ms	10 µs	15 ms
signal bandwidth (e.g. FMCW chirp span)	2 GHz	2 GHz	14 GHz	14 GHz
cycle time	5 ms	250 ms	5 ms	250 ms

4.2.5 Interference criterion

As in ECC Report 334 [i.2], the present document assumes a minimum SNR of 15 dB is required.

4.3 Applicable regulations

The technical conditions applicable to vehicular radars operating in 122,25 - 130 GHz and 134 - 148,5 GHz are specified in ECC Decision (22)03 [i.3], Annexes 2.7 and 2.8, and are copied here in Table 4 (EVR) and Table 5 (IVR) for completeness.

Table 4: Technical requirements for Exterior Vehicular Radars (EVR)

Designated frequency band	Front radars		Corner and short/ultra-short range radars	
	Maximum mean EIRP (note)	Maximum peak EIRP	Maximum mean EIRP (note)	Maximum peak EIRP
122,25 - 130 GHz	32 dBm		9 dBm	
134 - 141 GHz	32 dBm		9 dBm	
141 - 148,5 GHz	-6 dBm within 1 GHz	-1 dBm within 1 GHz	-6 dBm within 1 GHz	-1 dBm within 1 GHz
NOTE: Maximum mean EIRP is the radiated mean power during transmitter signal repetition time as defined in ETSI EN 303 883-1 [i.6], clause 5.3.1 with the effect of duty cycle included.				

Table 5: Technical requirements for In-cabin Vehicular Radars (IVR)

Designated frequency band	Maximum mean EIRP density	Maximum mean EIRP over the bandwidth	Maximum peak EIRP over the bandwidth
122,25 - 130 GHz	-30 dBm/MHz	3 dBm	16 dBm
134 - 148,5 GHz	-30 dBm/MHz	3 dBm	16 dBm
NOTE: Maximum mean EIRP is the radiated mean power during transmitter signal repetition time as defined in ETSI EN 303 883-1 [i.6], clause 5.3.1 and Annex C meaning the effect of duty cycle is included.			

5 Related work

5.1 Situation in current 76 - 81 GHz frequency bands

Exterior vehicular radars are today deployed in most countries within the 76 - 81 GHz frequency band. In certain regions regulations describe only one 76 - 81 GHz block (e.g. USA [i.7], Canada [i.8], Brazil [i.9]). Other regions, like Europe, distinguish 76 - 77 GHz [i.10] from 77 - 81 GHz [i.11].

More specifically for Europe, ERC Recommendation 70-03 [i.10] defines a maximum mean EIRP of 50 dBm and a maximum peak EIRP of 55 dBm for 76 - 77 GHz. For 77 - 81 GHz, the current ECC Decision (04)03 [i.11] defines a maximum mean power density of -3 dBm/MHz EIRP It is for note that a new version of this ECC Decision is under development (in SE24 WI82), based on the SRDoc ETSI TR 103 593 [i.12] and ECC Report 350 [i.13].

5.2 Prior interference mitigation studies for 76 - 81 GHz

5.2.1 MOSARIM

The European funded project MOre Safety for All by Radar Interference Mitigation (MOSARIM) [i.14] started in January 2010 with the objective to investigate possible automotive radar interference techniques, using simulations and real-world tests. More than 20 mitigation strategies were investigated and 9 were identified as the most promising (see Figure 18 in [i.15]). Table 6 provides a summary of those, and an early indication whether such techniques could be re-used for 140 GHz radars. In the rightmost column, 'industry coordination' is used to indicate that this technique is only effective when implemented by most or all manufacturers.

Table 6: MOSARIM interference mitigation methods investigated

Mitigation strategy	Details	Efficacy	Concept potentially applicable to 140 GHz radars / industry coordination required
Constant False Alarm Rate (CFAR)	Adaptation of detection threshold, so that the false alarm rate stays at a certain value	10 - 20 dB	Yes / No
Detect interference and change transmit frequency	Changing transmit frequency range of FMCW chirps	up to infinity dB (note 2)	Yes / Yes
Randomize pause between chirps or pulses		few dB	Yes / No
Direction specific pre-defined frequency	Application of driving direction specific pre-defined frequency band separation	up to infinity dB (note 3)	Yes / Yes
Detect interference and repair	Signal processing type of repair at receive side	20 dB	Yes / No
Random sequence of chirp types (Up, Down, CW chirps)		few dB	No (note 1) / No
Detect interference and change timing of transmit chirp or pulses		large number of dB	Yes / TBD
NOTE 1: The industry seems to have settled to "up" chirps. NOTE 2: The efficacy depends on the occupied bandwidth/number of interferers. NOTE 3: It is for note that there is no mitigation margin when crossing traffic is considered.			

5.2.2 IMIKO

5.2.2.1 Overview

The Interference Mitigation by Cooperation in Radar for Autonomous Electric Cars (IMIKO-Radar) project [i.16] was a funded project by the German Federal Ministry of Education and Research which ran from 2018 to 2021. The first objective of this project was to understand and investigate the impact of mutual interference for automotive and road infrastructure radars. The second objective was the assessment of some interference mitigation cooperative methods (ranging from no cooperation measures to strong cooperation), both in terms of capability to meet system requirements as well as effectiveness. Table 7 provides a summary of the techniques studied, and an early indication whether such techniques could be re-used for 140 GHz radars. In the rightmost column, 'industry coordination' is used to indicate that this technique is only effective when implemented by most or all manufacturers.

Table 7: IMIKO interference mitigation methods investigated

Mitigation strategy type	Details	Applicability to 140 GHz radars	Industry coordination required
Detect and repair mechanisms (aka Healing)		Yes	No
Detect and avoid mechanisms	Adaptation of radar parameters (e.g. carrier frequency)	Yes	Partly
Resource allocation mechanisms	Random Hopping (Frequency, time, slope)	Yes	Yes
	(Sliding) Compass method	Yes	Yes
Strong Cooperation	Based on side channel communication (e.g. V2X)	Unlikely	N/A
	Central infrastructure orchestration	Unlikely	N/A

5.2.2.2 Random hopping

Random frequency hopping intends to average-out interference over time, by changing the starting frequency of the FMCW chirps. The sequences of starting frequencies may follow pseudo random patterns (which may be standardized). Effectively, the footprint of radars is spread over the whole spectrum. Hopping was considered at frame level in IMIKO, but could also be implemented at faster granularity (e.g. chirp-by-chirp). This technique may apply to all modulation schemes.

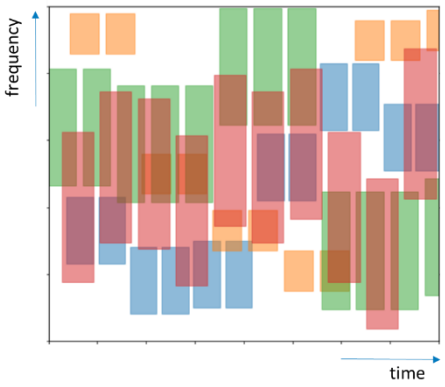


Figure 3: Example for Random Frequency Hopping (Source: Bosch, ERMTGUWB(24)000005)

Random time hopping intends to average-out interference over the time dimension, by randomizing the timing of FMCW chirp transmissions. This method was conceived primarily for FMCW type of radars. The objective is to avoid persistent specific interference over time cause by victim and interfere radars having the same (T_{on}, T_{off}) pattern. Introduction of randomness between FMCW chirps helps mitigate this issue.

Randomly varying of the chirp slope is an additional technique for mitigating persistent interference. This method applies only to FMCW type of radars.

5.2.2.3 Compass method

An early version of the Compass method was tested in MOSARIM [i.15] (referred to as "T3.4 Application of driving direction specific pre-defined frequency band separation"). In IMIKO [i.16] a refined version called "Sliding Compass Method" has been evaluated. The starting frequency is set according to the movement direction, according to equation (1):

$$f_c = f_{c,band} + \frac{\alpha}{2\pi}[f_{max} - f_{min} - B] \quad (1)$$

Where B is the signal bandwidth of the transmitted signal, α is the vehicle direction.

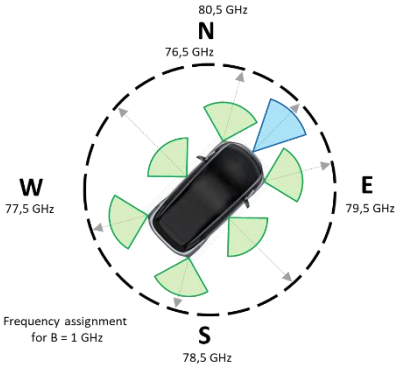


Figure 4: IMIKO, Sliding Compass method (Source: Bosch, ERMTGUWB(24)000005)

This method intends to address in priority the direct and high-power LOS interference (e.g. from vehicles coming in opposite direction). It requires harmonization within the vehicles to be efficient. This method applies to all modulation techniques.

A variation of the method called "channelized Compass" organized the 4 GHz between 77 GHz and 81 GHz into 4 channels of 1 GHz for use by the short and medium range radars, while the long-range front radars (with smaller signal bandwidth) are transmitted within 76 - 77 GHz.

One drawback of the Compass method is to reduce upfront the usable part of the medium for each radar, which may lead to performance degradation in case of un-balanced scenarios.

5.2.3 Other mitigation techniques

5.2.3.1 Splitting a band into several distinct blocks or defining subchannels

Some other interference mitigation techniques have been defined in other contexts or for some other applications but may also be relevant for the present document.

Splitting a band into several distinct blocks depending on device type or device class could be envisioned. This technique is relevant in case of strong interference between systems having different nature and parameterization, although at the expense of a reduced capacity for each system. This is not a recommended technique in general.

Another possibility is to define subchannels within a band, for example blocks of several hundred MHz. As an example, the ETSI ITS-G5 specification based on 3GPP LTE-V2X in ETSI TS 103 613 [i.27] defines that a 10 MHz channel is divided into 5 subchannels and each station can only use a given number of subchannels depending on the context.

5.2.3.2 Adapting transmission parameters depending on vehicle dynamics

These techniques are based on the hypothesis that some of the transmit parameters could be adapted depending on the context and/or vehicle dynamics.

For example, the ETSI ITS-G5 CAM generation trigger conditions as specified in ETSI EN 302 637-2 [i.17] depend on the vehicle dynamics, where a new message is generated in case of:

- Position: a change in position ≥ 4 m
- Heading: a change of direction ≥ ± 4°
- Speed: a change of speed ≥ 0,5 m/s
- Otherwise: generate a message after 1 second max

For instance, considering on the effect of the station's speed, the transmit rate increases as shown in Figure 5.

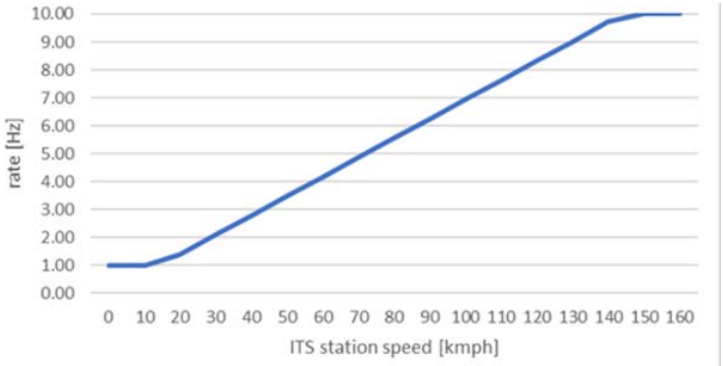


Figure 5: ITS-G5 CAM transmit rate vs ITS station's speed

For vehicular radars considered in the present document, similar techniques could be envisioned, for example by defining a maximum duty-cycle depending on the vehicle's speed.

5.2.3.3 Channel access and sensing mechanisms

Channel access and sensing mechanisms types of techniques are based on the hypothesis that before its transmission a radar device could sense the medium, and potentially adapt its parameters and/or defer its transmission until the medium is considered as 'free'.

For instance, ETSI ITS-G5 specification based on IEEE 802.11p in ETSI EN 302 663 [i.18] defines a channel access mechanisms based on CSMA-CA protocol (known as 'listen-before-talk'). A transmitter waits for the medium to appear as free, also based on randomness to avoid packets collisions.

6 Methodology

Two types of vehicular radars are considered in the present document: IVR and EVR. Considering that their respective application requirements (especially the targeted unambiguous range) and system parameters vastly differ, single-entry studies are conducted to assess the amount of interference from a potential IVR interferer to a victim EVR, and the other way around, as well as between two IVRs. In contrast, aggregate studies are conducted for assessment of interference between multiple EVRs. Table 8 provides a summary of the different types of assessment considered in the present document.

Table 8: Types of assessments considered in the present document

<table>
<tr><th colspan="2" rowspan="2">Victim
Interferer</th><th colspan="3">EVR</th><th rowspan="2">IVR</th></tr>
<tr><th>Front radar</th><th>Corner radar</th><th>Short/ultra-short radar</th></tr>
<tr><td rowspan="3">EVR</td><td>Front radar</td><td colspan="3" rowspan="3">Aggregate studies
(clause 8)</td><td rowspan="3">Single-entry study
(clause 7.5)</td></tr>
<tr><td>Corner radar</td></tr>
<tr><td>Short/ultra-short radar</td></tr>
<tr><td colspan="2">IVR</td><td colspan="3">Single-entry study
(clause 7.4)</td><td>Single-entry study
(clause 7.3)</td></tr>
</table>

7 Single-entry studies involving IVR

7.1 Introduction

Single-entry studies are conducted by means of spreadsheet calculators. They consider worst-case situations, where a single interferer is providing the maximum amount of interference. Computation assumptions include:

- Worst case in terms of geometry: position of the respective vehicles and subsequent radars antenna patterns
- Worst case in terms of transmit power, based on ECC Decision (22)03 [i.3]

System parameterization based on ECC Report 351 [i.1], as listed in clauses 4.1 and 4.2:

- Protection criteria of 15 dB SNR at the radar receiver
- Interference power assumed uncorrelated

7.2 IVR-EVR worst-case geometry

For the scenarios where interference between IVR and EVR are considered, the relative position which yields the highest coupling between the systems was investigated. The coupling is defined here as the combination of the EVR antenna patterns (either in transmit or receive position) and the free-space propagation path loss; both having a dependency on the distance between the two vehicles.

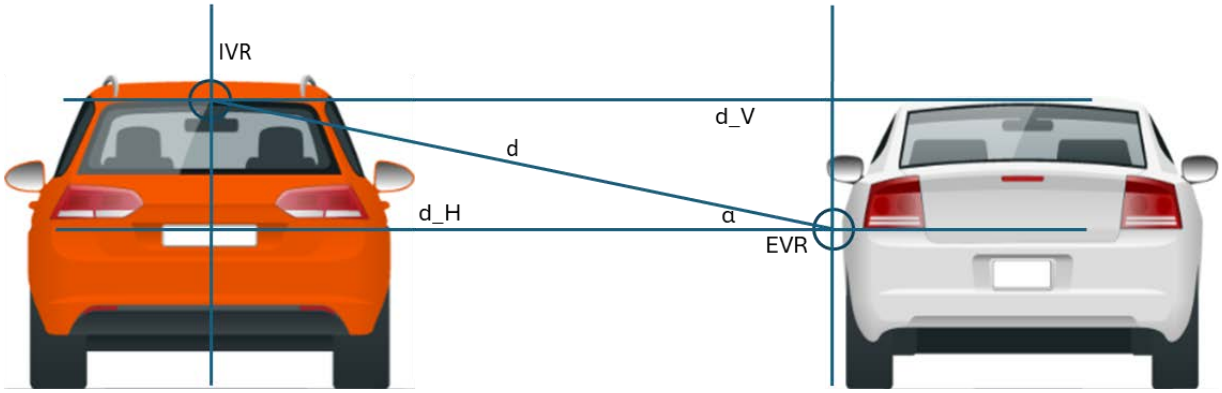


Figure 6: IVR-EVR geometry

The EVR antenna patterns are based on Recommendation ITU-R M.2057-1 [i.19], similarly as in ECC Report 351 [i.1]. It is assumed that the IVR is located at a height of 1,5 metre, while EVR is mounted at a height of 0,7 metre. Based on the horizontal distance (d_H) between the vehicles and thereby the angle of the direct line α, the attenuation of the link is calculated, as illustrated by Figure 6.

For type A (front-radar), Recommendation ITU-R M.2057-1 [i.19] specifies a peak gain of 30 dBi for both EVR in transmit or receive position. The maximum coupling (+4,8 dB) happens at a separation of 12,9 metres between the cars, as shown by Figure 7.

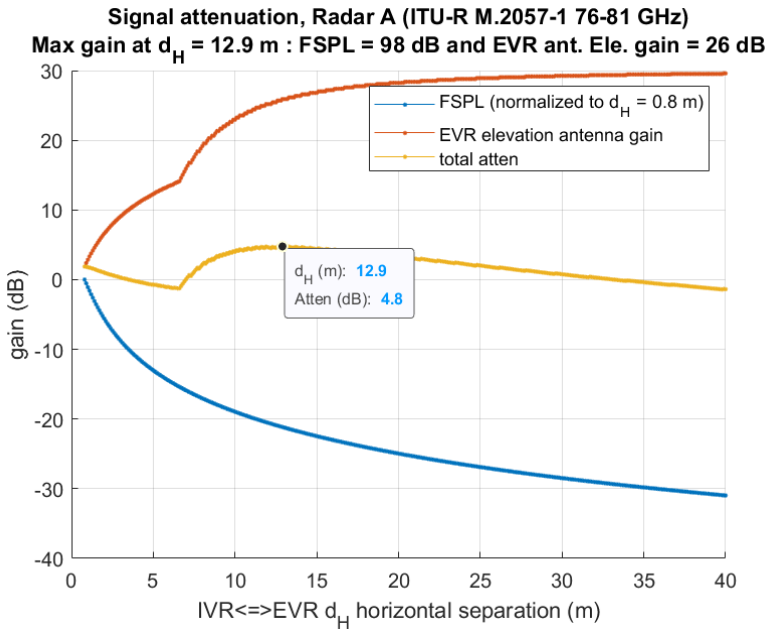


Figure 7: IVR-EVR front radar coupling versus distance

In accordance with Recommendation ITU-R M.2057-1 [i.19], a peak gain of 23 dBi is used for radar types C and E (corner and ultra-short radars, respectively) acting as a transmitter. The maximum coupling (+2,9 dB) happens at a distance of 7,2 metres, as shown in Figure 8.

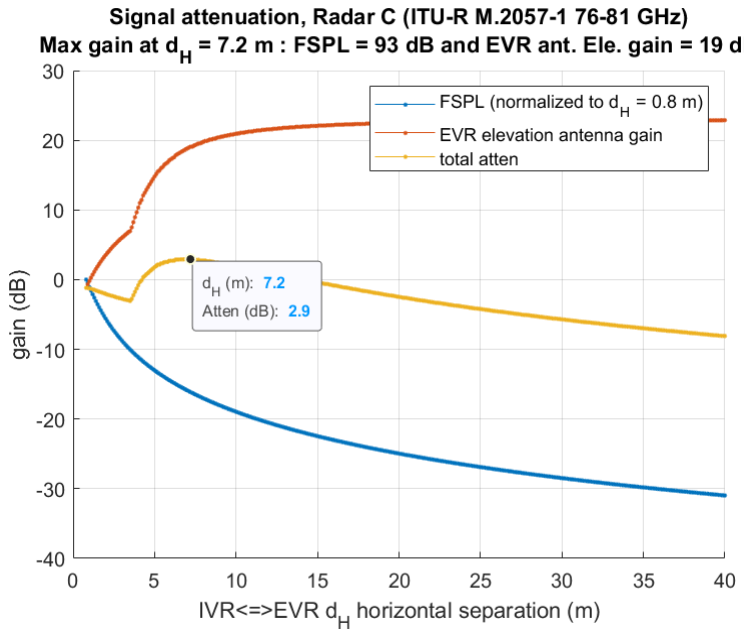


Figure 8: IVR-EVR corner and short/ultra-short radar TX coupling

When acting as a receiver, Recommendation ITU-R M.2057-1 [i.19] specifies a peak gain of 13 dBi for radar types C and E. The maximum coupling is reduced by 10 dB compared to Figure 8 (-7,1 dB), but still occurs at a distance of 7,2 metres, as shown in Figure 9.

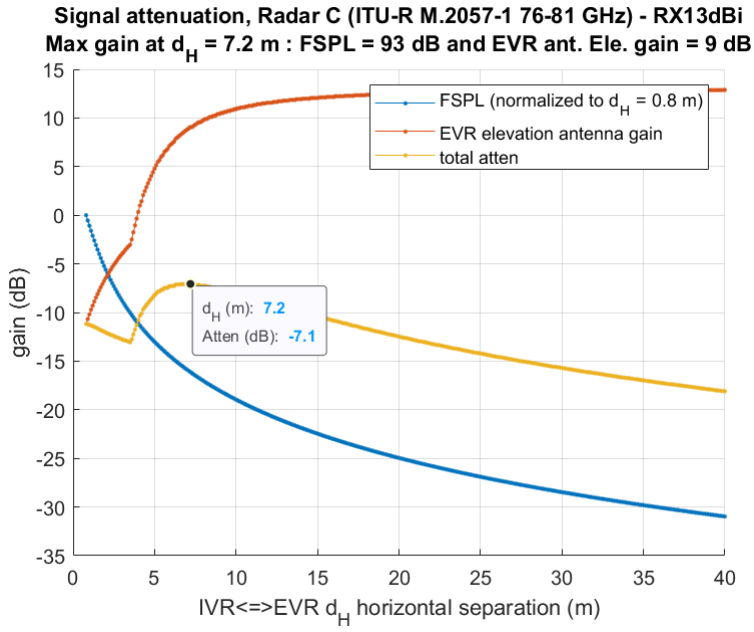


Figure 9: IVR-EVR corner and short/ultra-short radar RX coupling

7.3 IVR interference to IVR

There are four IVR configurations but 2 share the same signal bandwidth, meaning that 4 victim configurations need to be evaluated with 2 interferer signal bandwidths, resulting in 8 combinations. Details of the evaluation are contained in the attached Excel calculator, with an example detailed in Annex A.

Interference is evaluated in terms of Signal-to-Interference-plus-Noise Ratio (SINR). The results for the 8 combinations are listed in Table 9. As can be seen, for all combinations, the SINR has a large margin (e.g. at least 8,5 dB) compared to the +15 dB target. For the all the configurations, the impact of the IVR interferer on the IVR victim is negligible. The degradation due to interference is less than 0,2 dB, which is further verified by the I/N metric which yields values below -15 dB.

It is therefore concluded that even without mitigation, IVR systems will not be affected by interference from other IVR.

Table 9: IVR interference to IVR results

Victim config	Interferer bandwidth	SNR (without interference)	SINR (with interference)	Degradation due to interference	I/N
IVR config1 (2 GHz, 10 µs)	2 GHz	23,6 dB	23,5 dB	0,12 dB	-15,4 dB
	14 GHz	23,6 dB	23,6 dB	0,02 dB	-23,9 dB
IVR config2 (2 GHz, 15 ms)	2 GHz	40,8 dB	40,7 dB	0,12 dB	-15,4 dB
	14 GHz	40,8 dB	40,8 dB	0,02 dB	-23,9 dB
IVR config3 (14 GHz, 10 µs)	2 GHz	23,6 dB	23,5 dB	0,12 dB	-15,4 dB
	14 GHz	23,6 dB	23,5 dB	0,12 dB	-15,4 dB
IVR config4 (14 GHz, 15 ms)	2 GHz	40,8 dB	40,7 dB	0,12 dB	-15,4 dB
	14 GHz	40,8 dB	40,7 dB	0,12 dB	-15,4 dB

7.4 IVR interference to EVR

With 2 IVR signal bandwidths configurations and 6 EVR radars configurations (the 3 radar types configured first as per ECC Report 351 [i.1] representative of maximum signal bandwidth, and 3 additional configurations representative of alternative settings such as lower signal bandwidth and/or different antenna gain), 12 combinations have to be investigated. Details of the evaluation are contained in the attached Excel calculator, with an example detailed in Annex B.

Interference is evaluated in terms of Signal-to-Interference-plus-Noise Ratio (SINR). The results for the 12 combinations are listed in Table 10. For the all the configurations, the impact of the IVR interferer on the EVR victim is negligible. The degradation due to interference is less than 0,05 dB, which is further verified by the I/N metric which yields values below -20 dB. This is due to the low perceived RX interferer power (-89,6 dBm to -108,2 dBm) compared to the higher RX noise power (-69 dBm to -76 dBm). The EVR performance is limited by the noise but not by the interference.

It is therefore concluded that EVR will not be affected by interference from IVR systems.

Table 10: IVR interference to EVR results

EVR type	EVR BW	IVR BW	SNR (without interference)	SINR (with interference)	Degradation due to interference	I/N
front	1 GHz	2 GHz	21,0 dB	20,9 dB	0,04 dB	-20,2 dB
corner	4 GHz	2 GHz	15,1 dB	15,1 dB	0,01 dB	-27,8 dB
short	7 GHz	2 GHz	17,8 dB	17,8 dB	0,01 dB	-27,4 dB
front	1 GHz	14 GHz	21,0 dB	21,0 dB	0,01 dB	-28,6 dB
corner	4 GHz	14 GHz	15,1 dB	15,1 dB	0,00 dB	-33,3 dB
short	7 GHz	14 GHz	17,8 dB	17,8 dB	0,01 dB	-30,4 dB
front	0,15 GHz	2 GHz	21,0 dB	20,9 dB	0,04 dB	-20,2 dB
corner	1,5 GHz	2 GHz	17,6 dB	17,6 dB	0,02 dB	-23,7 dB
short	3 GHz	2 GHz	17,8 dB	17,7 dB	0,02 dB	-22,4 dB
front	0,15 GHz	14 GHz	21,0 dB	21,0 dB	0,01 dB	-28,6 dB
corner	1,5 GHz	14 GHz	17,6 dB	17,6 dB	0,00 dB	-35,1 dB
short	3 GHz	14 GHz	17,8 dB	17,8 dB	0,00 dB	-30,4 dB

7.5 EVR interference to IVR

With 4 IVR configurations and 3 EVR radar types, 12 combinations have to be assessed. Details of the evaluation are contained in the attached Excel calculator, with an example detailed in Annex C.

Interference is evaluated in terms of Signal-to-Interference-plus-Noise Ratio (SINR). The results for the 12 combinations are listed in Table 11. For 11 of the 12 cases, the 15 dB SINR target is met without the need for any mitigation. Corner and short-range EVR meet the criterion with 8,5 dB margin or more. EVR front radar causes more challenges, especially when the IVR has short chirps (configurations 1 and 3). In such cases, removal of the interfered samples ("option 2") improves significantly the results and the 15 dB SINR target is met with more than 6 dB of margin.

It is therefore concluded that the IVR will need to reject anomalous high-powered samples, and that with such rejection mechanism implemented, the IVR will not be affected by interference from EVR systems.

Table 11: EVR interference to IVR results

IVR config	EVR		Option 1: receiver chain without removal of the interfered samples			Option 2: receiver chain with removal of the interfered samples	
		SNR (without interference)	SINR (with interference)	Degradation due to interference	I/N	SNR (with samples removal)	Degradation due to interference
config 1 (2 GHz, 10 µs)	front	23,6 dB	11,9 dB	11,7 dB	11,4 dB	20,6 dB	3,01 dB
	corner		23,1 dB	0,5 dB	-9,6 dB		
	short		23,3 dB	0,3 dB	-12,0 dB		
config 2 (2 GHz, 15 ms)	front	40,8 dB	26,3 dB	14,6 dB	14,4 dB	37,8 dB	3,01 dB
	corner		40,0 dB	0,9 dB	-6,5 dB		
	short		40,3 dB	0,5 dB	-9,0 dB		
config 3 (14 GHz, 10 µs)	front	23,6 dB	17,1 dB	6,5 dB	5,4 dB	23,3 dB	0,32 dB
	corner		23,3 dB	0,2 dB	-12,6 dB	22,1 dB	1,46 dB
	short		23,3 dB	0,2 dB	-12,6 dB	20,6 dB	3,01 dB
Config 4 (14 GHz, 15 ms)	front	40,8 dB	26,3 dB	14,6 dB	14,4 dB	40,5 dB	0,32 dB
	corner		39,3 dB	1,6 dB	-3,5 dB	39,4 dB	1,46 dB
	short		39,3 dB	1,6 dB	-3,5 dB	37,8 dB	3,01 dB

7.6 Recommendations

From the results obtained in Table 11 and discussed in clause 7.5, it is recommended that IVR implement a mechanism to reject anomalous high-powered samples to avoid interference originating from high-power EVR front-radars, in case of IVR using configurations similar or close to config 1 (short signal bandwidth (e.g. 2 GHz) coupled with fast sweep (e.g. 10 µs).

8 Aggregate studies for EVR

8.1 Traffic modelling

8.1.1 Introduction

For EVR-to-EVR interference analysis, a simulation environment comprising different building blocks is used. The SUMO traffic simulator [i.20] is used to generate the distribution and movements of vehicles for different road scenarios. A MATLAB-based simulation environment, WiLabVIsim [i.22], exploits the positions of the vehicles derived from the SUMO road traffic simulator to calculate the direct and first reflection potential interferers. The MATLAB simulator and SUMO are connected through the TraCI interface [i.23].

8.1.2 SUMO traffic simulator

The road scenario characteristics (highway or urban, number of lanes, etc.), the distribution and movements of the vehicles is obtained by using the microscopic SUMO traffic simulator [i.20] as explained in [i.21] and illustrated by Figure 10.

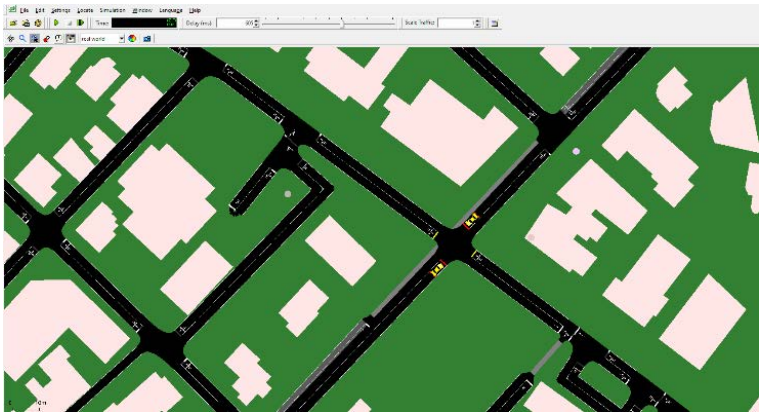


Figure 10: Example of an urban environment modelled by the SUMO traffic simulator

The use of SUMO guarantees that the positions and movements of the vehicles are realistically reproduced. Each vehicle is modelled as a rectangle of a given length and width. A number of vehicles are instantiated at the beginning of the simulation, corresponding to the desired density; then, a wrap-around principle is used to keep the number of vehicles always constant during the simulation. SUMO output granularity is typically set to 0,1 s.

Several scenarios are considered, based on the MOSARIM and IMIKO descriptions.

8.1.3 Road scenarios used in this study

8.1.3.1 Highway

This scenario intends to depict a situation with high radar density. A 2×3 lanes highway is considered, as illustrated by Figure 11.

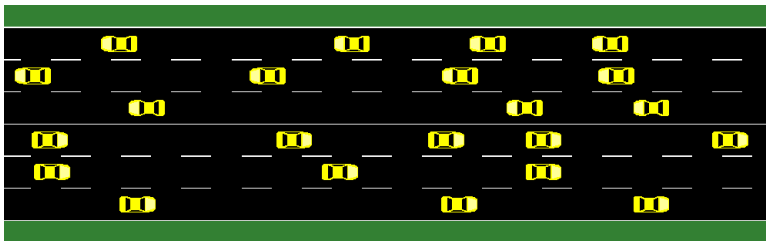


Figure 11: Highway scenario, example SUMO view

The vehicle density is set to one of the following values: 30, 60, 90, 120, 150, 210, 270 veh/km.

8.1.3.2 Asymmetric Highway

This scenario intends to depict a situation with high radar density on one direction, and low-density on the opposite direction. A one-side-congested 2×3 lanes highway is considered, as illustrated by Figure 12. This scenario is expected to be a worst-case for front radars.

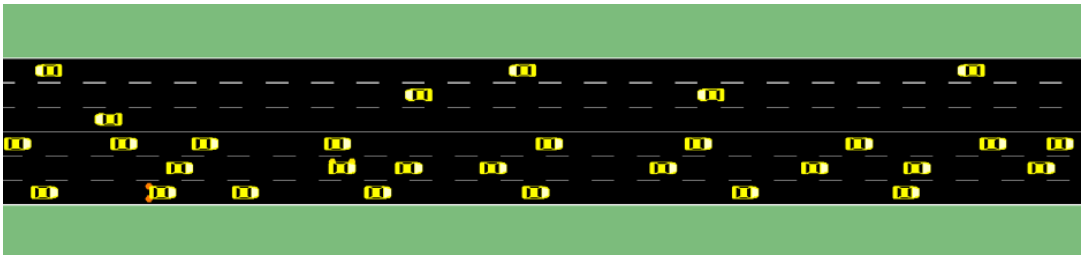


Figure 12: Asymmetric Highway scenario, example SUMO view

The density in the lightly loaded direction is fixed at 30 veh/km, while the one on the opposite and heavily loaded direction can be set to one of the following values: 30, 120 or 240 veh/km (leading to combined densities of 60, 150 and 270 veh/km).

8.2 Modulation agnostic interference analysis

8.2.1 Introduction

Before testing interference mitigation methods, a simulation environment is used to get some preliminary insights and a high-level understanding of the magnitude of the interference problem, such as how many interferers can be perceived from a victim radar, depending on road scenarios and their geometry. Numerical simulations are conducted in a MATLAB-based simulation environment, WiLabVIsim [i.22], that exploits the positions of the vehicles derived from the SUMO road traffic simulator to calculate the direct and first reflection potential interferers. The MATLAB simulator and SUMO are connected through the TraCI interface [i.23].

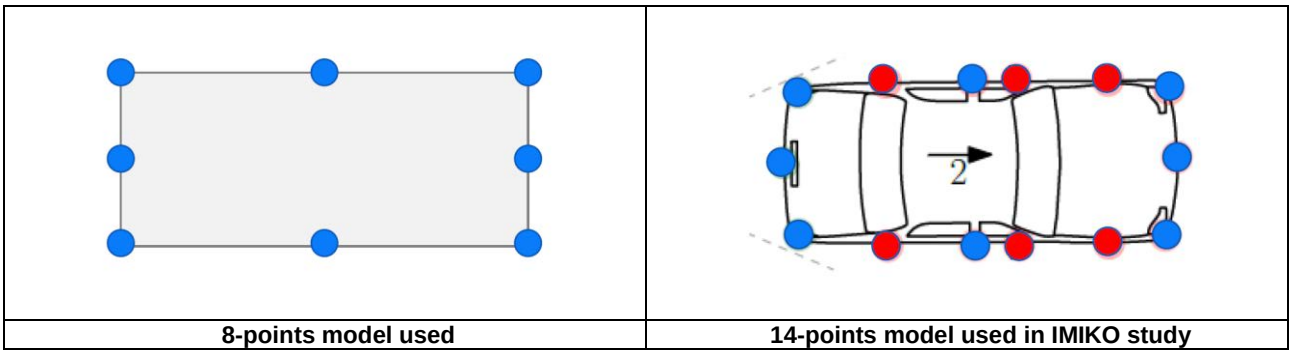
The simulator identifies the presence of potential interferers, calculating the characteristics of the geometric paths taken by the possibly interfering signals. The output of WiLabVIsim is independent of the specific radar signal settings, such as carrier frequency, waveform, bandwidth, signal processing algorithms used, or methods to reduce the effect of interference. A "potential interferer" situation is defined as situation where a radar sensor could interfere with the victim radar, should they use the same frequencies at the same time.

For the present study, it is assumed that all the vehicles are equipped with one or a set of sensors with the same characteristics. More specifically, each sensor is identified by a certain position relative to the car, a direction, and a Field Of View (FOV).

The simulator calculates the presence of signals in the receive field of view of the various sensors that either:

- Are in Line Of Sight (LOS) conditions and thus come directly from another sensor, called direct interferers; or
- Are in Non-Line Of Sight (NLOS) conditions but reach the sensor after a single reflection, called first reflection interferers.

An 8-points model is used for the analysis, in a fairly similar manner as used in the IMIKO study with 14 points (as suggested in [i.24]), as illustrated in Figure 13.



8-points model used	14-points model used in IMIKO study

Figure 13: Scattering modelling

The simulator considers FOV of 30° for front radars, 60° for corner radars. Figure 14 provides an exemplary view of the simulator output for a highway scenario at a specific instant. In this example, all the vehicles are equipped with corner radars; the vehicles coloured in blue have at least one of their radars potentially interfering directly with one of those of the tagged (red) vehicle; the vehicles coloured in green have at least one of the radars potentially interfering, not directly but after one reflection, with one of the radars of the tagged (red) vehicle. In this example, the tagged vehicle (in red) is thus potentially directly interfered by 5 vehicles, and potentially interfered by 9 vehicles by first reflection signal.

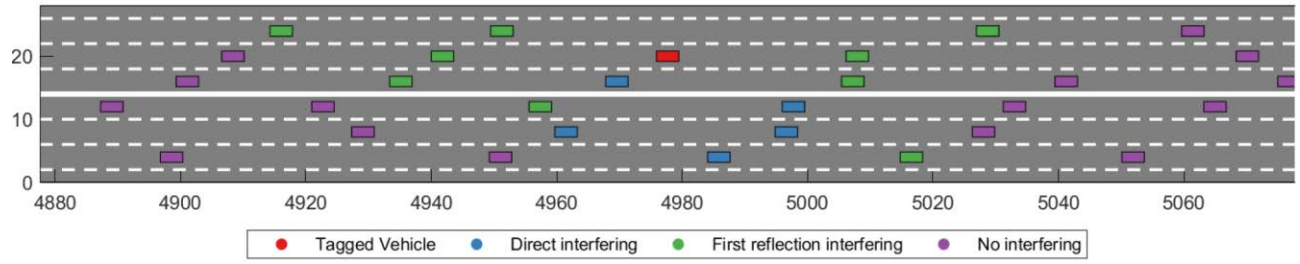


Figure 14: Example simulator output for a highway scenario with 150 veh/km² (corner radars)

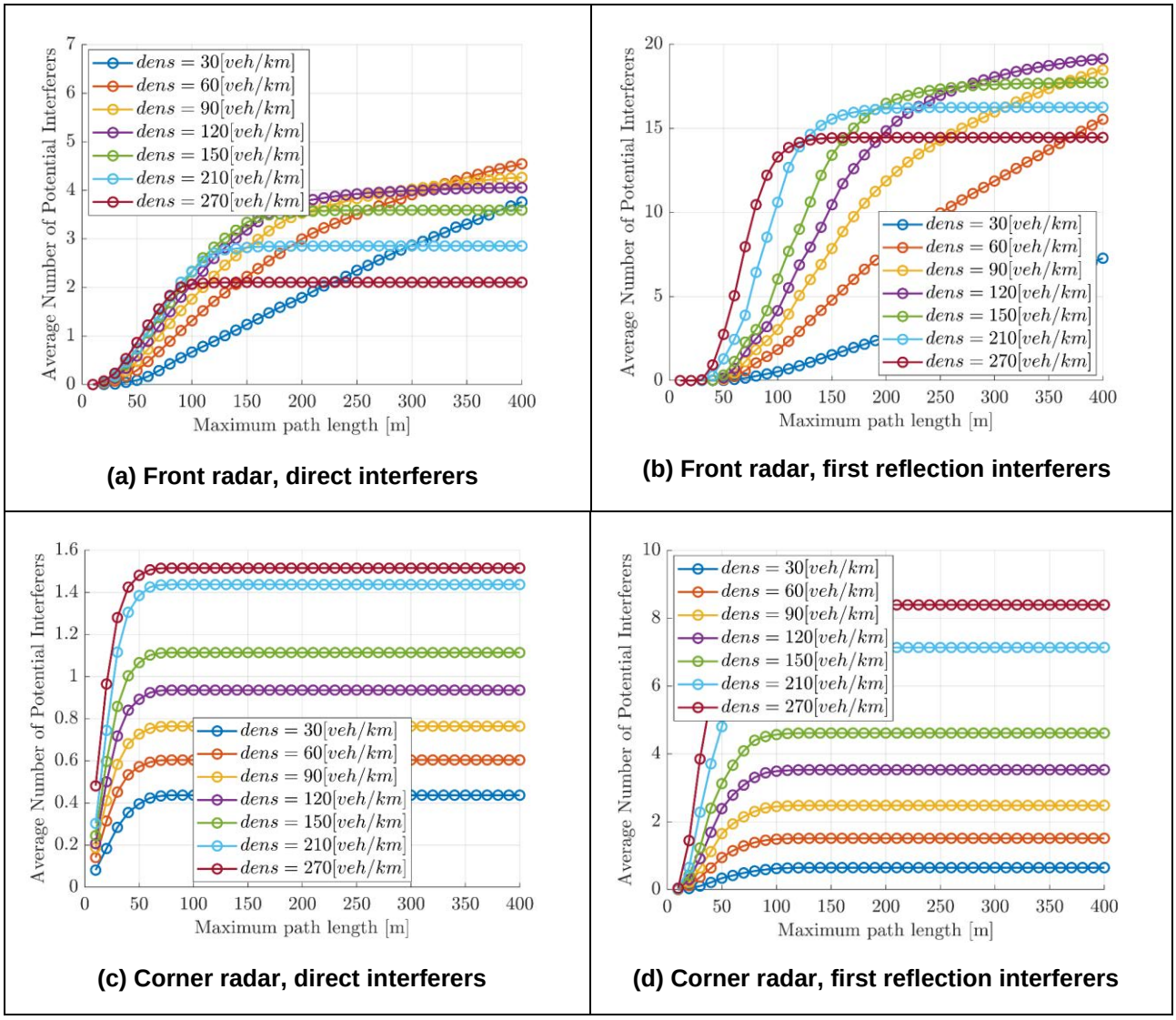
8.2.2 Metrics definition: "average number of potential interferers" and "distribution of the number of potential interferers"

The metric "average number of potential interferers" is computed from the SUMO simulation traffic logs and the scattering models. It represents the average, over the whole simulation time, and over all vehicles, of the number of the surrounding radars that can interferer with a victim radar.

It is usually shown as a function of the maximum path length. Unless otherwise specified, the average number of potential interferers is computed assuming all vehicles are exclusively equipped with either front or corner radars, and results are provided separating direct and first reflection interferers. The maximum path length restricts the potential interferers to those for which the path covered by the interfering signal does not exceed that value.

If the interference is direct, it means that the distance from the interfering radar to the interfered radar is no longer than the maximum path length. If the interference comes from a reflection, it means that the sum of the distance from the interfering radar to the point of reflection and the distance from the point of reflection to the interfered radar is no longer than the maximum path length.

Figure 15 illustrates this metric and provides the average number of potential interferers for a symmetrical highway scenario. Different curves are shown for different vehicle densities.



(a) Front radar, direct interferers

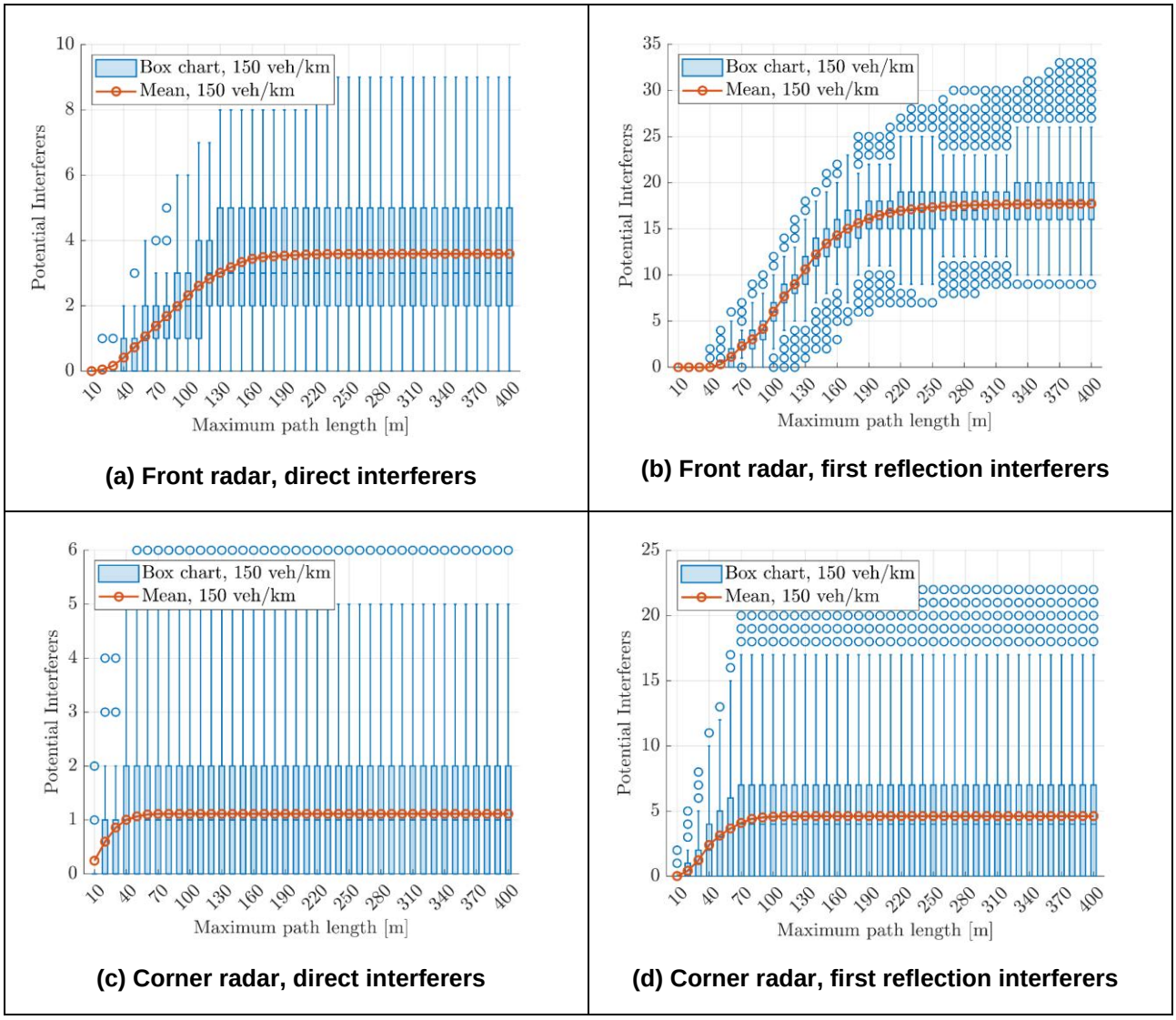
(b) Front radar, first reflection interferers

(c) Corner radar, direct interferers

(d) Corner radar, first reflection interferers

Figure 15: Average number of potential interferers varying the maximum path length

In addition to the "average number of potential interferers" metric, the distribution of the "number of potential interferers" can also be investigated to determine the highest numbers of potential interferers observed throughout the simulation. This type of visualization provides insights on the possible worst-case number of interferers that a radar will have to cope with. Such statistics are obtained using the `boxchart` function of MATLAB, and typically shown including the median, the lower and upper quartiles, any outliers, and the minimum and maximum values.



(a) Front radar, direct interferers

(b) Front radar, first reflection interferers

(c) Corner radar, direct interferers

(d) Corner radar, first reflection interferers

NOTE: Highway with 150 vehicles/km.

Figure 16: Cumulative distribution of the potential interferers

As an example, Figure 16 focuses on the scenario with 150 vehicles/km, again separating front from corner radars and direct from first reflection interference. It can be seen that up to 9 and 6 potential direct interferers can be noted for front radars and corner radars, respectively, while up to 33 and 3 potential first reflection interferers can be noted for front radars and corner radars, respectively.

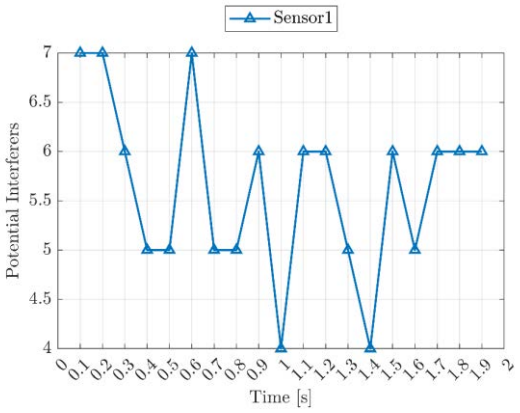
Annex D provides all the potential interferer analysis figures.

8.2.3 Variability in time of number of potential interferers

In addition to the "average number of potential interferers" and the "distribution of the number of potential interferers" metrics, the time variability of the number of potential interferers is further investigated, although it requires focusing on a specific time instant.

As examples, Figure 17 and Figure 18 show the time variability of the number of potential interferers for the same symmetrical highway scenario with density of 150 vehicles/km.

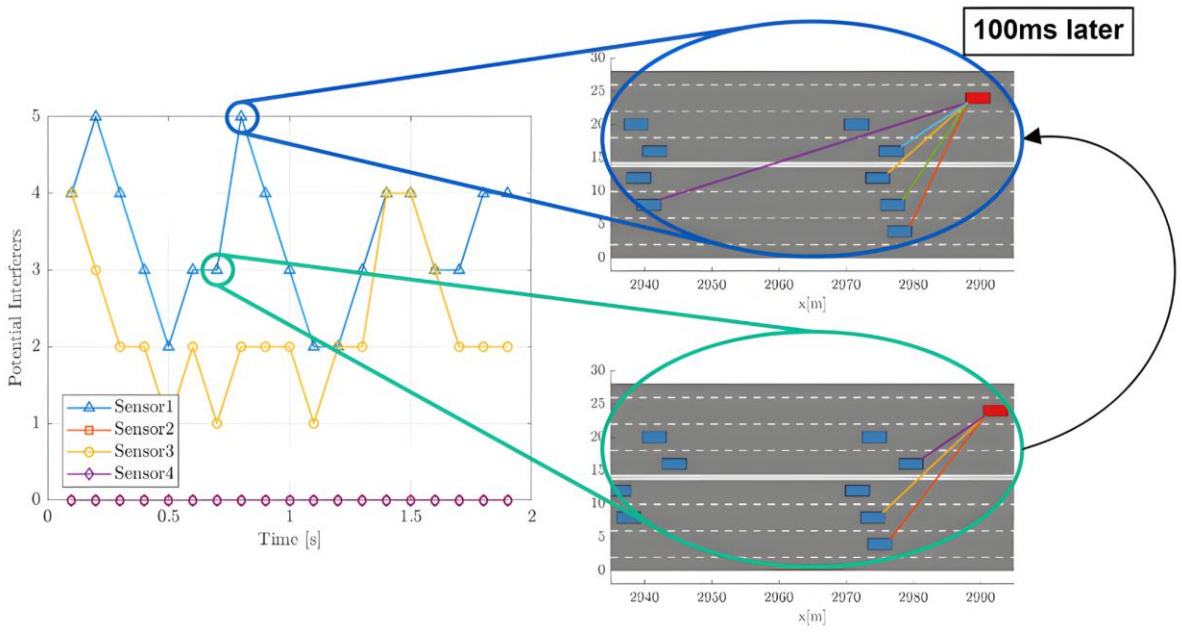
In Figure 17, the test car is equipped with one front radar suffering from direct interference with a maximum path length of 200 m. In that example, it can be seen that within 100 ms, the number of potential interferers can suddenly change by ±2, for instance going from 5 to 7 from time = 0,5 s to time = 0,6 s.



NOTE: Maximum path length of 200 m.

Figure 17: Variability in time of the number of direct potential interferers of one example front radar

In Figure 18, the test car is in the rightmost lane, equipped with four corner radars suffering from direct interference with a maximum path length of 100 m.

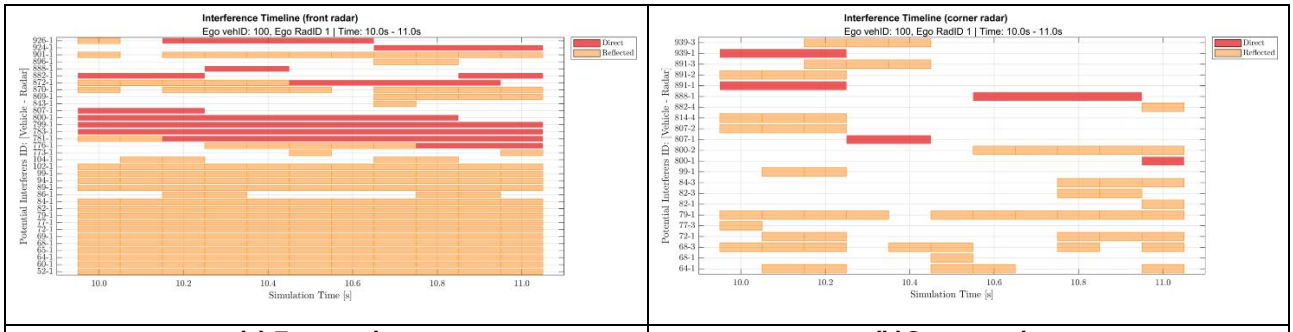


NOTE: Maximum path length of 100 m.

Figure 18: Variability in time of the number of direct potential interferers of the four corner radars of an example car located in the rightmost lane

It can be observed that also with corner radars, the number of potential interferers suddenly increases from 3 to 5 in 100 ms. This figure also allows to appreciate that the sensors on the right of the car (sensors 2 and 4) do not have any potential interferers because there are no other cars or obstacles at the right side of the car. This observation also means that, depending on the position of the generic vehicle on the road, the average number of potential interferers it perceives can be higher or lower than the general average, especially in the case of corner radars. From this example, it can be seen that a high variability of the number of interferers (at least ±2, maybe more in some other examples) can be expected in very short time (e.g. 100 ms).

Figure 19 is an extract from the simulation environment and shows the evaluation of the potential interferers over time for front and corner radars, respectively in subfigures (a) and (b). For a fixed ego vehicle and target (e.g. vehicle Id = 100, radar Id = 1), the figures show the number of potential interferers over time, based on the IDs of the interfering vehicles and their radars. For instance, when a line on the y-axis shows "52-1", it indicates that the potential interferer is the radarID 1 mounted on vehicleID 52.



(a) Front radars | **(b)Corner radars**

Figure 19: Dynamicity of potential interferers over time

It shows very clearly that for corner radars the environment is much more dynamic than for front radars. For front radars, the interfering radars stay active a long time for most of them (even with a continuous line for the whole time), while for corner radars the duration of the interference is visually much less, with most interfering lines staying active only for a short time duration, for example over a time window of three consecutive transmissions (3 × 80 = 240 ms). This means that for corner radars interference is less predictable, as the set of potential interferers changes frequently.

8.2.4 Definition of "equivalent distance"

Previous clause 8.2.1 defined the concept of "potential interferers" and clause 8.2.2 provided statistics from potential interferers as a first step to understand the magnitude of radar interference and its challenges, derived from the positions of the vehicles, and distinguishing between direct and first-path potential interference.

However, it is clear that direct and first-path interferers will yield different characteristics, for example in terms of signal power levels. It is therefore desirable to establish a comprehensive metric that helps comparing the effect of these two different types of interference. A new metric is thus introduced: the *equivalent distance* d_{ref}.

The *equivalent distance* d_{ref} is defined as the distance of an interfering radar generating an interfering signal perceived at the receiver with the same power as if it was sent without reflections from a source at distance d_{ref}. This implies that:

- A direct interferer (with LOS to the victim) at a physical distance d_{ref} is also a potential interferer at the equivalent distance d_{ref}.
- A first-reflection interferer at the same physical distance d_{ref} is a potential interferer at the equivalent distance d_{ref} for which:

$$d_{\text{ref}} = \sqrt{\frac{4\pi {d_1}^2 {d_2}^2}{\sigma}}$$

Where d_1 is the path length covered by the signal from the interferer to the point of reflection, d_2 is the one from the point of reflection to the victim, and σ is the Radar Cross-Section (RCS). More theoretical background about the "potential interferers" can be found in [i.28], section III.A.

8.2.5 Potential interferers up to a given "equivalent distance"

Leveraging the definition of the "equivalent distance", the "number of potential interferers up to a given equivalent distance *d*" can also be defined as being the number of potential interferers which have an equivalent distance smaller or equal to *d*. This number of potential interferers can include signals originating both from a direct interferer or from a first-reflection interferer.

8.2.6 Radar system settings

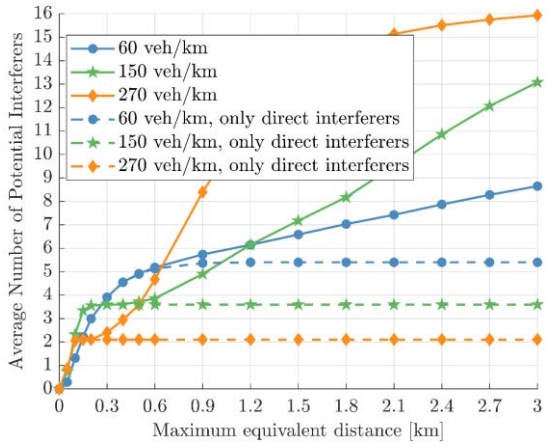
Aligned with the system values depicted in earlier Table 1, which are based on ECC Dec(22)03 [i.3] and ECC Report 351 [i.1], Table 12 provides the radar system settings used for the equivalent distance computations.

Table 12: radar system settings

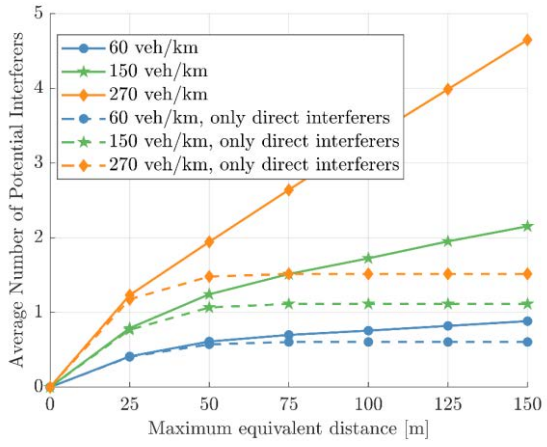
Parameter	Symbol	Front	Corner
Maximum radiated mean power during T_{on}	EIRP	35 dBm	15 dBm
Rx antenna gain (as per Recommendation ITU-R M.2057-1 [i.19])	G_r	30 dBi	23 dBi
Field of View	FoV	30°	60°
Target radar cross section	σ	10 m²	

8.2.7 Highway scenario results

Figure 20 provides results for a highway symmetric traffic, for vehicles densities of 60, 150 and 270 vehicles per km, representative of low, medium, and high congestion levels. The figure shows the average number of potential interferers as function of the maximum equivalent distance. Figure 20(a) shows results for the front-looking radars, and Figure 20(b) shows results for the corner-looking radars. The solid lines represent the total interference (stemming from both direct interferers and first-reflection interferers), while the dotted lines show the contribution coming only from the direct interferers.



(a) Symmetric case, front radar.



(b) Symmetric case, corner radar.

Figure 20: Average number of potential interferers with highway symmetric traffic, as function of the maximum equivalent distance

It can be observed from Figure 20 that for each density and both types of radars, the number of potential interferers increases rapidly for small values of the maximum equivalent distance, and then the increase becomes less steep, sometimes exhibiting a plateau.

Comparing the solid lines and the dotted lines, it appears that for small values of the maximum equivalent distance the number of potential interferers increases mainly due to direct interferers, which then have a very limited additional contribution for large values of the maximum equivalent distance. This trend is due to the following considerations:

1) The power attenuation due to the reflection makes negligible the probability to have a potential reflected interferer at a short equivalent distance.

2) When the source of the direct interferer is far from the victim radar, there is a high probability that there is at least another vehicle in between acting as obstacle and therefore blocking the signal path.

3) For corner radars, the maximum distance of a direct interferer is limited by the geometry of the scenario, since in the worst case the interferer and victim are located in the rightmost lanes of the two directions (e.g. it can be seen that the maximum is achieved already at 50 m).

Comparing the different vehicles densities, for front radars in Figure 20(a), a larger density yields a lower maximum number of potential interferers coming from direct interferers only (dotted lines). As the density of vehicles increases, in fact, direct interference decreases due to more vehicles acting as obstacles. When considering potential interferers coming from both direct interferers and first-reflection (solid lines), different configurations alternatively leading for the left part of the chart, within 100 to 600 meters. After 600 meters, the situation is clearer and it becomes clear that a higher vehicles density yields a higher total number of equivalent interferers.

For corner radars in Figure 20(b), homogenous trends are observed when analysing the total interference and only direct interference. It can be seen that a higher vehicles density yields a higher number of equivalent interferers. In the case of corner radars, the effect of the vehicles acting as obstacles is limited due to the few lanes between the attackers and the victims and a larger density causes an increase of the number of potential direct interferers.

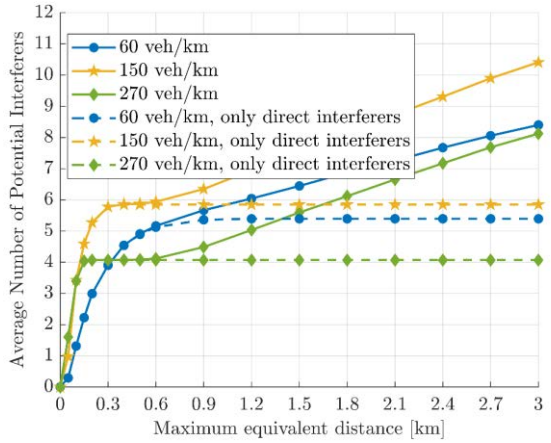
8.2.8 Asymmetric Highway scenario results

For asymmetric highway, the density in the lightly loaded direction is fixed at 30 veh/km, while the one on the opposite and heavily loaded direction varies (30, 120 or 240 veh/km, leading to combined densities of 60, 150 and 270 veh/km).

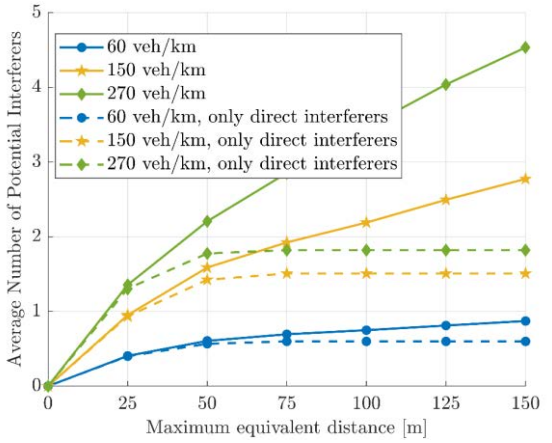
Figure 21 provides results distinguishing between the case of the victim travelling in the lightly loaded direction or in the heavily loaded direction. In both cases, interference may originate from any direction.

It shows the average number of potential interferers as function of the maximum equivalent distance. Figure 21(a) shows results for the front-looking radars with victim in the lightly loaded direction, Figure 21(b) shows results for the corner-looking radars with victim in the lightly loaded direction, Figure 21(c) shows results for the front-looking radars with victim in the heavily loaded direction and Figure 21(d) shows results for the corner-looking radars with victim in the heavily loaded direction.

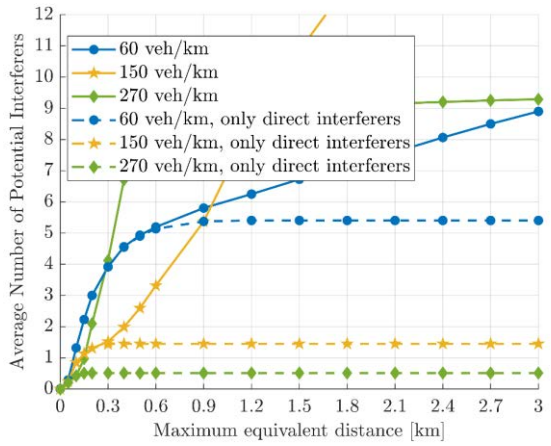
The solid lines represent the total interference (stemming from both direct interferers and first-reflection interferers), while the dotted lines show the contribution coming only from the direct interferers.



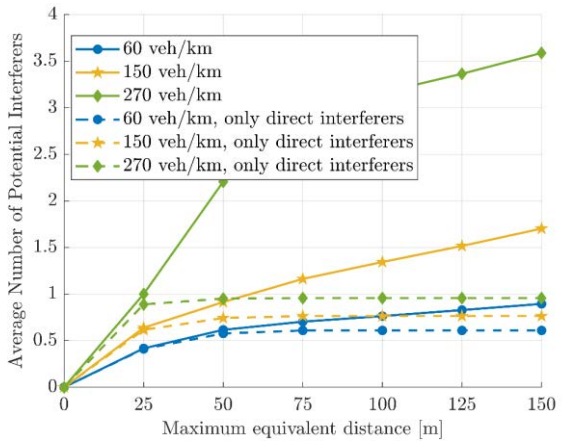
(a) Asymmetric case, front radar, victim in the lightly loaded direction.



(b) Asymmetric case, corner radar, victim in the lightly loaded direction.



(c) Asymmetric case, front radar, victim in the heavily loaded direction.



(d) Asymmetric case, corner radar, victim in the heavily loaded direction.

Figure 21: Average number of potential interferers with highway asymmetric traffic, as function of the maximum equivalent distance

For front radars with victim in the lightly loaded direction (a), the number of potential interferers increases from 60 to 150 veh/km but decreases for 270 veh/km.

Comparing with the symmetric highway case, the number of potential interferers is higher at short distances with asymmetric traffic, meaning that the increased traffic in the opposite direction without obstructions in the same direction increases the number of potential interferers, as was anticipated by construction of the asymmetric scenario. At larger distances, there is a lower contribution of the reflected interferers and therefore the overall number of potential interferers is lower with asymmetric traffic than with symmetric traffic.

For front radars with victim in the heavily loaded direction (c), the trends are somewhat similar to the symmetric highway case with lower values, except with 150 veh/km at large distance and 270 veh/km at short distances, where the impact of reflected interferers increases the overall number of potential interferers.

For corner radars, both in the case of the victim being in the lightly loaded direction (b) and the case of the victim being in the heavily loaded direction (d), very similar trends than the symmetric highway case are observed.

8.2.9 Conclusion

8.2.9.1 Preliminary analysis from potential interferers

Results from potential interferers investigation are agnostic of the modulation parameterization, as they are solely derived from the geometry of the scenario. Influence of several 'dimensions' is analysed as follows:

- **Maximum path length**. For low to medium densities, increasing the maximum path length in general increases the average number of potential interferers. However, for corner radars and/or high vehicles densities, the average number of potential interferers saturates quickly, after a relatively low value of the maximum path length, as shown for example in Figure D.3(a). This behaviour is due to the geometry of the scenario (for example for corner radars, the longest path is from a car in the rightmost lane in one direction to a car in the rightmost lane of the opposite direction).

- **Penetration rate**. It can be observed that the penetration rate impacts the average numbers of potential interferers roughly linearly, but that the distribution of the number of interferers is not linearly impacted. As observable, already 30 % penetration yields a significant number of potential interferers in some cases. As an example, the maximum number of interferers focusing on the front radar and direct interference is 7 with 30 % penetration and 9 with 100 % penetration.

- **Vehicle density**. Increasing the density does not necessarily increase the average number of potential interferers, as for example shown in Figure D.3 where the highest values of average number of potential interferers are obtained for densities of 60 or 120 veh/km in conjunction with long maximum path length. Often for the high vehicle densities, the average number of potential interferers saturates after a certain value of the maximum path length. The increased number of radars is compensated by an increased number of vehicles obstructing the potential interference. For example, for front radar with direct interference, as in Figure D.3(a), the maximum of the average number of potential interferers decreases when the vehicle density increases.

- **Road scenario (symmetric vs asymmetric highway)**. The asymmetric highway case is overall worse when looking at direct interference and better when looking at first reflection interferers.

- **Radar type (front radar vs corner radar)**. The number of potential interferers decreases from front to corner radars (because of geometrical considerations).

- **Direct vs first reflection**. The number of potential interferers increases from direct to first reflection. Note, however, that the same maximum path length should not be considered in the same way in the four combinations, as the power received from front radars may be higher because of a higher transmission power, and as the power received from reflected interference is lower because it is attenuated by the reflection.

In addition, the following observations can be made:

- **Variability in time**. High variability of number of potential interferers can be expected, with suddenly changes of ±2 within 100 ms.

- **Worst case number of potential interferers**. The intention is not to find the exact number, because the simulation space is not exhaustive, but rather the order of magnitude of how many interferers can be expected in average, or in maximum, for the worst scenarios:

    - **Average number of potential interferers**
      From the results obtained and summarized in Table 13, it can be seen that the worst case for front radars in terms of direct interference is yield from the asymmetric highway road scenario with up to 6,3 potential such interferers in average, while the worst case in terms of first reflection interference is yield from the (symmetric) highway road scenario with up to 19,1 potential such interferers in average. For corner radars, the worst case in terms of direct interference is yield from the (symmetric) highway road scenario with up to 1,5 potential such interferers in average, while the worst case in terms of first reflection interference is also yield from the (symmetric) highway road scenario with up to 8,4 potential such interferers.

Table 13: Worst case average number of potential interferers

		Penetration (%)	Density (veh/km)	Road Scenario	Maximum number of average number of potential interferers	Figure
Front radar	direct interference	N/A (100 %)	120	asymmetric highway	6,3	Figure D.3(c)
	first reflection interference	N/A (100 %)	120	highway	19,1	Figure D.3(b)
Corner radar	direct interference	N/A (100 %)	270	highway	1,5	Figure D.5(a)
	first reflection interference	N/A (100 %)	270	highway	8,4	Figure D.5(b)

- **Maximum number of potential interferers (distribution)**
  From the results obtained and summarized in Table 14, it can be seen that for front radars, up to 14 potential direct interferers were seen in Figure D.4(a) (highway, 150 veh/km) and up to 34 potential first reflection interferers were seen in Figure D.4(b) (highway, 150 veh/km).
  For corner radars (fewer tests), up to 6 potential direct interferers were seen in Figure D.2(c) (highway, 150 veh/km) and up to 22 first reflection direct interferers were seen in Figure D.4(b) (highway, 150 veh/km).

Table 14: Worst case maximum number of potential interferers

		Penetration (%)	Density (veh/km)	Road Scenario	Maximum number of maximum number of potential interferers	Figure
Front radar	direct interference	N/A (100 %)	150	highway	14	Figure D.4(a)
	first reflection interference	N/A (100 %)	150	highway	34	Figure D.4(b)
Corner radar	direct interference	N/A (100 %)	150	highway	6	Figure D.2(c)
	first reflection interference	N/A (100 %)	150	highway	22	Figure D.4(b)

8.2.9.2 Analysis from "equivalent distance" potential interferers

Reflecting on the results from clause 8.2.7 and 8.2.8, the following trends can be highlighted:

- The number of potential interferers increases rapidly for small values of the maximum equivalent distance due to the contribution from direct interferers. This means that the likelihood of having a short-distance reflected interferer is very low.
- The increase at higher equivalent distance is in contrast originating from the reflected signals. This can be clearly observed as the dotted-line curves, representing the contribution from the direct interferers flattens out quickly. This is due to the high probability that other vehicles block far away signal. This means that the likelihood of having a very long-distance direct interferer is very low, either being blocked by obstacles or due to very low density. Furthermore, such increase at higher equivalent distance correlates well with higher vehicles densities.
- Asymmetric highway yields the worst-case situation for front radars in terms of potential interferers at short distances, confirming the original intuition, while at large equivalent distance it is less severe than the (symmetric) highway scenario due to fewer first-reflection interferers.

8.3 FMCW focused interference mitigation

8.3.1 Interference metrics

The model evaluates how frequently the radar system faces issues due to interference, and derives the average occurrence of system-level failures.

The model considers the following definitions and assumptions:

- All radars in the system use FMCW with the same parameters.
- The chirp bandwidth is B_{ch}.
- The Analog to Digital Conversion (ADC) bandwidth is B_{ADC}.
- Each chirp is sent within an interval denoted as T_{rch}, which includes the chirp transmitting time itself, denoted as T_{ch}, but also the time to reset the oscillator and possibly an interval at the beginning when the signal is not sent, to introduce some variability in the transmission time.
- It is assumed that the time gap from chirp to chirp slightly varies randomly. The time gap is part of T_{rch}.

The above definitions are illustrated by Figure 22.

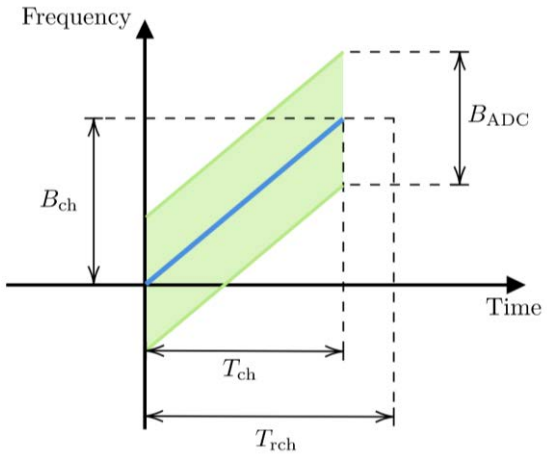


Figure 22: Radar system parameters definition

- The total available bandwidth is B_{TOT}.
- Each radar sends a train of chirps, called frame, followed by an empty interval. The number of chirps in a frame is denoted as N_{ch} and the time ratio between active (i.e. the frame) and inactive states is called *duty cycle* and denoted as δ.
- The frame repetition time, including the inactive time before the beginning of the next frame, is denoted as T_{rf} and can be calculated as: $T_{rf} = \frac{T_{rch} \cdot N_{ch}}{\delta}$.
- Unslotted use of frequency and time domains is assumed, which means that there are no restrictions about when a chirp starts in time and frequency, apart from the constraints related to the total available bandwidth and the chirp and frame duration
- There is one target radar, called 'victim', and N_{int} interfering radars, called 'attackers' (the terminology used is for clarity and does not mean that there is intention to reduce the effectiveness of victim radar sensing).

The above definitions are illustrated by Figure 23, showing one 'victim' radar interfered by one 'attacker'.

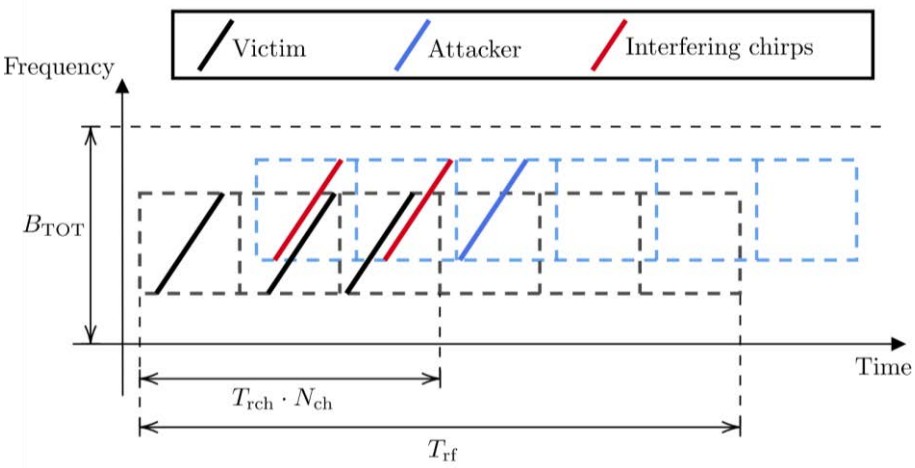


Figure 23: Radar system interference parameters definition

- A chirp from the attacker is considered to cause a collision with the chirp from the victim radar if at least x_{chirp} percentage (x_{chirp} is always set to 0,5) of the attacker's chirp falls inside the Analog to Digital Conversion (ADC) bandwidth of the victim. This is illustrated by Figure 24, where the victim chirp, ADC bandwidth and corresponding area are represented. One other chirp is sent by a nearby radar and overlaps more than x_{chirp} = 0,5 with the ADC area and is therefore considered as a collision (attacker 1). Another chirp is sent by another nearby radar and overlaps less than x_{chirp} = 0,5 and is therefore not considered as a collision (attacker 2).

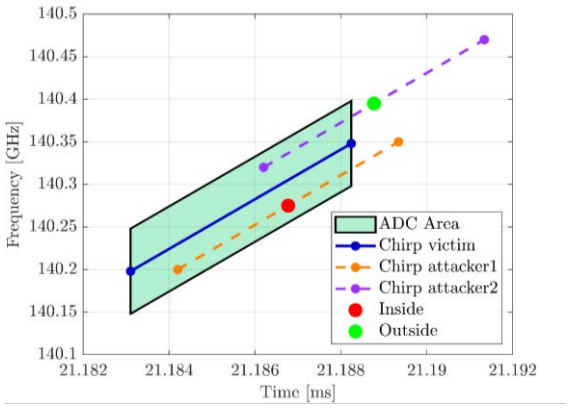


Figure 24: Example of other radars' chirps classification as collision or not collision

- One frame of the victim is lost if K_{ch} or more chirps are lost.
- The radar system has a failure if M_f or more consecutive frames are lost.

8.3.2 Definition system failure

A system failure is defined as a situation where multiple consecutive frames (M_f) are considered lost, due to having each more than a given percentage (K_{ch}) chirps considered lost (due to interference). Such situation may trigger a notification to the driver, for instance indicating that some functionalities (e.g. ACC) are temporarily disabled or not fully functioning as intended. While the actual values may vary from system to system, the present document defines a system failure as a loss of more than 20 % of the chirps in a frame (K_{ch}=20 % or more) for at least 3 consecutive frames (M_f =3):

- P_{fail} is defined as the probability of having a system failure (probability that M_f consecutive frames are lost). It is computed from the following steps:

  - First, an indicator filled with 0 or 1 values, indicating for the (*k*)-th radar if the frames from *(i)* to *(i+M-1)* where lost:

$$1_{failure}^{(k)}(i) = \begin{cases} 1 \text{ if frames from i to } (\mathrm{i} + \mathrm{M}_f - 1) \text{are lost} \\ 0 \text{ otherwise} \end{cases}$$

    Its length is $N^{(k)} - M_f + 1$, where $N^{(k)}$ is the total number of frames transmitted by the k-th radar.

    For example if the k-th radar transmits $N^{(k)}$=12 frames during the simulation and M_f=3, the maximum number of possible consecutive system failures is 10.

  - The number of system failures for the k-th radar is computed as: $N_{sys-fail}^{(k)} = \sum_{i=1}^{N^{(k)}-M_f+1} 1_{failure}^{(k)}(i)$

  - And thus P_{fail} is computed as a summation over all radars (k) in the simulation:

$$P_{fail} = \frac{\sum_{k=1}^{K} N_{sys-fail}^{(k)}}{\sum_{k=1}^{K}(N^{(k)}-M_f+1)}$$

- T_{fail} is finally computed as the average time between two system failures. It is the inverse of the P_{fail} probability and is an indication of how frequently the event occurs in terms of frames. It can be calculated as: $T_{fail} = T_{rf} \cdot \frac{1}{P_{fail}}$

Figure 25 illustrates these definitions and metrics by means of an example showing the average time between failures T_{fail} while varying the total bandwidth B_{TOT}. The scenario is a highway with 60 vehicles/km. Results are shown for a 'baseline' method where radars are not changing their band of operation, which is randomly set at the beginning of each unitary simulation. More details are provided in clause 8.3.5.1. Following the definition above, it is assumed that a frame of the victim is lost if K_{ch}=20 % or more of the chirps are lost, and that a system failure occurs when three consecutive frames are lost (M_f =3).

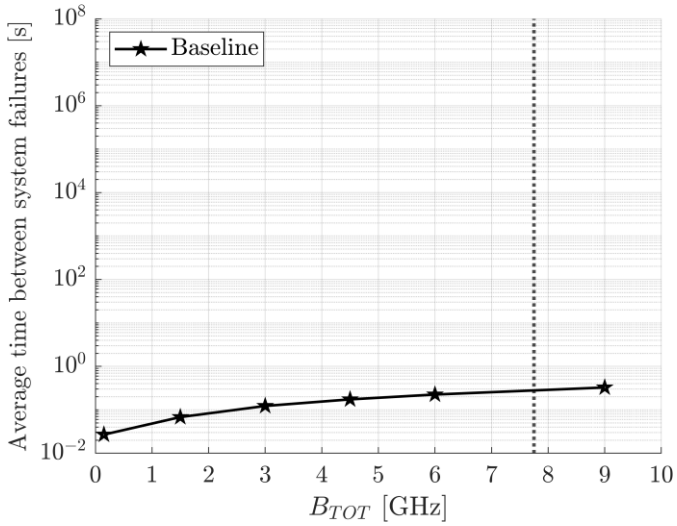


Figure 25: Average time between two system failures metric example

8.3.3 Overview of the mitigation and sharing mechanisms studied

Table 15 lists the mitigation and sharing mechanisms studied for mitigation of the EVR-EVR interference observed in aggregate studies. The table also provides information whether some form of standardization would be required.

Table 15: mitigation and sharing mechanisms studied for mitigation of the EVR-EVR interference

Method	Acronym	Standardization required	Comment
"Baseline"		No	
"Frame-by-frame" frequency hopping	FbF	No	
Variant of the "Frame-by-frame" frequency hopping using discrete subbands	FbF-DSH	No	
"Chirp-by-chirp" frequency hopping	CbC	No	
Compass	FbF compass CbC compass FbF, 2 sect FbF, 4 sect CbC, 2 sect CBC, 4 sect	Yes	Need to standardize the number of quadrants, and the relationship between chirp frequency and quadrant
Frame-by-frame frequency hopping with collision detection	FbF-CD	Partly	Need to ensure some form of stability in resource usage, especially in absence of interference, so that the outcome of sensing or collision detection remains useful for the next frames(s)
Frame-by-frame frequency hopping with collision avoidance	FbF-CA	Partly	

8.3.4 Common parameterization of the radar

Table 16 provides the radar system settings used for the equivalent distance computations.

Table 16: radar system settings

Parameter	Symbol	Front	Corner
Maximum radiated mean power during T_{on}	EIRP	35 dBm	15 dBm
Duty cycle	N/A	50 %	25 %
Rx antenna gain (as per Recommendation ITU-R M.2057-1 [i.19])	G_r	30 dBi	23 dBi
Field of View	FoV	30°	60°
Target radar cross section	σ	10 m²	

8.3.5 Mitigation and sharing mechanisms

8.3.5.1 Baseline

This method is intended to provide a reference point, and can be considered as "no interference mitigation mechanism". It is assumed that the starting frequency of each radar is randomly selected within B_{TOT} - B_{ch} with uniform distribution, and stays with this configuration throughout the simulation. This is illustrated by Figure 26 which shows a snapshot of a numerical simulation.

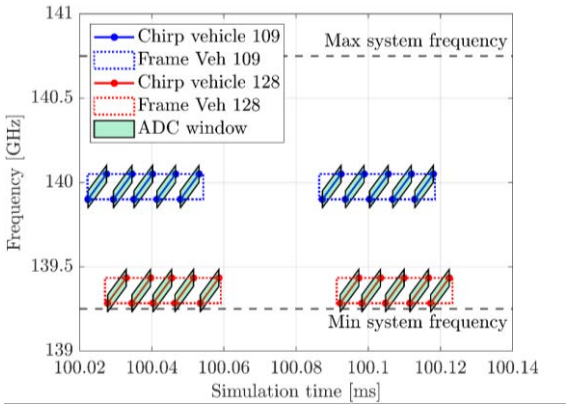


Figure 26: Example of the baseline approach

8.3.5.2 "Frame-by-frame" frequency hopping

With this method, the radar selects randomly a different starting frequency for each radar frame. The starting frequency is randomly selected at the beginning of the radar frame, within B_{TOT} - B_{ch} with uniform distribution, and stays with this configuration throughout the radar frame. Figure 27 provides and exemplary view, as this method is implemented in the simulation environment.

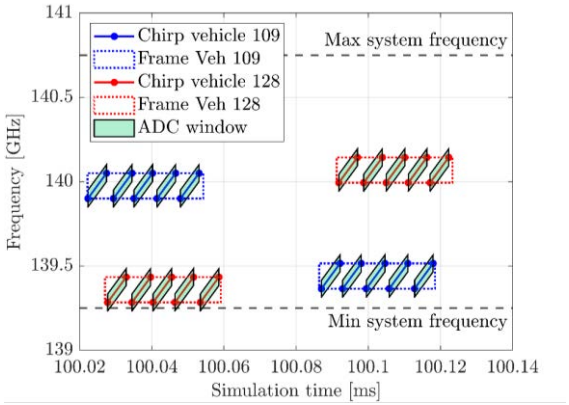


Figure 27: Example of the "Frame-by-frame" frequency hopping

8.3.5.3 "Frame-by-frame" frequency hopping with Discrete Subband Hopping (DSH)

This method is a variation of the "Frame-by-frame" frequency hopping. The starting frequency is still randomly selected at the beginning of the radar frame, but not as any value within B_{TOT} - B_{ch}, instead restricting it to a discretized interval (e.g. defining possible steps of several hundreds of MHz).

This can be realized by defining subband (i.e. subchannels), as introduced in clause 5.2.3.1. The radar selects randomly a subchannel for every frame. The starting frequency of the chirps within a radar frame is defined as the lower bound of that subchannel. Figure 28 provides and exemplary view of this method as implemented in the simulation environment.

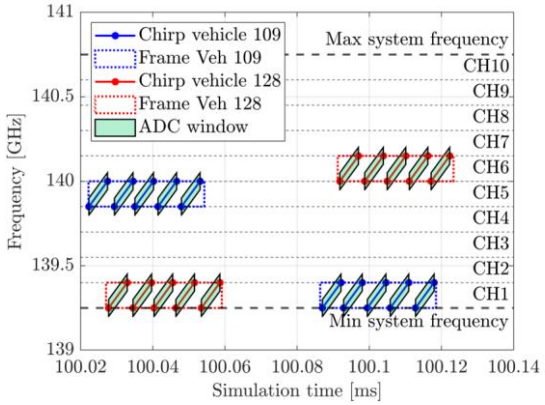


Figure 28: Example of the Frame-by-frame DSH

The tests conducted and showed in the subsequent clauses of the present document assume a subchannel granularity equal to the effective ADC processing bandwidth (and not the FMCW chirps signal bandwidth).

8.3.5.4 "Chirp-by-chirp" frequency hopping

With this method, the radar selects randomly a different starting frequency for each radar chirp within a frame. The starting frequency is randomly selected within B_{TOT} - B_{ch} with uniform distribution. Figure 29 provides and exemplary view, as this method is implemented in the simulation environment.

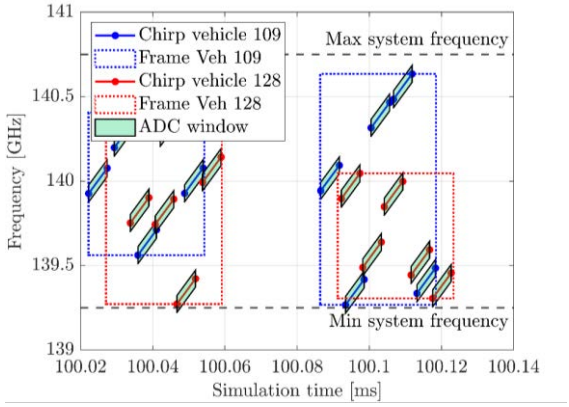


Figure 29: Example of the "Chirp-by-chirp" frequency hopping

8.3.5.5 Compass

As explained in clause 5.2.2.3, the Compass method has been studied in the IMIKO project. Its primary purpose is to reduce the number of interferers coming from the opposite direction, meaning primarily direct type of interferers at high power level.

IMIKO included a version of Compass where the bandwidth is divided into distinct blocks. Building on this description, the Compass method has been implemented in a parameterizable way, where the system bandwidth is divided into multiple sectors (or subchannels), assigning a specific sector to each radar based on its orientation. The number of sectors is set to 2 for front radars, and can be set to 2 or 4 for corner radars. Note that increasing the number of sectors reduces the bandwidth per sector, which could be counterproductive in certain cases, when interference from radars in similar direction is strong.

Considering usage for the highway or asymmetric highway scenarios with the road going from west to east, vehicles can travel from left to right (direction 0°) or from right to left (direction -180°). Setup of the Compass method are illustrated by Figure 30 for front radars and by Figure 31 for corner radars.

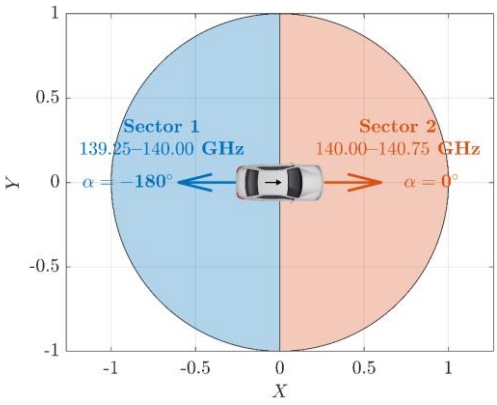
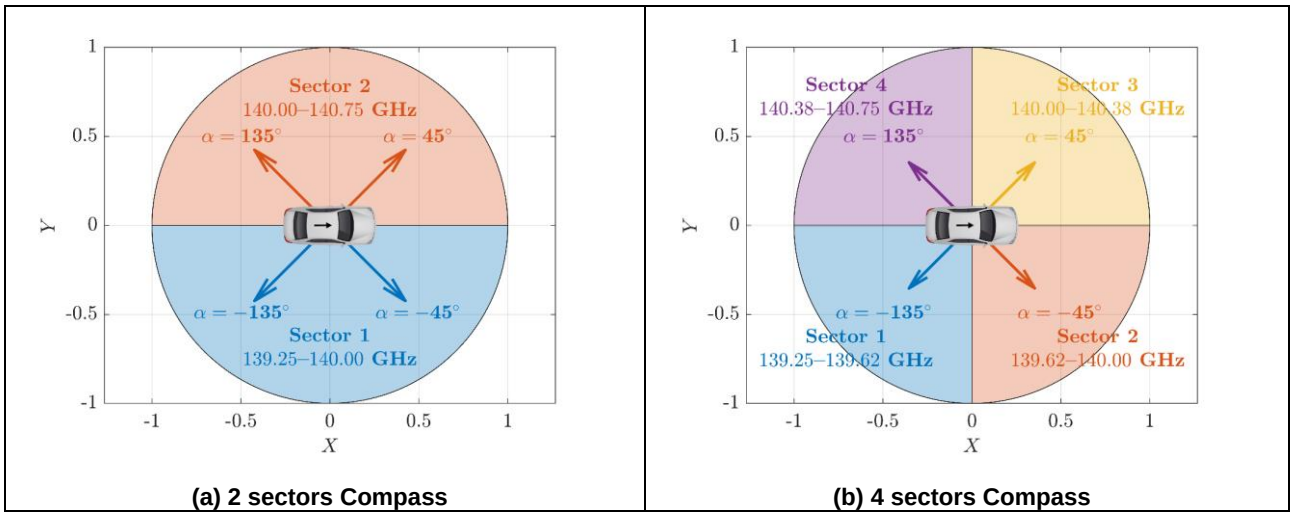


Figure 30: Setup of the Compass method with 2 sectors for front radars



(a) 2 sectors Compass **(b) 4 sectors Compass**

Figure 31: Setup of the Compass method for corner radars

It is important to remark that usage of the Compass method is not mutually exclusive with usage of other mitigation methods. For example, inside each sector, the above-mentioned mitigation and sharing mechanisms such as "Baseline", "Frame-by-frame" frequency hopping or "Chirp-by-chirp" frequency hopping can be used. An example is shown with Figure 32, a scenario with front radars and 2 sectors, and vehicles 14 and 109 travelling in opposite directions.

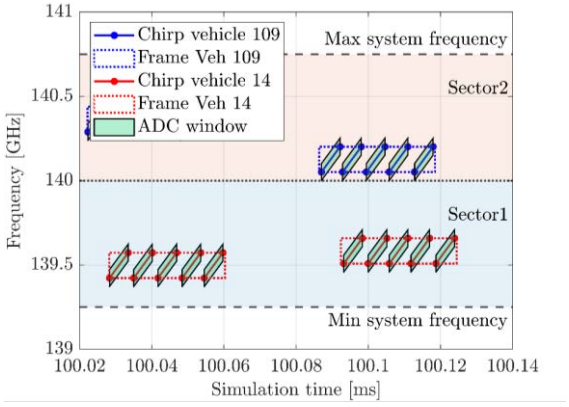


Figure 32: Example of Compass + "Frame-by-frame" frequency hopping

8.3.5.6 Frame-by-Frame frequency hopping with Collision Detection

The Frame-by-Frame frequency hopping with Collision Detection (FbF-CD) technique leverages the capability of the radar to identify that it suffered interference. It counts the number of consecutive chirps *K* which are interfered, and when above a threshold, denoted K_{CD}, it triggers a change for the starting frequency of the next frame, which is randomly reset within the whole band. The collision detection approach is not relevant with a chirp-by-chirp frequency change.

Figure 33 provides an example of this technique as implemented in the simulator with $K_{CD}=2$; the first frame on the left of the figure contains two interfered chirps and thus the starting frequency of the second frame is changed, while that starting frequency remains constant over the second and third frames as no interference is perceived.

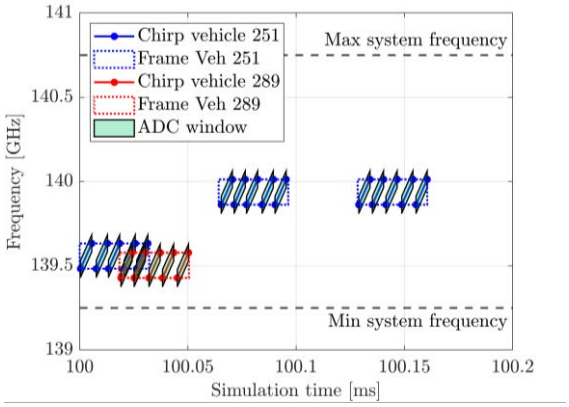


Figure 33: Example of Frame-by-Frame frequency hopping with Collision Detection (FbF-CD) technique

It was found by simulations that optimal setting for K_{CD} is when it is equal to K_{ch}, the number of collided chirps per frame beyond which the frame is considered lost, as defined in clause 8.3.2 (typically K_{ch} = 20 % of the number of chirps within the frame). Results therefore correspond to an optimal setting of the threshold.

8.3.5.7 Frame-by-Frame frequency hopping with Collision Avoidance

The Frame-by-Frame frequency hopping with Collision Avoidance (FbF-CA) technique leverages the capability of the radar to identify possible interference during the idle time between two frame transmissions. As introduced in clause 5.2.3.3, in fact, it might be possible for some EVR to perform some form of energy sensing of the medium prior to transmitting a frame. This can help to identify the portions of the spectrum which are under use by other radars and thus avoid those. This technique is referred to as Collision Avoidance (CA). The power threshold beyond which the frequency chunk is qualified as occupied is denoted P_{int}^{th}. The collision avoidance approach is not relevant with a chirp-by-chirp frequency change.

Figure 34 illustrates the behaviour of this mitigation technique for $P_{int}^{th} = -25\ dBm$. Every time the detected interference power (leftmost vertical axis) gets higher than this threshold (horizontal dotted lines), it triggers a change in the starting frequency (rightmost vertical axis).

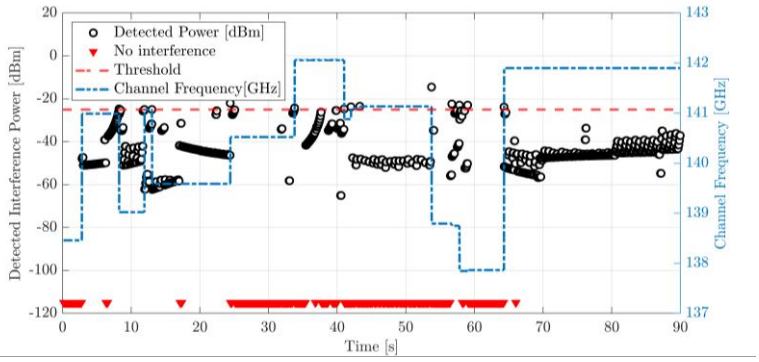


Figure 34: Example of Frame-by-Frame frequency hopping with Collision Avoidance (FbF-CA) technique

It was found by simulations that optimal setting for P_{int}^{th} is when set to lower values, such as -100 dBm. It does trigger more often changes in the starting frequency, but appears beneficial overall.

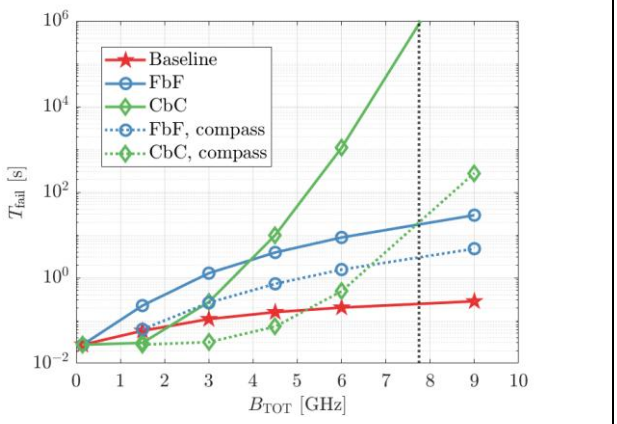
8.3.6 Mitigation and sharing mechanisms results

Annex E provides all the detailed simulation results.

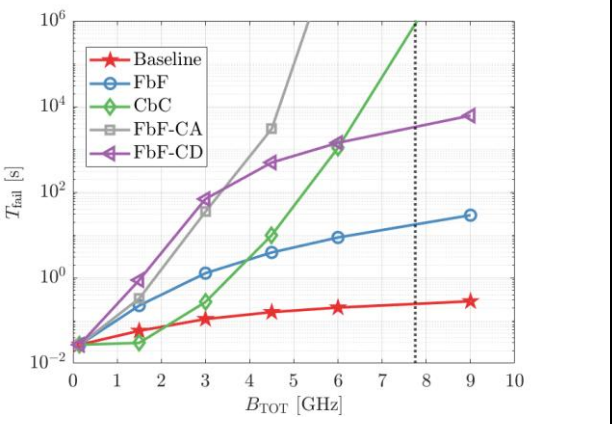
For front radars, it can be observed that:

- The Frame-by-Frame frequency hopping method improves the results by 1 or 2 orders of magnitude compared to the Baseline method.
- The Discrete Subband Hopping (DSH) variation of the Frame-by-Frame frequency hopping method does not improve performance.
- The Compass method does not provide improvement, even arguably degraded for the most challenging use cases.
- The Collision Avoidance (CA) and Collision Detection (CD) methods provide significant improvement, especially CA for large B_{TOT}.
- The Chirp-by-Chirp frequency hopping method yields bad results for low B_{TOT}, but good for large B_{TOT} (second best after CA in most cases, and best for symmetric highway high density).

Figure 35 reproduces results from the annex for the most challenging case for front radars, that is asymmetrical highway heavy traffic direction.



(a) comparison of FbF, CbC, Compass



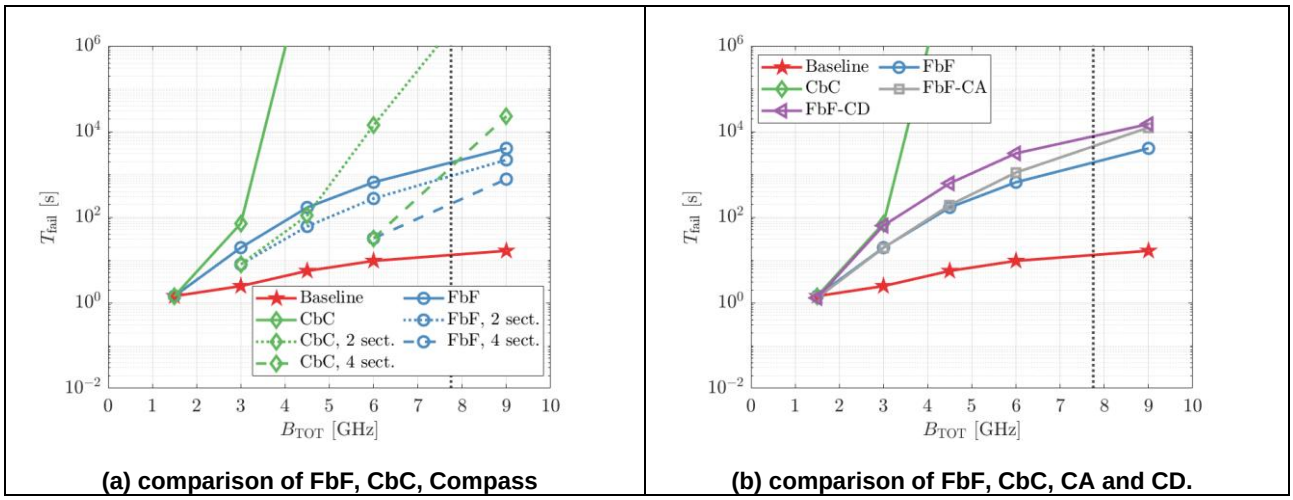
(b) comparison of FbF, CbC, CA and CD.

Figure 35: Summary average time between system failures as function of the total available bandwidth, for front radars, asymmetrical highway heavy traffic direction

For corner radars, it can be observed that:

- The Frame-by-Frame frequency hopping method improves the results by 2 or 3 orders of magnitude compared to the Baseline method.
- The Discrete Subband Hopping (DSH) variation of the Frame-by-Frame frequency hopping method does not improve performance.
- The Compass method does not provide improvement, even arguably degraded for the most challenging use cases.
- The Collision Avoidance (CA) and Collision Detection (CD) methods provide only little improvement over the Frame-by-Frame.
- The Chirp-by-Chirp frequency is by far the best performing method for corner radars.

Figure 36 reproduces results from the annex for the most challenging case for corner radars, that is symmetrical highway high density.



(a) comparison of FbF, CbC, Compass

(b) comparison of FbF, CbC, CA and CD.

Figure 36: Summary average time between system failures as function of the total available bandwidth, for corner radars, symmetrical highway high density

Comparing the performance of the mitigation techniques used for front radars and corner radars, it appears that while front radars could benefit from the most advanced techniques like Collision Avoidance (CA) and Collision Detection (CD) that intend to select the resource in a smart way, corner radars rather benefit from the techniques which randomize the resource usage as much as possible. These results echo the observations from Figure 19 which show that corner radars yield a much more dynamic environment, with interferers that can appear and disappear much faster than for front radars, causing the advanced techniques, such as Collision Avoidance (CA) in particular, to perform only slightly better than the frame by frame randomization for corner radars.

For front radars, it can be concluded that Collision Avoidance (CA) in particular, and Collision Detection (CD) to a lesser degree, provided the best results. As explained in clause E.1.1.3, these results should be treated as an upper bound of the performance of these methods, as in the simulation environment all the radars are assigned the same technique and the same frame structure. Nonetheless, these results show that there is a collective incentive in preserving a form of stability in resource usage (when not interfered), to allow the other radars to sense the medium and purposely select different resources, in contrast to blindly randomizing the resource usage and being un-predictable to others.

For corner radars, it can be concluded that the environment appears to be too dynamic for the advanced techniques aiming at selecting the radar resource in an optimized way to be highly effective. In that case, the Chirp-by-Chirp randomization yields the best performance, followed by the Collision Detection (CD).

Overall, it can be concluded that the Collision Avoidance (CA) technique is particularly effective when the ratio between the total bandwidth B_{TOT} and the chirp bandwidth is sufficiently large and the number of potential interferers is limited. Otherwise, the Chirp-by-Chirp frequency hopping method and Collision Detection (CD) provide the best performance.

9 Conclusions

The present document assesses the risk of self-interference for the Exterior Vehicular Radars (EVR) and In-cabin Vehicular Radars (IVR), for the case of transmissions in the bands studied in ECC Report 351 [i.1] (122,25 - 130 GHz, 134 - 148,5 GHz). The risk of interference from EVR towards IVR, from IVR towards EVR and from IVR towards IVR has been assessed by means of single-entry studies, while the risk of interference from EVR to EVR has been assessed by means of aggregate studies, for which several mitigation and medium sharing mechanisms have been tested.

From all the single-entry studies conducted, only the case of IVR being interfered by EVR could lead to harmful interference, under specific configurations. This concerns IVR in configuration 1 (short signal bandwidth of 2 GHz combined with fast sweep time of 10 µs) in conjunction with EVR being a high-power front radar. However, it was shown that if the IVR receiver can detect and discard part of the signal which was interfered, even coarsely, IVR recovers sufficient SNR to perform as intended. It is thus recommended that IVR implement a mechanism to reject anomalous high-powered samples to avoid interference originating from high-power EVR front-radars, to be tested with IVR using configurations similar or close to config 1 (short signal bandwidth (e.g. 2 GHz) coupled with fast sweep (e.g. 10 µs).

The EVR to EVR aggregate study results show a risk of radar self-interference, potentially highly detrimental and yielding radar system level error failures in absence of interference mitigation measures. However, the effectiveness of several mitigation and medium sharing mechanisms in reducing significantly the interference is also demonstrated in the present document, with important differences between front radars and corner radars. For corner radars, the Chirp-by-Chirp random hopping method appeared as the best performing one, aiming at randomizing as much as possible the usage of the medium. This technique clearly does not require standardization. For front radars, prediction-based techniques such as Collision Avoidance (CA) primarily and Collision Detection (CD) interference mitigation mechanisms yielded the best results. To be effective, these prediction-based techniques require some form of stability in the frequency resource usage, to enable a reliable reciprocal assessment of the resource occupancy, allowing radars to select interference-free resource. At this stage of market development and given future technical innovation, standardization of an access mechanism is not desirable.

Table 17 provides a summary of the assessments considered in the present document and their conclusion.

Table 17: Summary of the assessments considered in the present document

<table>
<tr><th colspan="2" rowspan="2">Victim
Interferer</th><th colspan="3">EVR</th><th rowspan="2">IVR</th></tr>
<tr><th>Front radar</th><th>Corner radar</th><th>Short/ultra-short radar</th></tr>
<tr><td rowspan="3">EVR</td><td>Front radar</td><td colspan="3" rowspan="3">Aggregate studies (clause 8)
Result: risk of interference

Mitigation: several mechanisms are effective:
• For corner and short/ultra-short radar: Chirp-by-Chirp random hopping, Collision Detection (CD) to a lesser extent
• For front radars: Collision Avoidance (CA) and Collision Detection (CD)</td><td rowspan="3">Single-entry study (clause 7.5)
Results: light risk of interference (in very specific configuration)

Mitigation: IVR receiver to detect and discard the part of the signal which was interfered</td></tr>
<tr><td>Corner radar</td></tr>
<tr><td>Short/ultra-short radar</td></tr>
<tr><td colspan="2">IVR</td><td colspan="3">Single-entry study (clause 7.4)
Results: no risk of interference</td><td>Single-entry study (clause 7.3)
Results: no risk of interference</td></tr>
</table>

Annex A:
Details calculation IVR interference to IVR

There are four IVR configurations but 2 share the same signal bandwidth, meaning that 4 victim configurations need to be evaluated with 2 interferer signal bandwidths, resulting in 8 combinations.

As an example, the methodology for victim IVR configuration 1 (2 GHz bandwidth, 10 µs sweep time) is detailed here. The full derivation for all combinations is contained in the Excel calculator attached to the present document. In the calculator, input cells are coloured, while derived calculations are in the grey cells.

The first section contains the parameters of the desired IVR transmission. The maximum radiated mean power comes from the regulations, while the sweep time, signal bandwidth and cycle time come from the IVR configurations in ECC Report 351 [i.1]. The pulse repetition interval calculation assumes the chirp occupies 90 % of the repetition interval. The chirp slope is calculated as the ratio of the signal bandwidth to the sweep time. The frame duration is assumed to last 50 % of the cycle time. The maximum beat frequency is calculated from the chirp slope and the entered distance to the target and also determines the minimum sampling frequency. The next line allows the user to enter an assumed sample frequency, taking into account the calculated minimum. The number of chirps and number of samples per chirp are calculated.

victim config --> interferer config		IVR config1 2 GHz	IVR config1 14 GHz
parameter name	**unit**	**value**	**value**
Maximum radiated mean power (EIRP)	dBm	3	3
Chirp Period (sweep time)	s	1,00E-05	1,00E-05
signal bandwidth	Hz	2,00E+09	2,00E+09
cycle time	s	5,00E-03	5,00E-03
Maximum radiated mean power (EIRP)	W	0,002	0,002
PRI (based on chirp period = 90 % of PRI)	s	1,11E-05	1,11E-05
chirp slope		2,00E+14	2,00E+14
frame duration	s	2,50E-03	2,50E-03
Max Beat Frequency	MHz	2,67	2,67
minimum ADC Sample Frequency	MHz	5,34	5,34
RX ADC Sample Frequency	MHz	10	10
Number of Samples per Chirp (Fast Time)		112	112
Number of Chirps (Slow Time)		225	225

The next section contains the parameters of the interfering IVR system. The values entered are based on Tables 4, 5 and 6 in ECC Report 351 [i.1], selecting the worst case each time.

victim config --> interferer config		IVR config1 2 GHz	IVR config1 14 GHz
parameter name	**unit**	**value**	**value**
Maximum radiated mean power (EIRP)	dBm	3	3
interferer signal bandwidth (maximum)	Hz	2,00E+09	1,40E+10
interferer duty cycle	%	100 %	100 %

The third section calculates the signal propagation for the desired link based on the entered target distance. The total attenuation is a combination of the free space path loss and the gain stemming from the -18 dBm^2 radar cross section.

victim config --> interferer config		IVR config1 2 GHz	IVR config1 14 GHz
target distance (1-way)	m	2	2
free space path loss	dB	162,8	162,8
total attenuation	dB	116,4	116,4

Similarly, the fourth section calculates the signal attenuation between the interfering transmit antenna and the receiving antenna. In this case, it is assumed that two IVR antennas cannot be closer than 2 metre to each other, mounted in the centre of each vehicle. If they are, that probably means they are on cars next to each other, and mounted in the car roof pillars, but in that case, they would most likely be pointing inwards and therefore away from each other. The free space path loss is calculated based on the entered distance. In addition, there will be attenuation from the two car bodies. No antenna pattern is taken into account for IVRs.

victim config --> interferer config		IVR config1 2 GHz	IVR config1 14 GHz
parameter name	**unit**	**value**	**value**
distance IVR-EVR	meters	2	2
free space path loss	dB	81,4	81,4
attenuation due to car bodies (x2)	dB	36	36
attenuation due to interferer Tx antenna pattern	dB	0	0
total attenuation	dB	117,4	117,4

Interference is evaluated in terms of Signal-to-Interference-plus-Noise Ratio (SINR). First, the signal-to-noise ratio without interference is calculated. The 15 dB noise figure is based on the assumptions in ECC Report 334 [i.2]. The noise factor is simply the linear expression of that value. The noise level is computed as the noise spectral density (which is the product of the Boltzmann constant and the system noise temperature) divided by the PRI, as the reference EIRP level is an average power over the whole cycle time, as in Blake's coherent radar equation [i.25].

The desired signal power level is based on the transmit EIRP, the calculated signal propagation and the antenna gain. Following ECC Report 334 [i.2], 6 dB overall IF signal loss is taken into account. The signal processing gain is calculated according to the formula in III(A) in [i.26]. The SNR in the absence of interference is then calculated as the ratio of the signal power, plus the signal processing gain minus the losses, and the noise level at the receiver. Based on ECC Report 334 [i.2], it is assumed that at least 15 dB SNR is required. The following lines simply verify that the wanted target is detectable.

victim config --> interferer config		IVR config1 2 GHz	IVR config1 14 GHz
parameter name	**unit**	**value**	**value**
RX antenna gain (no elevation, for target signal)	dBi	10	10
RX antenna gain (with elevation, for interferer signal)	dBi	10	10
Noise Figure	dB	15	15
Radiated EIRP (on transmit time)	dBm	3,46	3,46
Noise Factor		31,62	31,62
Rx noise level	dBm	-88,98	-88,98
Rx Target signal power level	dBm	-102,9	-102,9
Losses	dB	6	6
signal processing gain	dB	43,5	43,5
SNR	dB	23,6	23,6
SNR threshold	dB	15	15
Detectability (yes/no)		yes	yes
SNR (verif, without proc gain)		1,01E-02	1,01E-02
SNR (verif, with proc gain)	dB	23,6	23,6
Verification		OK	OK

Finally, the SINR is calculated. This calculation takes into account that the victim may not see all the energy transmitted by the interferer if its bandwidth is smaller. The interfering power level at the victim receiver antenna is then calculated from the transmitted EIRP, taking into account the propagation loss and antenna gain, and corrected for the potential bandwidth and duty cycle effects. As long as the combined SINR stays above the minimum SNR ratio of 15 dB, it is assumed that the receiver will not be affected by the interfering signal. SINR degradation and I/N metrics are also provided for easier read and to complement the analysis.

victim config --> interferer config		IVR config1 2 GHz	IVR config1 14 GHz
parameter name	**unit**	**value**	**value**
Different BW effect: victim smaller than interferer		0,0	-8,5
Effect of interferer duty cycle	dB	0,0	0,0
Rx Interferer power level	dBm	-104,4	-112,8
SINR	dB	23,5	23,6
Detectability (yes/no)		yes	yes
SINR degradation	dB	0,12	0,02
I / N ratio	dB	-15,4	-23,9

Annex B:
Details calculation IVR interference to EVR

With 2 IVR signal bandwidths configurations and 6 EVR radars configurations (the 3 radar types configured first as per ECC Report 351 [i.1] representative of maximum signal bandwidth, and 3 additional configurations representative of lower signal bandwidth settings), 12 combinations have to be investigated.

As an example, the methodology for 2 GHz IVR signal bandwidths (configurations 1 and 2) is detailed here. The full derivation for all combinations is contained in the Excel calculator attached to the present document. In the calculator, input cells are coloured, while derived calculations are in the grey cells.

In a first section, the Excel calculator considers the parameters of the EVR victim. The maximum radiated power follows from the regulations. Based on the EVR application requirements, the unambiguous range and speed, as well as speed resolution are set to the values listed. The pulse repetition interval calculation assumes the chirp occupies 90 % of the repetition interval. The signal bandwidth is either the maximum from regulations or the bandwidth required to resolve the unambiguous speed, whichever is the smallest. The minimum chirp period is calculated from the unambiguous range. The chirp slope follows from the bandwidth and the PRI, while the frame duration is calculated to provide the speed resolution. The maximum beat frequency then follows from the chirp slope and frame duration and determines the minimum ADC frequency. Finally, the number of samples in fast and slow time are calculated.

Interferer IVR config **victim EVR config**		**IVR config1&2**		
		front	**corner**	**short**
parameter name	**unit**	**value**	**value**	**value**
Maximum radiated mean power (EIRP)	dBm	32	9	9
duty cycle	%	50 %	25 %	25 %
Maximum radiated mean power (EIRP)	dBm	35	15	15
Maximum radiated mean power (EIRP)	W	3,1622777	0,0316228	0,0316228
Unambiguous Range	m	350	65	50
Unambiguous speed	km/h	300	300	200
Unambiguous speed	m/s	83,3	83,3	55,6
Range Resolution	m			
Speed Resolution	m/s	0,08	0,05	0,08
PRI	s	6,42E-06	6,42E-06	9,64E-06
signal bandwidth	Hz	1,00E+09	4,00E+09	7,00E+09
Min Chirp Period	s	2,33E-06	1,33E-06	6,67E-07
Chirp Period	s	5,14E-06	5,14E-06	7,71E-06
chirp slope		1,56E+14	6,23E+14	7,26E+14
frame duration	s	1,28E-02	1,28E-02	1,28E-02
Max Beat Frequency	MHz	363,45	830,74	484,60
ADC Sample Frequency	MHz	726,89	1 661,47	969,19
Number of Samples (Fast Time)		4 670	10 674	9 340
Number of Samples (Slow Time)		2 000	2 000	1 334

The second section of the calculator considers the IVR parameters. The maximum radiated mean power and signal bandwidth come from the regulations. The resulting power spectral density is calculated to verify that it remains within the allowed -30 dBm/MHz limit.

Interferer IVR config **victim EVR config**		**IVR config1&2**		
		front	**corner**	**short**
parameter name	**unit**	**value**	**value**	**value**
Maximum radiated mean power (EIRP)	dBm	3	3	3
interferer signal bandwidth (maximum)	Hz	2,00E+09	2,00E+09	2,00E+09
PSD	dBm/MHz	-30,0	-30,0	-30,0

Next, the signal propagation loss for the desired EVR link is calculated. It consists of the free space path loss combined with the RCS gain of a 10 dBm^2 target.

Interferer IVR config / victim EVR config		front	IVR config1&2 corner	short
parameter name	**unit**	**value**	**value**	**value**
free space path loss	dB	252,5	242,8	230,7
total attenuation	dB	178,1	168,4	156,3

Similarly, the signal propagation from the interferer to the EVR receiver needs to be calculated. The distances are taken from the worst-case geometry calculations reported above. The total propagation loss consists of the free space path loss and the car body attenuation.

Interferer IVR config / victim EVR config		front	IVR config1&2 corner	short
parameter name	**unit**	**value**	**value**	**value**
distance IVR-EVR	meters	12,9	7,2	7,2
attenuation due to distance	dB	97,6	92,5	92,5
attenuation due to car body	dB	18	18	18
total attenuation	dB	115,6	110,5	110,5

Interference is then evaluated in terms of Signal-to-Interference-plus-Noise Ratio (SINR).

First, the signal-to-noise ratio without interference is calculated. The antenna gain of the EVR in the direction of the target is assumed to be the maximum gain. The gain in the direction of the interferer is calculated based on the worst-case geometry assumptions from above. The 15 dB noise figure is based on the assumptions in ECC Report 334 [i.2]. The noise factor is simply the linear expression of that value. The noise level is computed as the noise spectral density (which is the product of the Boltzmann constant and the system noise temperature) divided by the PRI, as the reference EIRP level is an average power over the whole cycle time, as in Blake's coherent radar equation [i.25] The desired signal power level is based on the transmit EIRP, the calculated signal propagation and the antenna gain. Following ECC Report 334 [i.2], 6 dB overall IF signal loss is taken into account. The signal processing gain is calculated according to the formula in III(A) in [i.26]. The SNR in the absence of interference is then calculated as the ratio of the signal power, plus the signal processing gain minus the losses, and the noise level at the receiver. Based on ECC Report 334 [i.2], it is assumed that at least 15 dB SNR is required. The following lines simply verify that the wanted target is detectable.

Interferer IVR config / victim EVR config		front	IVR config1&2 corner	short
parameter name	**unit**	**value**	**value**	**value**
RX antenna gain (no elevation, for target signal)	dBi	30	13	13
RX antenna gain (with elevation, for interferer signal)	dBi	26	9	9
Noise Figure	dB	15	15	15
Radiated EIRP (on transmit time)	dB	35,98	15,97	15,97
Noise Factor		31,62	31,62	31,62
Rx noise level	dBm	-69,39	-70,68	-71,15
Rx Target signal power level	dBm	-112,1	-119,9	-115,3
Losses	dB	6	6	6
signal processing gain	dB	69,7	70,3	67,9
SNR	dB	21,0	15,1	17,8
SNR threshold	dB	15	15	15
Detectability (yes/no)		yes	yes	yes
SNR (verif, without proc gain)		1,33E-05	3,00E-06	9,56E-06
SNR (verif, with proc gain)	dB	21,0	15,1	17,8
Verification		OK	OK	OK

Finally, the SINR is calculated. This calculation takes into account that the victim may not see all the energy transmitted by the interferer if its bandwidth is smaller. The interfering power level at the victim receiver antenna is then calculated from the transmitted EIRP, taking into account the propagation loss and antenna gain, and corrected for the potential bandwidth effect. As long as the combined SINR stays above the minimum SNR ratio of 15 dB, it is assumed that the receiver will not be affected by the interfering signal. SINR degradation and I/N metrics are also provided for easier read and to complement the analysis.

Interferer IVR config victim EVR config		IVR config1&2		
		front	corner	short
parameter name	**unit**	**value**	**value**	**value**
Different BW effect (victim smaller than interferer)	dB	-3,0	0,0	0,0
Rx Interferer power level	dBm	-89,6	-98,5	-98,5
SINR	dB	20,9	15,1	17,8
Detectability (yes/no)		yes	yes	yes
SINR degradation	dB	0,04 dB	0,01 dB	0,01 dB
I / N ratio	dB	-20,2 dB	-27,8 dB	-27,4 dB
I / N ratio	dB	-20,6 dB	-35,5 dB	-38,0 dB

Annex C:
Details calculation EVR interference to IVR

With 4 IVR configurations and 3 EVR radar types, 12 combinations have to be assessed.

As an example, the methodology for IVR configurations 1 (2 GHz signal bandwidth, 10 µs sweep time) is detailed here. The full derivation for all combinations is contained in the Excel calculator attached to the present document. In the calculator, input cells are coloured, while derived calculations are in the grey cells.

The first section contains the parameters of the desired IVR transmission. It is the same as for the IVR interference to IVR and detailed in that section.

Victim IVR config / Interferer EVR config		IVR config1 (2 GHz, 10 µs)		
		front	corner	short
parameter name	**unit**	**value**	**value**	**value**
Maximum radiated mean power (EIRP)	dBm	3	3	3
Chirp Period (sweep time)	s	1,00E-05	1,00E-05	1,00E-05
signal bandwidth	Hz	2,00E+09	2,00E+09	2,00E+09
cycle time	s	5,00E-03	5,00E-03	5,00E-03
Maximum radiated mean power (EIRP)	W	0,002	0,002	0,002
PRI (based on chirp period = 90 % of PRI)	s	1,11E-05	1,11E-05	1,11E-05
chirp slope		2,00E+14	2,00E+14	2,00E+14
frame duration	s	2,50E-03	2,50E-03	2,50E-03
Max Beat Frequency	MHz	2,67	2,67	2,67
minimum ADC Sample Frequency	MHz	5,34	5,34	5,34
RX ADC Sample Frequency	MHz	10	10	10
Number of Samples per Chirp (Fast Time)		112	112	112
Number of Chirps (Slow Time)		225	225	225

The second section details the parameters of the EVR interferers, based on the worst-case values from Tables 3, 4 and 5 in ECC Report 351 [i.1].

Victim IVR config / Interferer EVR config		IVR config1 (2 GHz, 10 µs)		
		front	corner	short
parameter name	**unit**	**value**	**value**	**value**
Maximum radiated mean power (EIRP)	dBm	35	15	15
interferer signal bandwidth (maximum)	Hz	1,00E+09	4,00E+09	7,00E+09
interferer duty cycle	%	50 %	25 %	25 %

The third section calculates the signal propagation for the desired link based on the entered target distance. The total attenuation is a combination of the free space path loss and the gain stemming from the -18 dBm^2 radar cross section.

Victim IVR config / Interferer EVR config		IVR config1 (2 GHz, 10 µs)		
		front	corner	short
parameter name	**unit**	**value**	**value**	**value**
target distance (1-way)	m	2	2	2
attenuation due to distance	dB	162,8	162,8	162,8
total attenuation	dB	116,4	116,4	116,4

Similarly, the signal propagation from the interferer to the IVR receiver needs to be calculated. The distances and EVR antenna gain are taken from the worst-case geometry calculations reported above. The total propagation loss consists of the free space path loss, EVR antenna gain and the car body attenuation.

	Victim IVR config / Interferer EVR config	IVR config1 (2 GHz, 10 µs)		
		front	corner	short
parameter name	**unit**	**value**	**value**	**value**
distance IVR-EVR	meters	12,9	7,2	7,2
attenuation due to distance	dB	97,6	92,5	92,5
attenuation due to car body	dB	18	18	18
attenuation due to interferer Tx antenna pattern	dB	4	4	4
total attenuation	dB	119,6	114,5	114,5

Interference is then evaluated in terms of Signal-to-Interference-plus-Noise Ratio (SINR).

First, the signal-to-noise ratio without interference is calculated. The antenna gain of the EVR in the direction of the target is assumed to be the maximum gain. The gain in the direction of the interferer is calculated based on the worst-case geometry assumptions from above. The 15 dB noise figure is based on the assumptions in ECC Report 334 [i.2]. The noise factor is simply the linear expression of that value. The noise level is computed as the noise spectral density (which is the product of the Boltzmann constant and the system noise temperature) divided by the PRI, as the reference EIRP level is an average power over the whole cycle time, as in Blake's coherent radar equation [i.25]. The desired signal power level is based on the transmit EIRP, the calculated signal propagation and the antenna gain. Following ECC Report 334 [i.2], 6 dB overall IF signal loss is taken into account. The signal processing gain is calculated according to the formula in III(A) in [i.26]. The SNR in the absence of interference is then calculated as the ratio of the signal power, plus the signal processing gain minus the losses, and the noise level at the receiver. Based on ECC Report 334 [i.2], it is assumed that at least 15 dB SNR is required. The following lines simply verify that the wanted target is detectable.

	Victim IVR config / Interferer EVR config	IVR config1 (2 GHz, 10 µs)		
		front	corner	short
parameter name	**unit**	**value**	**value**	**value**
RX antenna gain (no elevation, for target signal)	dBi	10	10	10
RX antenna gain (with elevation, for interferer signal)	dBi	10	10	10
Noise Figure	dB	15	15	15
Radiated EIRP (on transmit time)	dBm	3,46	3,46	3,46
Noise Factor		31,62	31,62	31,62
Rx noise level	dBm	-88,98	-88,98	-88,98
Rx Target signal power level	dBm	-102,9	-102,9	-102,9
Losses	dB	6	6	6
signal processing gain	dB	43,5	43,5	43,5
SNR	dB	23,6	23,6	23,6
SNR threshold	dB	15	15	15
Detectability (yes/no)		yes	yes	yes
SNR (verif, without proc gain)		1,01E-02	1,01E-02	1,01E-02
SNR (verif, with proc gain)	dB	23,6	23,6	23,6
Verification		OK	OK	OK

Finally, the SINR is calculated. This calculation takes into account that the victim may not see all the energy transmitted by the interferer if its bandwidth is smaller. The interfering power level at the victim receiver antenna is then calculated from the transmitted EIRP, taking into account the propagation loss and antenna gain, and corrected for the potential bandwidth and duty cycle effects. As long as the combined SINR stays above the minimum SNR ratio of 15 dB, it is assumed that the receiver will not be affected by the interfering signal. SINR degradation and I/N metrics are also provided for easier read and to complement the analysis.

	Victim IVR config Interferer EVR config	IVR config1 (2 GHz, 10us) front	 corner	 short
parameter name	**unit**	**value**	**value**	**value**
Option1: receiver chain without removal of samples				
Different BW effect (victim smaller than interferer)		0,0	-3,0	-5,4
Effect of interferer duty cycle	dB	-3,0	-6,0	-6,0
Rx Interferer power level	dBm	-77,6	-98,5	-101,0
SINR	dB	11,9	23,1	23,3
Detectability (yes/no)		no	yes	yes
SINR degradation	dB	11,7	0,5	0,3
I / N ratio	dB	11,4	-9,6	-12,0

In the example, it can be seen that the front radar affects the IVR and pushes the SINR below the 15 dB target. Since the front radar power will be very high, the IVR could exclude such anomalous samples from its further processing. The effect on the signal processing gain is modelled as a bandwidth change and the SINR is recalculated. Removing the interfering samples lift the SINR 23 dB above the 15 dB target.

	Victim IVR config Interferer EVR config	IVR config1 (2 GHz, 10us) front	 corner	 short
parameter name	**unit**	**value**	**value**	**value**
Option2: receiver chain with removal of samples				
Different BW effect (victim smaller than interferer)		-3,0		
SNR without interfered samples	dB	20,6		
Detectability (yes/no)		yes		
SINR degradation	dB	3,01		

Annex D:
Details potential interferers results

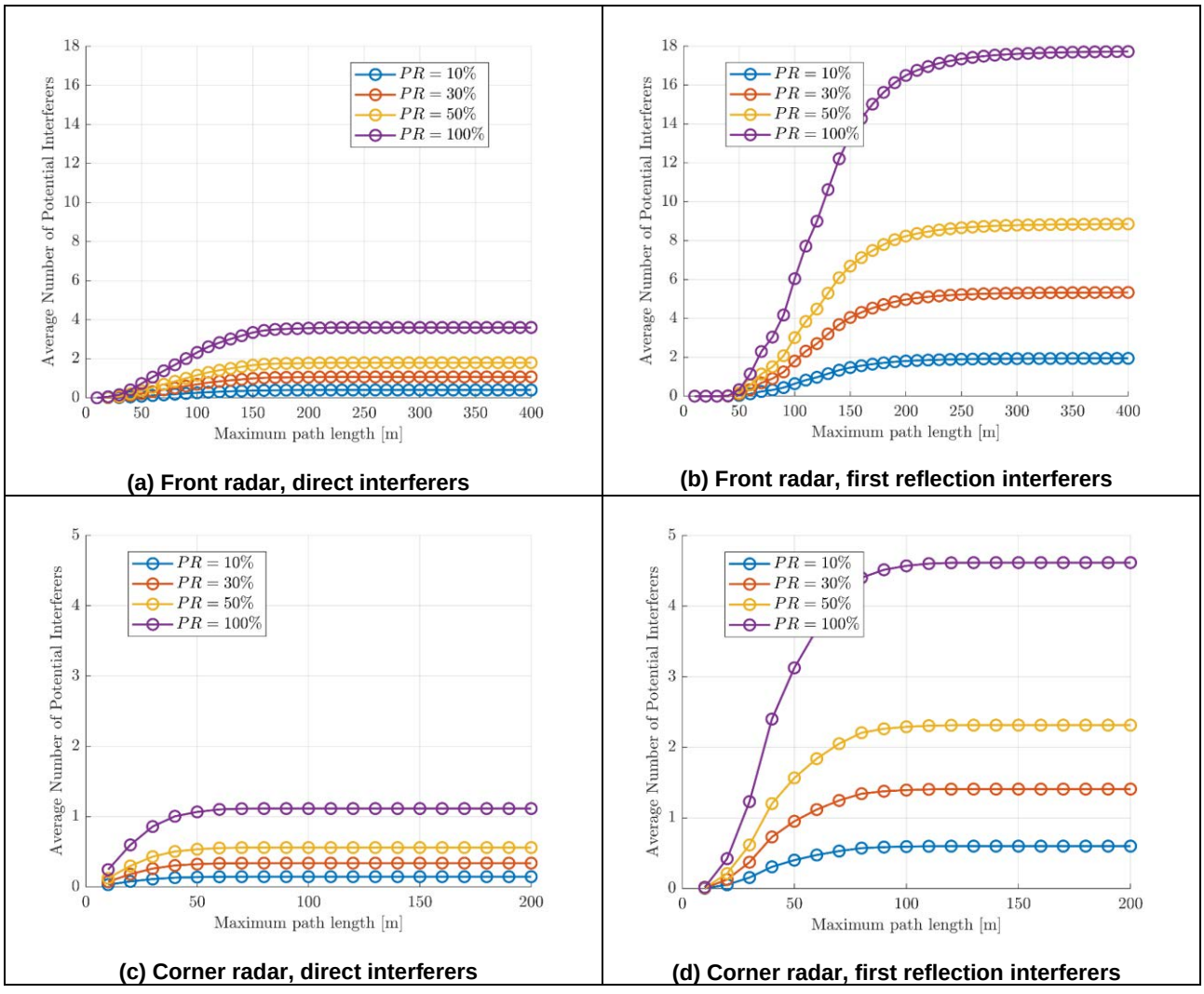
D.1 Results varying the penetration rate

D.1.1 Introduction

These simulations attempt to understand the expected interference level when the penetration of radar systems amongst all the vehicles increases. It should be noted that varying the radar penetration for a certain scenario with a given vehicle density is not equivalent to modifying the density. In fact, each vehicle that is not equipped with radar sensors acts as an potential obstacle and/or reflector, which on one hand may reduce the average number of direct interferers but on the other hand may increase the number of first reflection interferers. Hereafter, a variable percentage of vehicles is equipped with radars. Either front or corner radars, and either direct or first reflection interference are assumed and evaluated separately.

D.1.2 Highway scenario (150 veh/km)

Figure D.1 and Figure D.2 provide results for the "average number of potential interferers" and for the "distribution of the number of interferers" respectively.



(a) Front radar, direct interferers

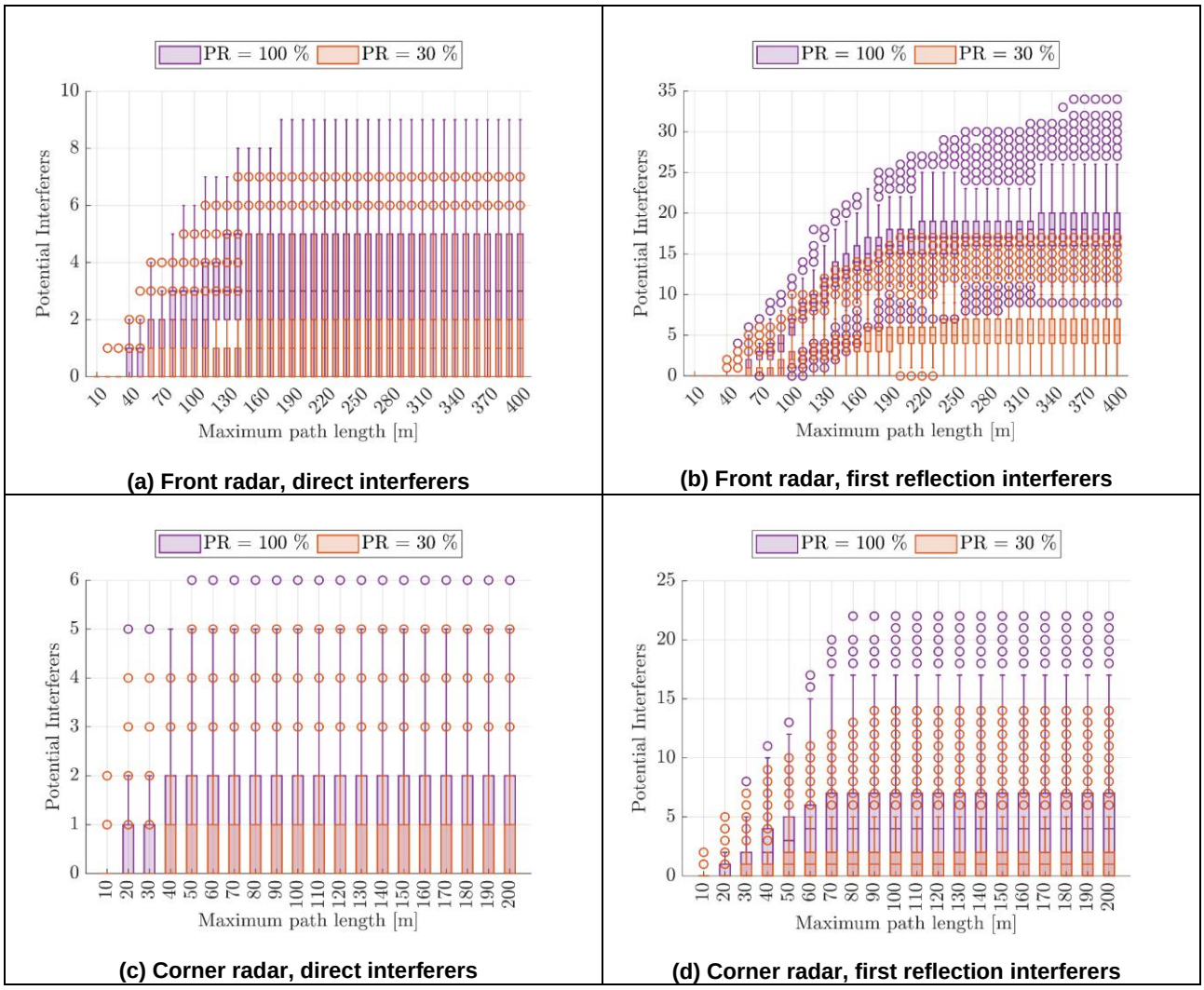
(b) Front radar, first reflection interferers

(c) Corner radar, direct interferers

(d) Corner radar, first reflection interferers

Figure D.1: Average number of potential interferers varying the penetration rate for highway

From Figure D.1, it can be observed that the penetration rate is linearly related to the average number of interferers. This is indeed reasonable, given that the interferers for a certain penetration rate are a random portion of those that would interfere if all vehicles were equipped with radars.



(a) Front radar, direct interferers

(b) Front radar, first reflection interferers

(c) Corner radar, direct interferers

(d) Corner radar, first reflection interferers

Figure D.2: Distribution of the number of potential interferers varying the penetration for highway

However, from Figure D.2, it can be observed that the actual number of interferers is not linearly related to the penetration rate. The box charts are shown for 30 % and 100 % penetration. As it can be observed, the worst case with a penetration of 30 % is much larger than the 30 % of the maximum value when 100 % vehicles are equipped. As an example, the maximum number of interferers focusing on the front radar and direct interference is 7 with 30 % penetration and 9 with 100 % penetration. As observable, 30 % penetration already yields a significant number of potential interferers in some cases.

D.2 Results varying the vehicle density

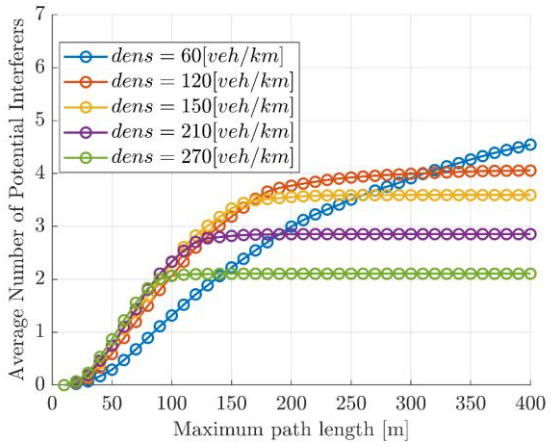
D.2.1 Introduction

These simulations attempt to understand the expected interference level when the vehicle density increases, assuming a penetration rate of 100 %. Either front or corner radars, and either direct or first reflection interference are assumed and evaluated separately.

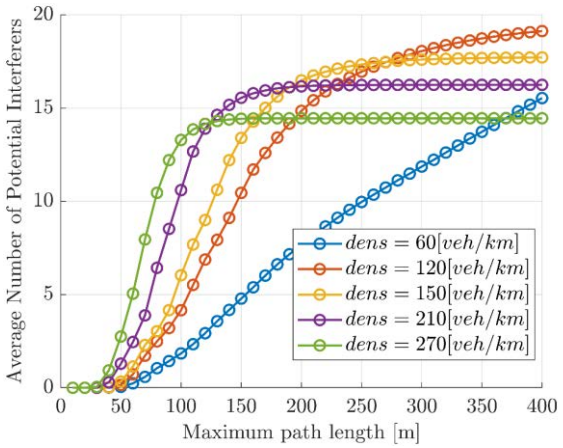
D.2.2 Highway vs asymmetrical highway scenario (front radars)

Figure D.3 and Figure D.4 provide results for front radars for the "average number of potential interferers" and for the "distribution of the number of interferers" respectively.

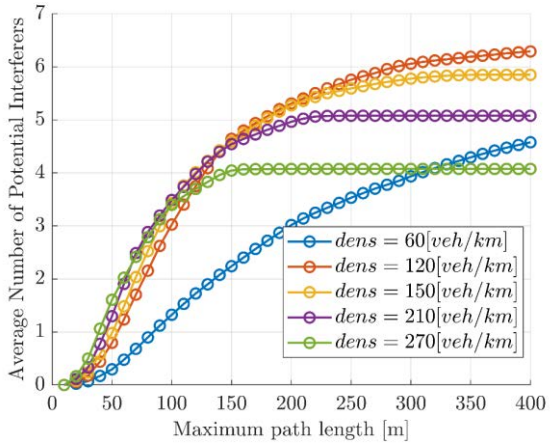
Both symmetric and asymmetric traffic densities have been simulated. In the latter case, a light density of 30 veh/km is considered in the studied direction, with a variable density in the opposite direction.



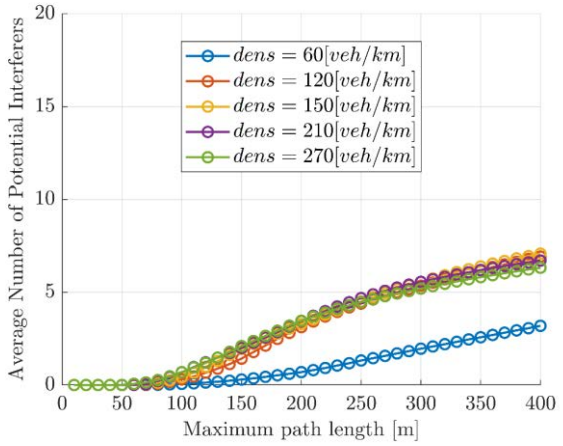
(a) Symmetric case. Front radar, direct interferers.



(b) Symmetric case. Front radar, first reflection interferers.



(c) Asymmetric case. Front radar, direct interferers.



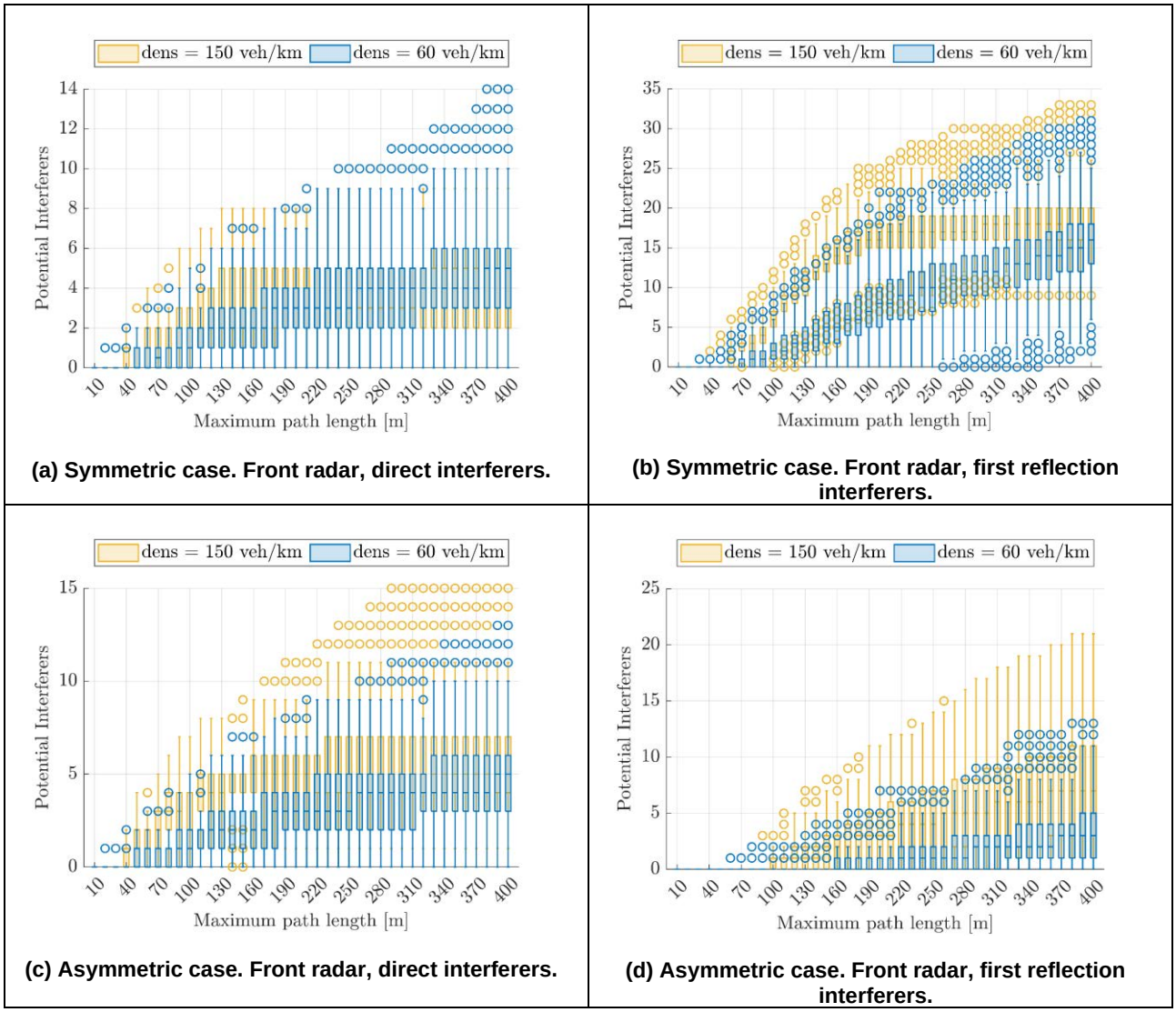
(d) Asymmetric case. Front radar, first reflection interferers.

Figure D.3: Average number of potential interferers varying the density for front radars, highway vs asymmetrical highway

From Figure D.3, similar trends can be seen for symmetrical and asymmetrical highway scenarios:

- The average number of potential interferers increases as the maximum path length increases.
- In some cases, the average number of potential interferers saturates after a certain value of the maximum path length, because the further away interferers are blocked by closer vehicles.
- Increasing the density does not always increase the average number of potential interferers. While the number of active radars increases with density, the risk of interference could be compensated for by the increased number of cars acting as obstacles. For example, in the symmetrical highway scenario, the maximum of the average number of potential interferers decreases when the vehicle density increases.

However, some differences exist. Compared to the symmetric scenario, the asymmetric case is overall worse when considering direct interference but better when considering first reflection interferers.



(a) Symmetric case. Front radar, direct interferers.

(b) Symmetric case. Front radar, first reflection interferers.

(c) Asymmetric case. Front radar, direct interferers.

(d) Asymmetric case. Front radar, first reflection interferers.

Figure D.4: Distribution of the number of potential interferers varying the density for front radars, highway vs asymmetrical highway

Figure D.4 shows that symmetrical and asymmetrical highway scenarios share similar trends:

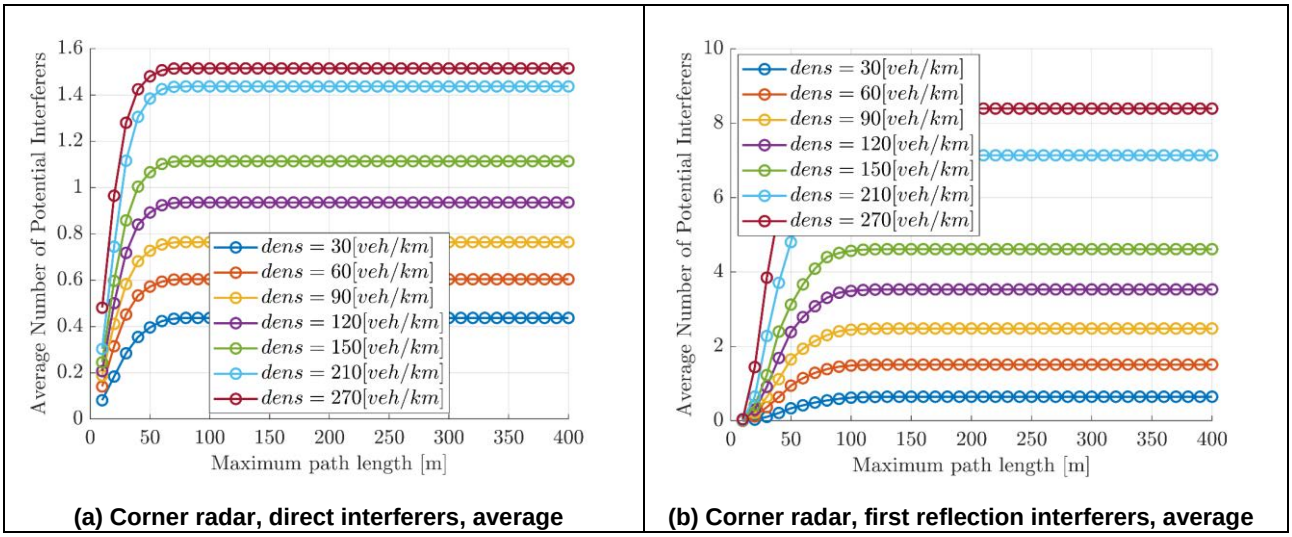
- Increasing the maximum path length increases the upper bound of the distribution.
- Increasing the density increases the upper bound of the distribution.

However, some differences between the two scenarios exist:

- The spread of the distribution increases linearly with the maximum path length for asymmetric case, while for symmetric traffic the increase seems to flatten out.
- In terms of upper bound, the asymmetric case is again overall worse when looking at direct interference and better when looking at first reflection interferers.

D.2.3 Highway scenario (corner radars)

Figure D.5 provides results for corner radars under symmetrical highway scenario.



(a) Corner radar, direct interferers, average

(b) Corner radar, first reflection interferers, average

Figure D.5: Average number of potential interferers of potential interferers varying the density for corner radars, symmetrical highway

It can be observed that the average number of direct potential interferers increases with the vehicle density and all the curves seem to reach a plateau at about 70 meters. This effect is due to the geometry of the road, with only six lanes in total. Furthermore it can be seen that the average number of direct potential interferers does not increase linearly as a function of the density: for instance there is very little difference between 210 and 270 veh/km.

The average number of first reflection potential interferers also increases with the vehicle density and also all the curves seem to reach a plateau, although at a larger distance, around 120 meters. The relationship between the average number of first reflection potential interferers and the vehicles density is more linear than for direct interferers, yielding much more spread in the number of first reflection potential interferers.

Annex E:
Detailed results mitigation and sharing mechanisms

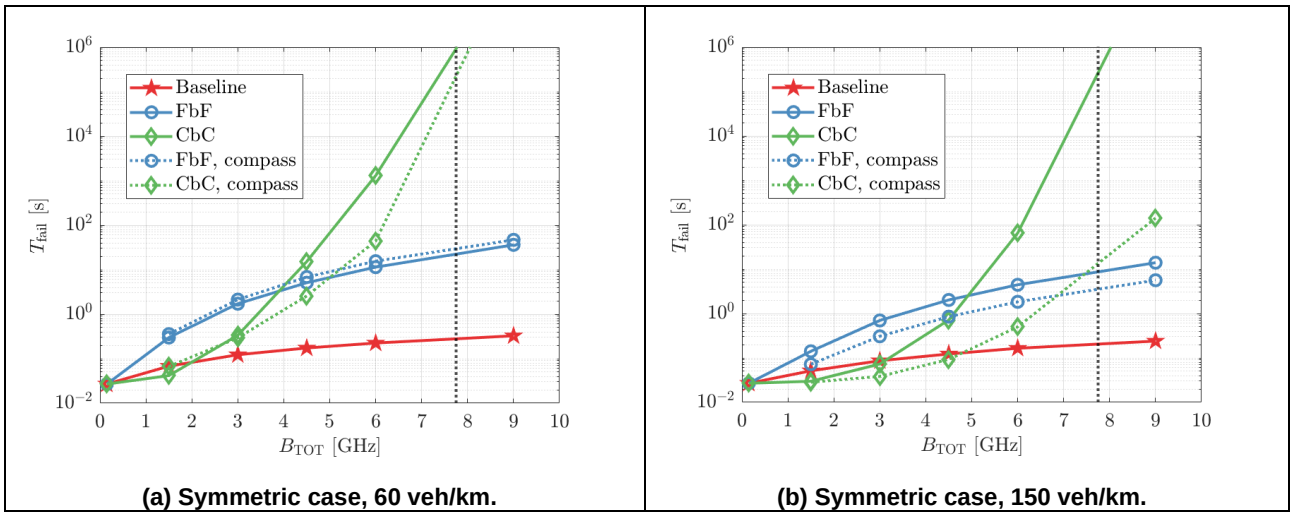
E.1 Front radars

E.1.1 Average time between system failures

E.1.1.1 Comparison of the main methods

Figure E.1 shows T_{fail}, the average time between system failures as function of the total available bandwidth, B_{TOT}, for a symmetric highway scenario, for densities of 60 and 150 veh/km.

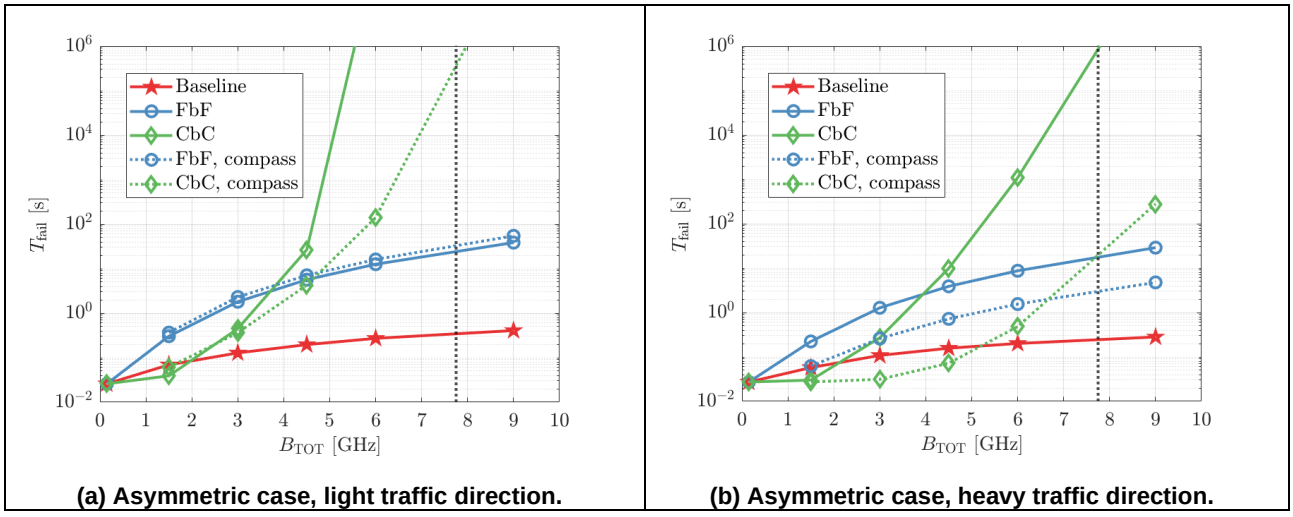
It should be noted that for all the figures in the present document showing T_{fail} as a function of B_{TOT}, the points for which less than 5 occurrences have been found are treated as not being statistically relevant, and have been assigned the value of 10^8. This means that values higher than 10^4 may not be statistically relevant and should be treated carefully.



(a) Symmetric case, 60 veh/km. **(b) Symmetric case, 150 veh/km.**

Figure E.1: Average time between system failures as function of the total available bandwidth, symmetrical highway scenario

Figure E.2 shows the average time between system failures as function of the total available bandwidth, for an asymmetric highway scenario set to 270 veh/km density, for the low-density and high-density directions separately.



(a) Asymmetric case, light traffic direction. **(b) Asymmetric case, heavy traffic direction.**

Figure E.2: Average time between system failures as function of the total available bandwidth, asymmetrical highway scenario

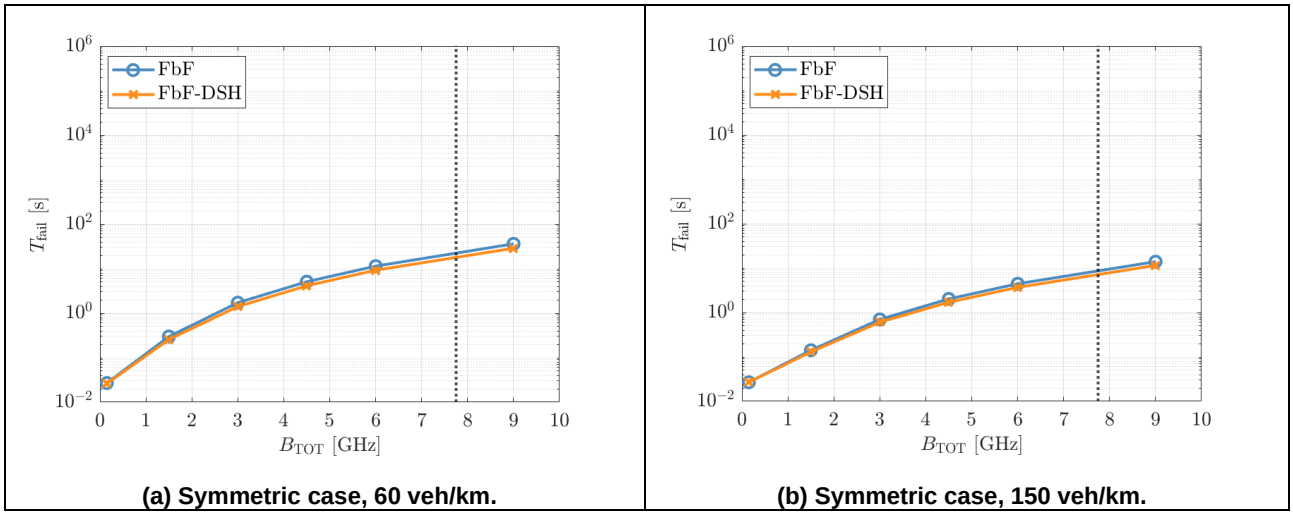
For front radars, it can be seen that the results for asymmetrical highway in Figure E.2 are remarkably similar to the ones obtained for the symmetrical highway (shown in the above Figure E.1), both in terms of trends and levels of Average time between system failures.

The following trends can be observed:

- The Baseline method provides the worst performance. Results improve with larger total available bandwidth in a logarithmic shape. The average time between total available failures stays below 0,3 - 0,4 s even for large total available bandwidths (e.g. 9 GHz).
- The Frame-by-Frame frequency hopping method manages to improve the results by 1 or 2 orders of magnitude compared to the Baseline method. Results also improve with larger total available bandwidth in a logarithmic shape. For 9 GHz they reach approximately 36 s and 14 s for symmetrical highway for 60 and 150 veh/km densities, respectively, and 38 s and 28 s for asymmetrical highway for the light and heavy traffic directions respectively.
- The Chirp-by-Chirp frequency hopping method instead yields an exponential shape. This method provides the worst results for the minimum total available bandwidth (e.g. 1 or 2 GHz), then matches the performance of the Frame-by-Frame frequency hopping method for 4 or 5 GHz of total available bandwidth, and exponentially improves for larger available bandwidths reaching average time between system failures of more than 10^6 s for 7 - 8 GHz.
- The Compass method yielded much worse performance than without compass when in conjunction with Chirp-by-Chirp Hopping for all scenarios (symmetrical and asymmetrical highway, low and high densities). When in conjunction of Frame-by-Frame Hopping, the Compass method yielded a very small improvement for symmetrical highway 60 veh/km and asymmetrical highway on the light traffic direction, and a severe degradation for symmetrical highway 150 veh/km and asymmetrical highway on the heavy traffic direction. It is therefore concluded that the Compass method is not useful.

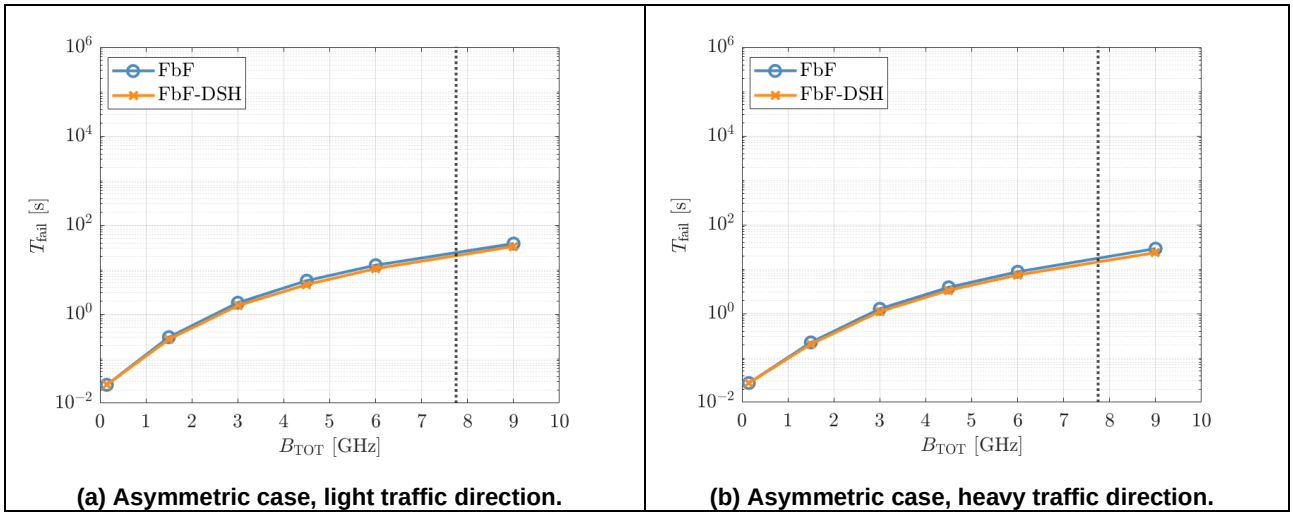
E.1.1.2 Variations of the frame-by-frame frequency hopping method

The following figures show performance of a variation of the frame-by-frame frequency hopping method, the DSH method as described in clause 8.3.5.3. Figure E.3 shows the average time between system failures as function of the total available bandwidth, for a symmetric highway scenario, for densities of 60 and 150 veh/km. Figure E.4 shows the same for an asymmetric highway scenario.



(a) Symmetric case, 60 veh/km. **(b) Symmetric case, 150 veh/km.**

Figure E.3: Average time between system failures as function of the total available bandwidth, for variations of the frame-by-frame frequency hopping method, symmetrical highway



(a) Asymmetric case, light traffic direction. **(b) Asymmetric case, heavy traffic direction.**

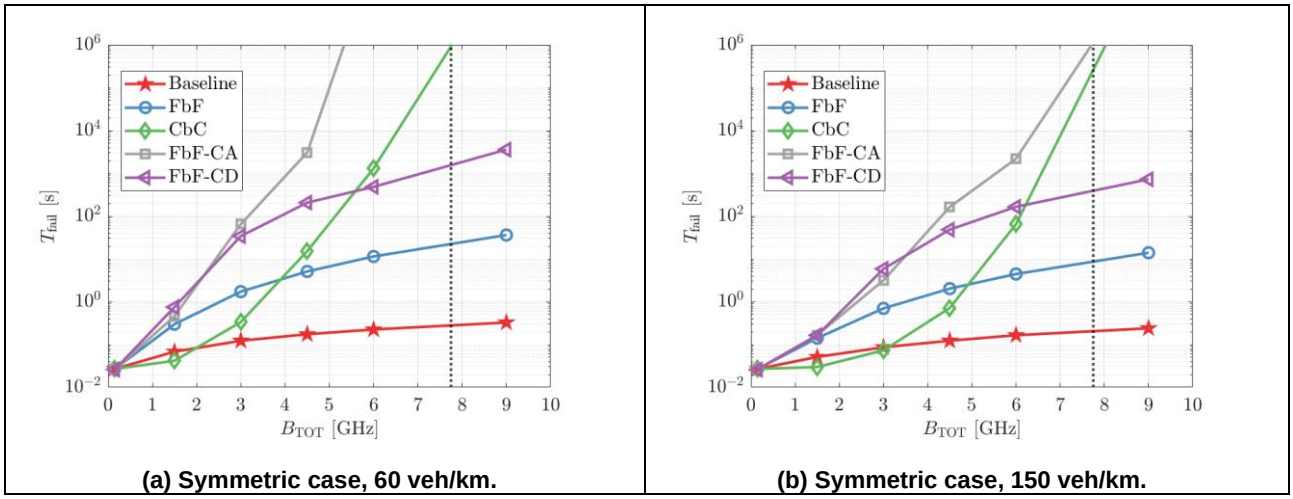
Figure E.4: Average time between system failures as function of the total available bandwidth, for variations of the frame-by-frame frequency hopping method, asymmetrical highway

It can be seen from both figures that the performance of the DSH variation is always very close to the one from the frame-by-frame frequency hopping method.

E.1.1.3 Collision Detection (CD) and Collision Avoidance (CA)

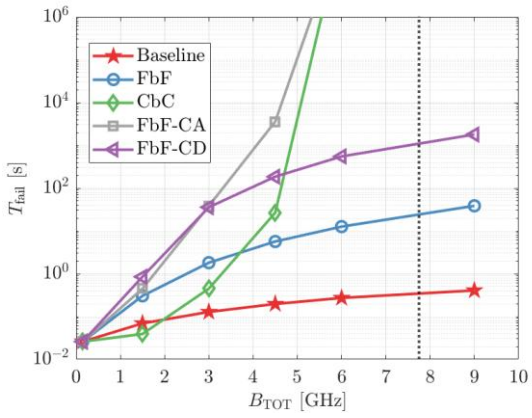
The following figures show performance of the Collision Detection (CD) and Collision Avoidance (CA) mitigation methods, as described in clauses 8.3.5.6 and 8.3.5.7. It should be noted that the simulation results presented here assume that all the radars of the simulation implement the same mechanism (either CA or CD) and have the same chirp plans (same power, same signal bandwidth, same number of chirps per frame, just a random starting time). Thus, the results presented below can be seen as the upper-bound performance of these mitigation mechanisms. If part of the radars would select their resources randomly, this would greatly reduce the efficacy of the CD or CA mechanisms.

Figure E.5 shows the average time between system failures as function of the total available bandwidth, for a symmetric highway scenario, for densities of 60 and 150 veh/km. Figure E.6 shows the same for an asymmetric highway scenario.

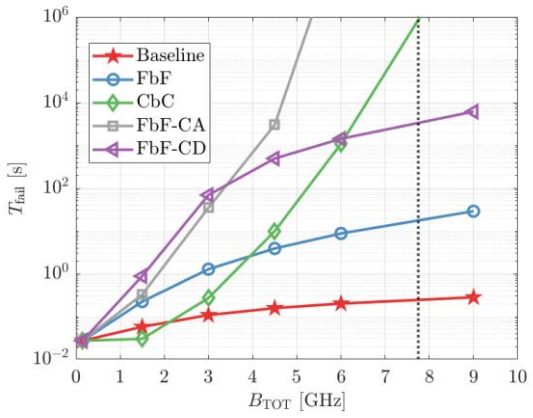


(a) Symmetric case, 60 veh/km. **(b) Symmetric case, 150 veh/km.**

Figure E.5: Average time between system failures as function of the total available bandwidth, for CA and CD methods, symmetrical highway



(a) Asymmetric case, light traffic direction.



(b) Asymmetric case, heavy traffic direction.

Figure E.6: Average time between system failures as function of the total available bandwidth, for CA and CD methods, asymmetrical highway

From both figures it is clear CA and CD yield significantly better performance than the default random frame-by-frame frequency hopping scheme. In all the four subplots, it can be observed that for low B_{TOT} values such as up to 3 GHz, the performance of CD is slightly better than CA, with both being substantially better than the default random frame-by-frame frequency hopping, while for larger B_{TOT}, CA outperforms CD and even more frame-by-frame frequency hopping.

Over the four subplots, it appears that the curves representative of CD have a logarithmic shape, and that the gap between CD and frame-by-frame is rather constant, between 2 and 3 orders of magnitude.

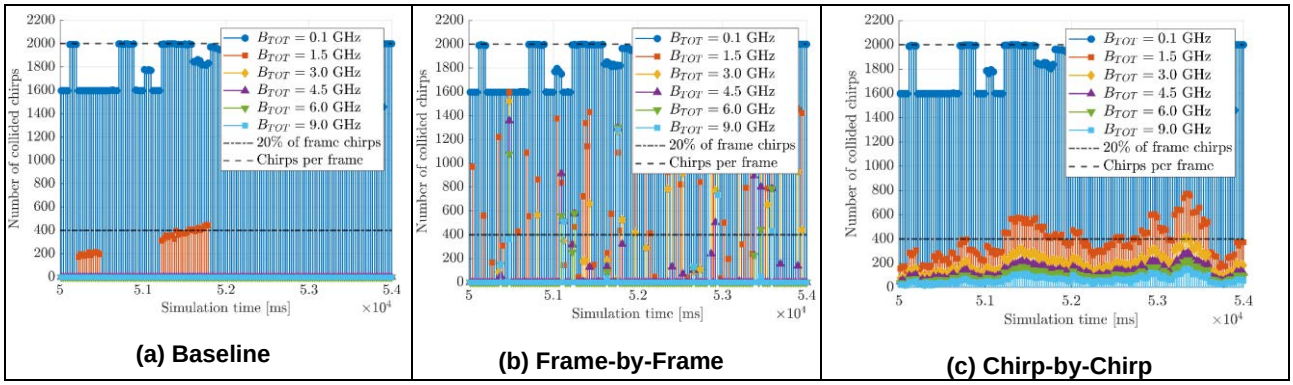
For CA the shape is exponential, similar to what was observed for the chirp-by-chirp frequency hopping method. Remarkably, the performance of CA is even better than the performance of the chirp-by-chirp frequency hopping method by about two orders of magnitude.

Overall, it can be concluded that Collision Detection (CD) generally provide best performance for small system bandwidths (e.g. B_{TOT} values $\leq$ 3 GHz) and that Collision Avoidance (CA) is the best performing method when sufficient spectrum is available (e.g. B_{TOT} values > 3 GHz). Chirp-by-chirp was the best performing method only when considering large bandwidths (e.g. B_{TOT} values > 6 GHz) in conjunction with dense symmetrical highway.

E.1.2 Number of collided chirps over the simulation time

It can be insightful to observe the number of chirps which suffered from interference for each frame throughout the simulation time. To this aim, the results shown hereafter focus on one specific vehicle, randomly selected in the scenario.

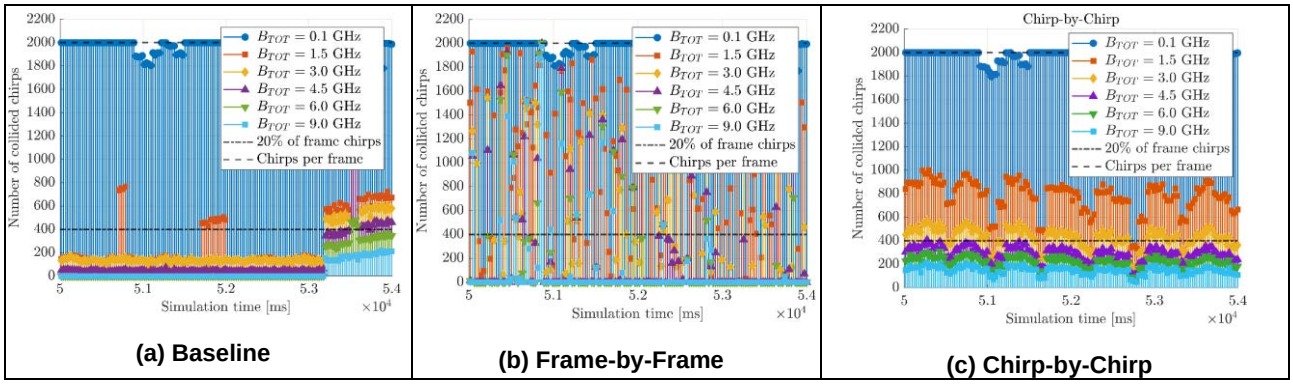
In particular, the following Figure E.7 shows an exemplary situation, with symmetrical highway scenario, with a focus on vehicle ID 153 during the time interval 50 - 54 seconds for the Baseline, Frame-by-Frame frequency hopping and Chirp-by-Chirp frequency hopping methods (in the sub-plots (a), (b) and (c), respectively) for various values of the total available bandwidth, and for a density of 60 veh/km.



(a) Baseline (b) Frame-by-Frame (c) Chirp-by-Chirp

Figure E.7: Number of collided chirps over the simulation time, symmetrical highway scenario, 60 veh/km

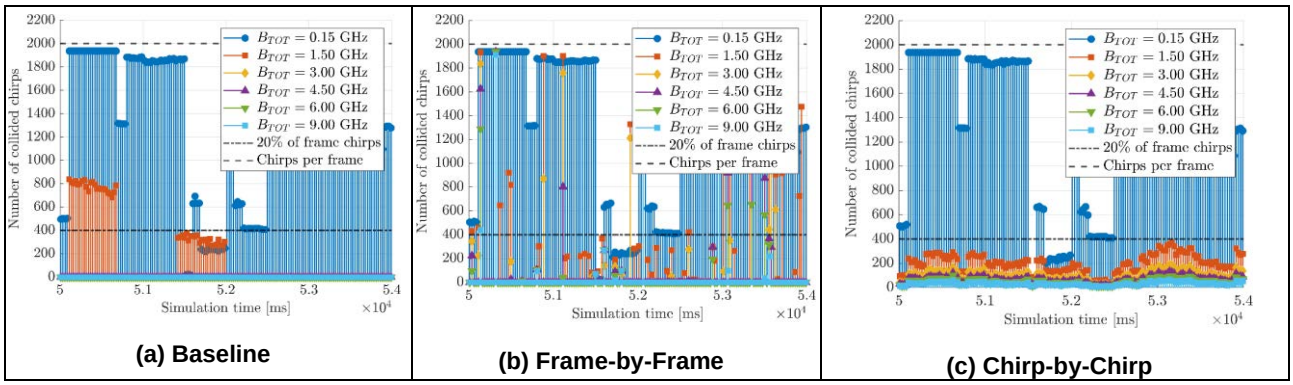
Figure E.8 shows the same scenario, and again with a focus on vehicle ID 153 during the time interval 50 - 54 seconds for the Baseline, Frame-by-Frame frequency hopping and Chirp-by-Chirp frequency hopping methods (in the sub-plots (a), (b) and (c), respectively) for various total available bandwidths, however for a higher density of 150 veh/km.



(a) Baseline (b) Frame-by-Frame (c) Chirp-by-Chirp

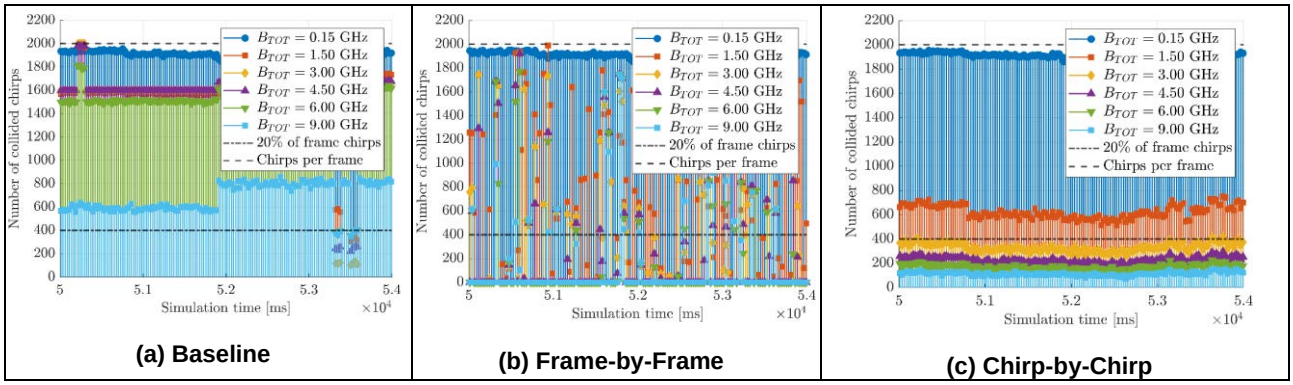
Figure E.8: Number of collided chirps over the simulation time, highway scenario, 150 veh/km

Similarly, Figure E.9 and Figure E.10 show another exemplary situation, with asymmetrical highway scenario, for light and heavy traffic directions, respectively.



(a) Baseline (b) Frame-by-Frame (c) Chirp-by-Chirp

Figure E.9: Number of collided chirps over the simulation time, asymmetrical highway scenario, light traffic direction



(a) Baseline (b) Frame-by-Frame (c) Chirp-by-Chirp

Figure E.10: Number of collided chirps over the simulation time, asymmetrical highway scenario, heavy direction

Throughout Figure E.7, Figure E.8, Figure E.9 and Figure E.10, some common patterns can be observed.

In all the subfigures(a), it can be observed that interference comes by bursts for the Baseline method, as for example between 5,13 and 5,18 seconds for B_{tot} = 1,5 GHz in Figure E.7, during which approximately 400 chirps per burst are impacted. In this particular example excerpt, there was no interference observed for larger total available bandwidth, e.g. B_{tot} = 3, 4,5, 6 or 9 GHz.

In all the subfigures(b), for the Frame-by-Frame method, a different pattern can be observed. Every timestep appears to be independent (no bursts), and all the total available bandwidth options exhibit impacted chirps, although it should be noted that the larger the total available bandwidth visually yield markers on the figure.

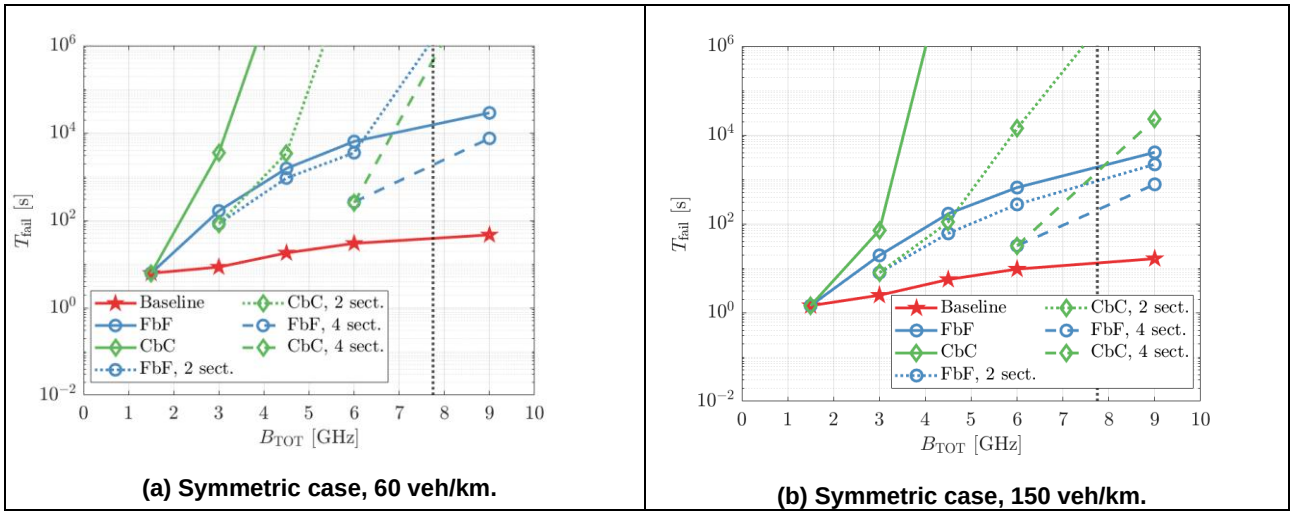
In all the subfigures(c), for the Chirp-by-Chirp method, yet another pattern appears. For every timestep, the number of collided chirps is always following the total available bandwidth options order. Also, the variation over time is relatively smooth; especially for the large total available bandwidths, there are no abrupt changes from one timestep to the next.

E.2 Corner radars

E.2.1 Average time between system failures

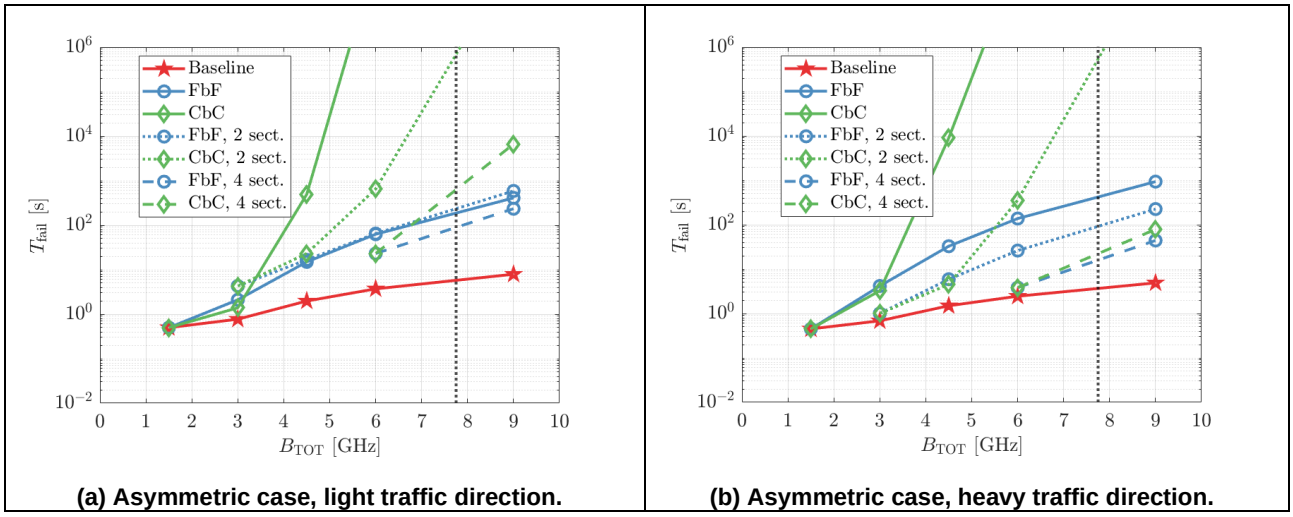
E.2.1.1 Comparison of the main methods

Figure E.11 shows the average time between system failures as function of the total available bandwidth, for a symmetric highway scenario, for densities of 60 and 150 veh/km.



(a) Symmetric case, 60 veh/km. (b) Symmetric case, 150 veh/km.

Figure E.11: Average time between system failures as function of the total available bandwidth, symmetrical highway scenario



(a) Asymmetric case, light traffic direction. **(b) Asymmetric case, heavy traffic direction.**

Figure E.12: Average time between system failures as function of the total available bandwidth, asymmetrical highway scenario

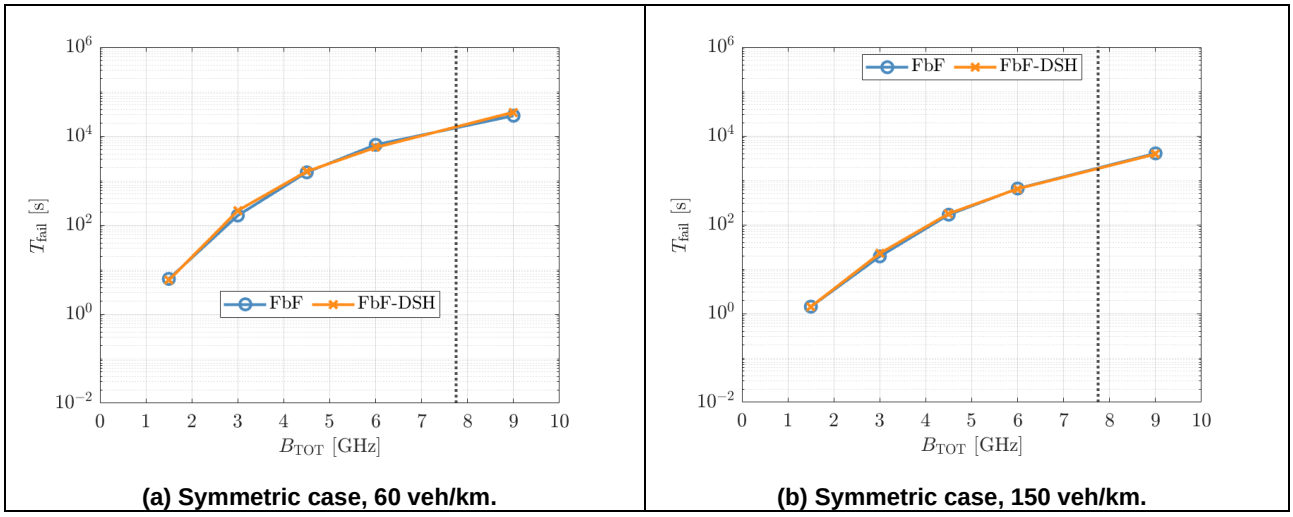
From Figure E.11 and Figure E.12, it can be observed that the trends are similar, that is:

- The Baseline method provides the worst performance. Results improve very slowly with larger total available bandwidth. For large total available bandwidths (e.g. 9 GHz), the average time between system failures is 47 and 16 seconds for symmetrical highway 60 and 150 veh/km, respectively; and 8 and 5 seconds for asymmetrical both light and heavy traffic directions, respectively.
- The Frame-by-Frame frequency hopping method manages to improve the results by 2 or 3 orders of magnitude compared to the Baseline method. Results also improve with larger total available bandwidth in a logarithmic shape. For 9 GHz, they reach approximately 29 000 s and 4 000 s for symmetrical highway 60 and 150 veh/km densities, respectively; and approximately 400 seconds and 900 for asymmetrical light and heavy traffic directions, respectively.
- The Chirp-by-Chirp frequency hopping method instead yields an exponential shape. This method provides the best results starting from the minimum total available bandwidth (e.g. 1 or 2 GHz), and exponentially improves for larger bandwidth reaching average time between system failures of more than 10^6 s for 4 - 5 GHz.
- The Compass method yielded much worse performance than without compass when in conjunction with Chirp-by-Chirp Hopping for all scenarios (symmetrical and asymmetrical highway, low and high densities, Compass with 2 or 4 sectors). When in conjunction of Frame-by-Frame Hopping, the Compass method yielded a very small improvement for symmetrical highway 60 veh/km and asymmetrical highway on the light traffic direction for high B_{TOT} values, and severe degradations for symmetrical highway 150 veh/km and asymmetrical highway on the heavy traffic direction. It is therefore concluded that the Compass method is not useful.

Comparing symmetrical highway with asymmetrical highway for corner radars, it can be seen that the overall trends are extremely similar. However, asymmetrical highway yield slightly more challenging environment with slightly lower average time between system failures (approximately 1 order of magnitude lower values).

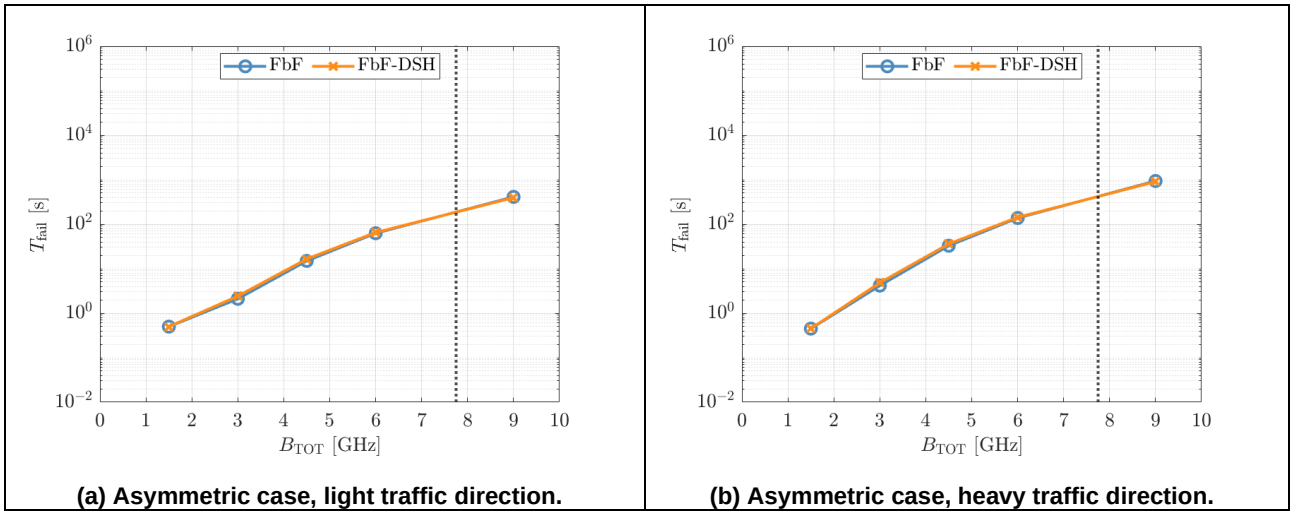
E.2.1.2 Variations of the frame-by-frame frequency hopping method

The following figures show performance of a variation of the frame-by-frame frequency hopping method, the DSH method as described in clause 8.3.5.3. Figure E.13 shows the average time between system failures as function of the total available bandwidth, for a symmetric highway scenario, for densities of 60 and 150 veh/km. Figure E.14 shows the same for an asymmetric highway scenario.



(a) Symmetric case, 60 veh/km. **(b) Symmetric case, 150 veh/km.**

Figure E.13: Average time between system failures as function of the total available bandwidth, for variations of the frame-by-frame frequency hopping method, symmetrical highway



(a) Asymmetric case, light traffic direction. **(b) Asymmetric case, heavy traffic direction.**

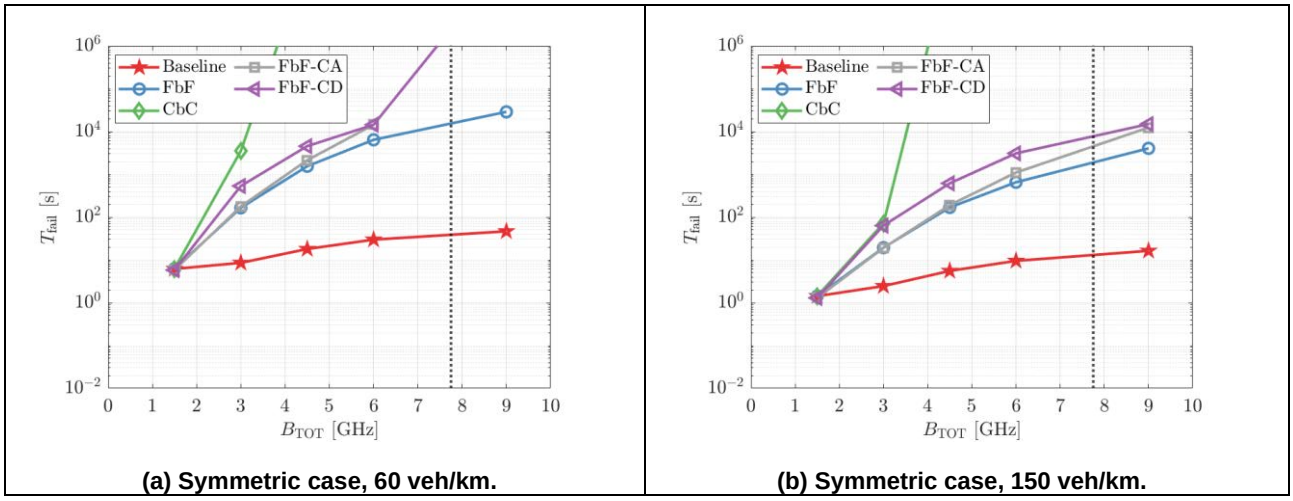
Figure E.14: Average time between system failures as function of the total available bandwidth, for variations of the frame-by-frame frequency hopping method, asymmetrical highway

It can be seen that the performance of the DSH variation is very close to the one from the frame-by-frame frequency hopping method.

E.2.1.3 Collision Detection (CD) and Collision Avoidance (CA)

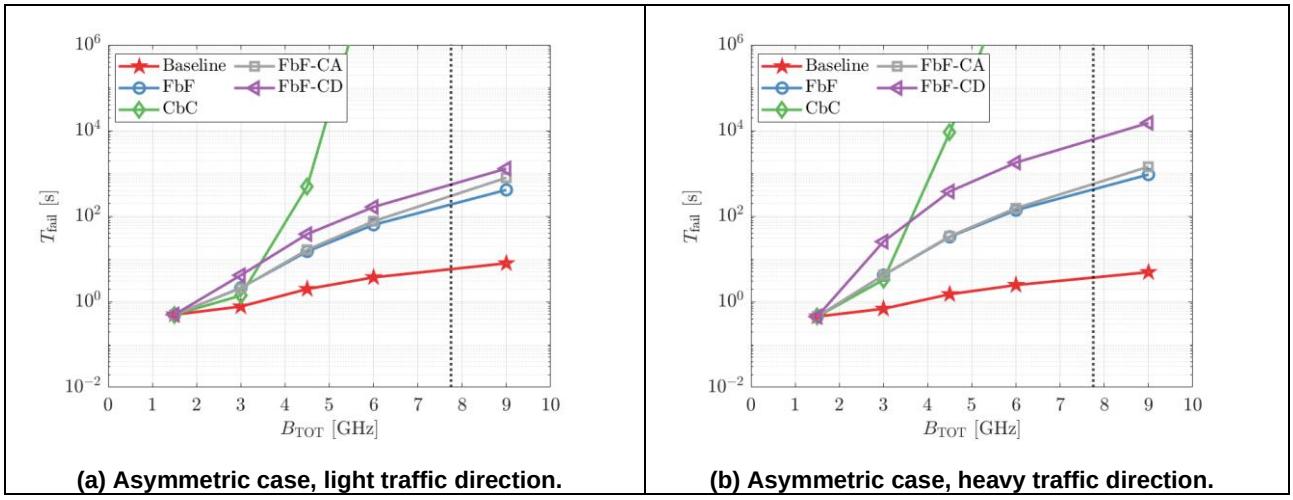
The following figures show performance of the Collision Detection (CD) and Collision Avoidance (CA) mitigation methods, as described in clauses 8.3.5.6 and 8.3.5.7.

Figure E.15 shows the average time between system failures as function of the total available bandwidth, for a symmetric highway scenario, for densities of 60 and 150 veh/km. Figure E.16 shows the same for an asymmetric highway scenario.



(a) Symmetric case, 60 veh/km. (b) Symmetric case, 150 veh/km.

Figure E.15: Average time between system failures as function of the total available bandwidth, for CA and CD methods, symmetrical highway



(a) Asymmetric case, light traffic direction. (b) Asymmetric case, heavy traffic direction.

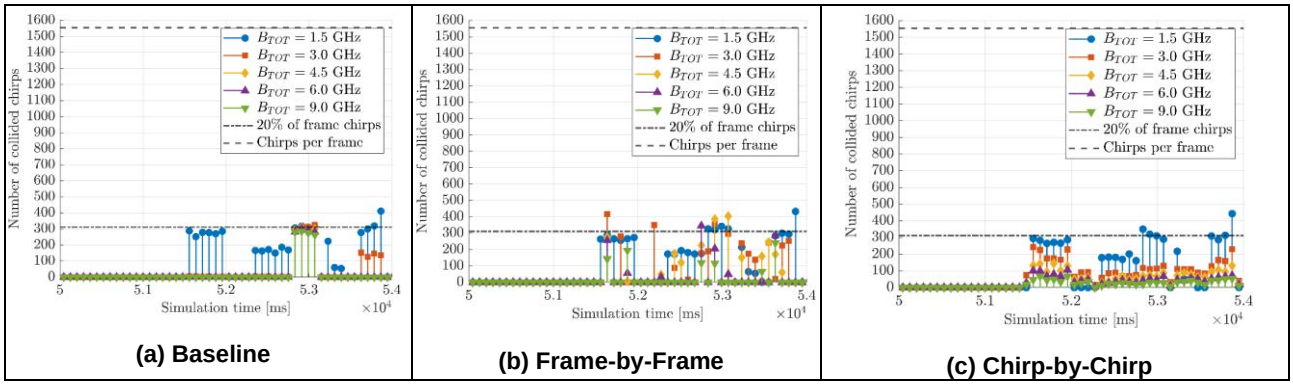
Figure E.16: Average time between system failures as function of the total available bandwidth, for CA and CD methods, asymmetrical highway

From both pictures, and in contrast with the figures for front radars, it can be seen that overall the improvement from CA and CD methods is much less pronounced compared to the default frame-by-frame frequency hopping method, especially for high density situations. It appears that CD helps improving the performance more constantly by approximately one order of magnitude, while CA is not providing as much improvement. For corner radars, chirp-by-chirp frequency hopping remains by far the best performing method for B_{TOT} values > 3 GHz.

E.2.2 Number of collided chirps over the simulation time

Also for corner radars, the number of chirps which suffered from interference for each frame throughout the simulation time are observed. To this aim, the results shown hereafter focus on one specific vehicle, randomly selected in the scenario.

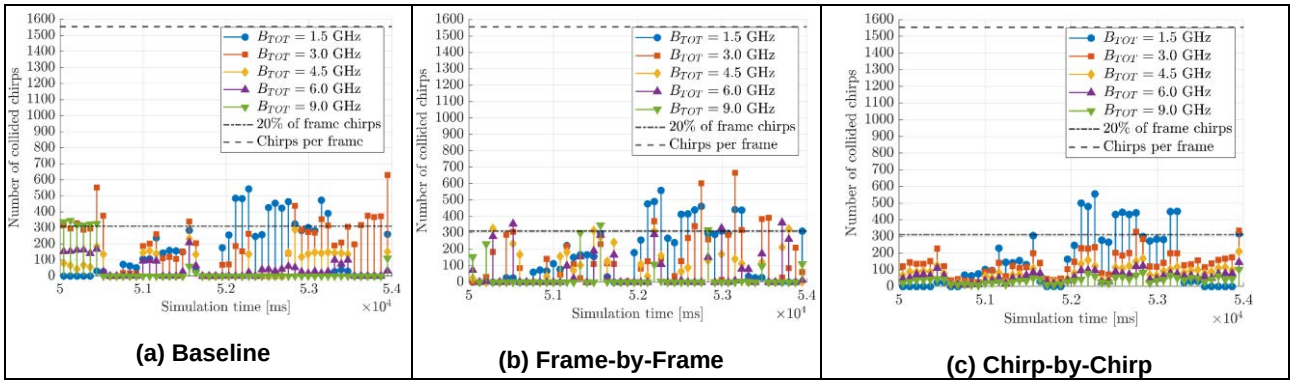
In particular, Figure E.17 shows an exemplary situation, with symmetrical highway scenario, with a focus on vehicle ID 153 during the time interval 50 - 54 seconds for the Baseline, Frame-by-Frame frequency hopping and Chirp-by-Chirp frequency hopping methods (in the sub-plots (a), (b) and (c), respectively) for various values of the total available bandwidth, and for a density of 60 veh/km. It should be noted that the same scale is used as for the front radar figures.



(a) Baseline (b) Frame-by-Frame (c) Chirp-by-Chirp

Figure E.17: Number of collided chirps over the simulation time, symmetrical highway scenario, 60 veh/km

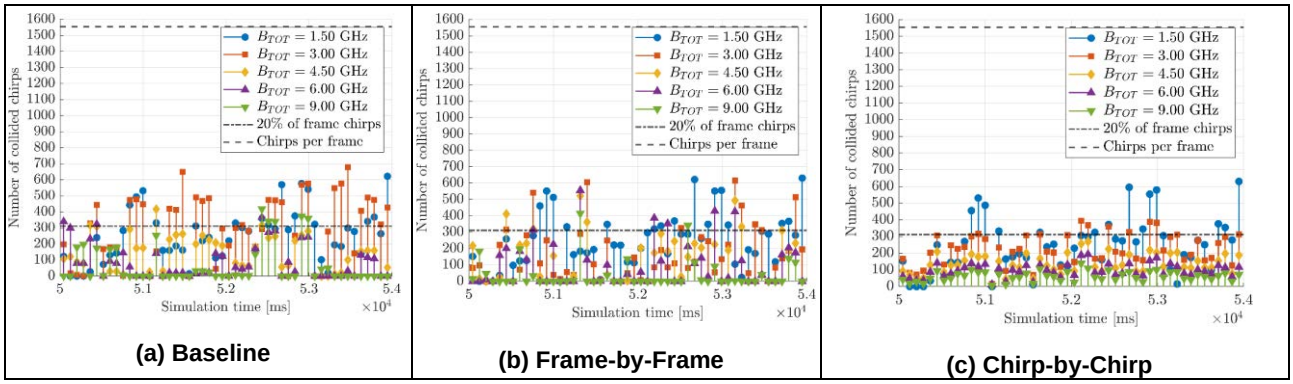
Figure E.18 shows the same scenario, and again with a focus on vehicle ID 153 during the time interval 50 - 54 seconds for the Baseline, Frame-by-Frame frequency hopping and Chirp-by-Chirp frequency hopping methods (in the sub-plots (a), (b) and (c), respectively) for various total available bandwidths, however for a higher density of 150 veh/km.



(a) Baseline (b) Frame-by-Frame (c) Chirp-by-Chirp

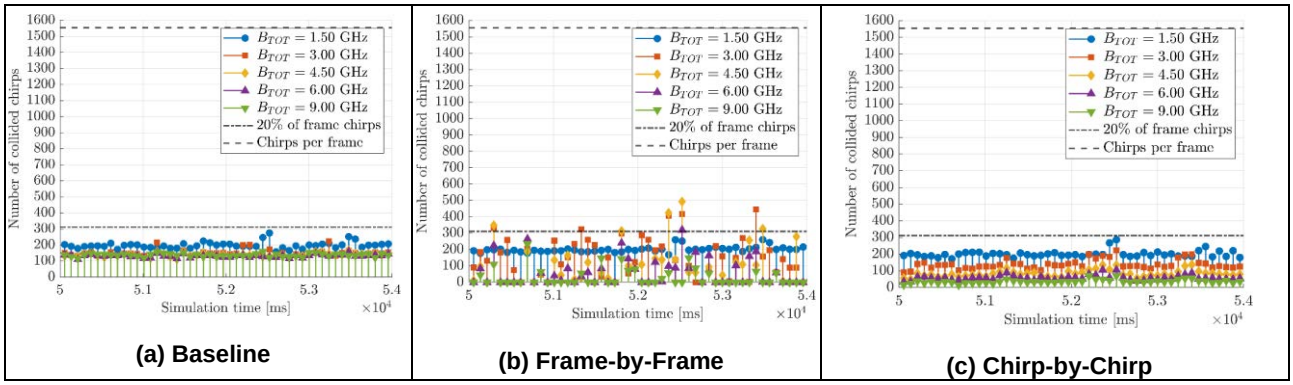
Figure E.18: Number of collided chirps over the simulation time, symmetrical highway scenario, 150 veh/km

Similarly, Figure E.19 and Figure E.20 show another exemplary situation, with asymmetrical highway scenario, for light and heavy traffic directions, respectively.



(a) Baseline (b) Frame-by-Frame (c) Chirp-by-Chirp

Figure E.19: Number of collided chirps over the simulation time, asymmetrical highway scenario, light traffic direction



(a) Baseline **(b) Frame-by-Frame** **(c) Chirp-by-Chirp**

Figure E.20: Number of collided chirps over the simulation time, asymmetrical highway scenario, heavy direction

In all the subfigures (a), it can be observed that interference comes by bursts for the Baseline method, as for example between 5,13 and 5,18 seconds for Figure 33 for B_{tot} = 1,5 GHz, during which approximately 300 chirps per burst are impacted. In this particular example excerpt, there was no interference observed for larger total available bandwidths, e.g. B_{tot} = 3, 4,5, 6 or 9 GHz.

In all the subfigures (b), for the Frame-by-Frame method, a different pattern can be observed. Every timestep appears to be independent (no bursts), and all the total available bandwidth options exhibit impacted chirps, although it should be noted that the larger the total available bandwidth visually yield markers on the figure.

In all the subfigures (c), for the Chirp-by-Chirp method, yet another pattern appears. For every timestep, the number of collided chirps is always following the total available bandwidth options order. Also, the variation over time is relatively smooth; especially for the large total available bandwidths, there are no abrupt changes from one timestep to the next.

E.3 Comparison between front and corner radars

For the average time between system failures, while the overall shape of the curve representative of each method is similar between corner and front radars, it can be noted that for corner radars, the Chirp-by-chirp method is better than the others from lowest bandwidth possible, while for front radars it needed 4 or 5 GHz of total available bandwidth to match the performance of the Frame-by-Frame frequency hopping method. Overall, it can clearly be observed that corner radars exhibit a higher average time between system failures compared to front radar, typically by 2 or 3 orders of magnitude, indicative of the interference effect being more pronounced for front radars. This aligns well with results seen for potential interferers, such as in Figure 20, where front radars may face up to 13 on average, while corner radars face up to 2 (with density equal to 150 veh/km).

For the number of collided chirps over the simulation time, the distribution patterns are similar for front and corner radars, and get amplified as the vehicles density increases. For the baseline method the interference comes by bursts. The frame-by-frame hopping method yields independent results from one frame to the next, and in average the number of collided chirps decreases with the total available bandwidth. For the chirp-by-chirp hopping method, the variation over time is relatively smooth and the number of collided chirps follows the total available bandwidth options order. Overall, similarly as for the average time between system failures, corner radars yield more favourable situations with fewer interference events.

History

Version	Date	Status
V1.1.1	October 2026	Publication